KNESER'S TETRATION CONTINUED AROUND THE CUSP $\mathrm{e}^{1/\mathrm{e}}$: SEPARATION FROM REGULAR ITERATION AND THE INVERSE HORN MAP OF $\mathrm{e}^u - 1$

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ABSTRACT. For $1 < b < \eta = \mathrm{e}^{1/\mathrm{e}}$ the equation $F(z+1) = b^{F(z)}$, $F(0) = 1$, has the regular solution R_b at the attracting fixed point. We prove that Kneser's real solution for $b > \eta$, continued analytically in the base around the cusp η, crosses the Shell–Thron boundary (jointly holomorphically in base and height, whatever the arithmetic of the neutral multiplier), extends single-valued to the whole upper half of the Shell–Thron region, and has boundary values K_b on $(1, \eta)$ along every path there. These are a solution K_b^W obtained by sewing the two regular charts, characterised, for b near η, by the position of its seam. Moreover, with a computer-assisted certificate, Kneser's family continues to a single-valued holomorphic family on the whole upper half b-plane: Kneser's own straight initial region survives on the entire exterior, and the family crosses every point of the open upper Shell–Thron boundary arc. So the continuation is path independent, it coincides with Paulsen's base-continued family (read as continuation along upper half-plane paths), and it is holomorphic across every real base in $(0, \eta) \setminus \{1\}$ (in particular at the neutral endpoint $\mathrm{e}^{-\mathrm{e}}$). Writing $K_b = R_b \circ (\mathrm{id} + \sum_{n \geq 0} c_n \mathrm{e}^{2\pi \mathrm{i} n z})$ and $\Lambda = \exp(4\pi^2/\log \lambda)$, we show $c_n/\Lambda^n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ as $b \to \eta^-$ for every n, where B_n are the Fourier coefficients of the inverse Écalle–Voronin horn map of $\mathrm{e}^u - 1$ and a is the Fatou coordinate of the base point; $B_1 \neq 0$. The proof combines a confluence theorem for the transition map between the two regular charts, a contraction in a weighted Wiener algebra, and a family of tilted initial regions whose quotients are uniformly quasiconformal to a flat cylinder. The corrections have an asymptotic expansion to all orders in the unfolding parameter with coefficients given by convergent parabolic series. Numerically, the sewn solution reproduces Paulsen's 120-digit value at $b = \sqrt{2}$ in all quoted digits.

1. INTRODUCTION

The tetration equation

$$F(z+1) = b^{F(z)}, \qquad F(0) = 1, \tag{1}$$

has, for real $b > \eta := \mathrm{e}^{1/\mathrm{e}}$, a distinguished real-analytic solution constructed by Kneser [16]. It tends to the two complex-conjugate fixed points of $w \mapsto b^w$ as $\operatorname{Im} z \to \pm\infty$. Its conformal uniformisation is uniquely characterised by the initial-region criterion of [31]. For $1 < b < \eta$ the fixed points are real: $L = \mathrm{e}^{\lambda}$ is attracting, with $0 < \lambda < 1$, and $L_2 = \mathrm{e}^{\lambda_2}$ is repelling, with $\lambda_2 > 1$. The Koenigs construction at L yields the regular solution R_b. This is the solution usually meant in that range [18]. Paulsen [25] proposed continuing Kneser's two-fixed-point construction through the upper half of the Shell–Thron region. If this base continuation $\mathcal{K}$ has a boundary value, denote it by

$$K_b = \lim_{y \downarrow 0} \mathcal{K}_{b + \mathrm{i}y}. \tag{2}$$

High-precision algorithms for the relevant pointwise constructions are developed in [17, 18, 30]. A different line of work constructs non-regular solutions by infinite compositions: Nixon [22] builds, for b in the interior of the Shell–Thron region, a family of solutions of period $2\pi\mathrm{i}/\lambda$ and

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compares them with the regular solution through a 1-periodic "θ-mapping" $F = R_b \circ (\mathrm{id}+\theta)$, the normal form we also use below. That work does not treat Kneser's solution for $b < \eta$, the limit $b \to \eta$, or continuation in the base, and for $b > \eta$ the link between its construction and Kneser's solution is left as a conjecture. Here the 1-periodic correction is determined by the transition map between the two regular charts, and its Fourier coefficients are shown to be asymptotic to $\Lambda^n B_n \mathrm{e}^{2\pi i n a}$ (notation below). We construct a two-sided sewing solution K_b^W independently and determine its difference from R_b. We then prove that the continuation of Kneser's solution through the upper half-disc about η, and then near the real axis, has boundary values $K_b = K_b^W$ (Theorem D), so every result about K_b^W holds for these boundary values.

Notation. Let $h = 2\pi/|\log\lambda|$ be the imaginary period of R_b, and put

$$\Lambda = \mathrm{e}^{-2\pi h} = \exp(4\pi^2/\log\lambda).$$

Both R and K^W are normalised by $F(0) = 1$. The map $P = R^{-1} \circ K^W - \mathrm{id}$ is 1-periodic with $P(0) = 0$. In the relevant region it has only non-negative modes, $P = \sum_{n\geq 0} c_n \mathrm{e}^{2\pi i n z}$, and the separation is

$$K_b^W(z) - R_b(z) = c_1 R_b'(z)(\mathrm{e}^{2\pi \mathrm{i} z} - 1) + O(c_2) + O(c_1^2).$$

At $b = \eta$ the map becomes the parabolic germ $f(u) = \mathrm{e}^u - 1$ in $w = \mathrm{e}(1+u)$. Let $\Phi_{\mathrm{att}}, \Phi_{\mathrm{rep}}$ be its Fatou coordinates, both normalised by the formal Abel function $\alpha(u) = -2/u + \frac{1}{3}\log(-u) + O(u)$. Let

$$\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}(z) = z + \sum_{n\geq 1} B_n \mathrm{e}^{2\pi i n z}$$

be the inverse horn map on the upper gate [8, 32], and $a = \Phi_{\mathrm{att}}(1/\mathrm{e} - 1)$ the attracting Fatou coordinate of the base point $w = 1$. Numerically

$$|B_1| = 0.0890584364122133\ldots, \quad \frac{B_2}{B_1^2} = -0.0253204931 + 1.5742190652\,\mathrm{i},$$
$$a = 3.0292972144180\ldots$$

(Section 2). The transition map between the two regular solutions is $T_b = R_b^{-1} \circ S_b$, where S_b is the regular solution at the repelling point. Write $T_b(z) - z = \sum t_n \mathrm{e}^{2\pi i n z}$ and $\tau_n = t_n \mathrm{e}^{-2\pi i n t_0}$ (Section 3).

Conventions. The following symbols have one meaning throughout the paper.

- $h = 2\pi/|\log\lambda|$ and $h_2 = 2\pi/\log\lambda_2$: the periods of R_b and S_b; $\Lambda = \mathrm{e}^{-2\pi h}$.
- $\varepsilon = 1 - \lambda$, where $\lambda = \lambda_1$ is the attracting multiplier; it is real in Sections 2–8 and a complex parameter from Section 10 on.
- $s = 1 - \mathrm{e}\log b$, the unfolding parameter; $p = |\log\lambda|\log\lambda_2$.
- $a = \Phi_{\mathrm{att}}(1/\mathrm{e} - 1)$; we write $\mathcal{L} = \log b$ for the logarithm of the base (continued along paths where stated).
- $\mathfrak{h}$: the horn map, $\mathfrak{h}^{-1} = \Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}$.
- σ, σ_{rep}: the Koenigs coordinates at the attracting and repelling fixed points.
- $T = R^{-1} \circ S$, its coefficients t_n, and the invariants $\tau_n = t_n \mathrm{e}^{-2\pi i n t_0}$; $T^\circ = T - \mathrm{id} - t_0$.
- c_n, $\hat{c}_n = c_n/\Lambda^n$, and the gauge-free ratios $\Xi_n = c_n/c_1^n$.
- $\kappa_j^{(n)}$: the coefficients of the p-expansion, $\kappa^{(n)} = \kappa_1^{(n)}$.
- $\mathcal{K}_b$: Kneser's solution continued in the base; K_b its boundary value and K_b^W the sewing solution.

Indexed families are separate symbols: $a_2, a_3, \ldots$ are Taylor coefficients, $s_*(b)$ is a normalising shift, and s_θ is the seam map of Section 10.10. Letters introduced inside a proof, and sans-serif letters such as r, are local to the proof or subsection in which they are defined.

Main results.

Theorem A (Confluence). *For every $n \geq 1$, $\tau_n(b) \to B_n e^{2\pi i n a}$ as $b \to \eta^-$. Moreover, $B_1 \neq 0$ and $t_n/t_1^n \to B_n/B_1^n$ for every $n \geq 1$.*

Theorem B (The sewing solution). *There is $b_1 < \eta$ such that for $b \in (b_1, \eta)$ the following holds. There is exactly one solution K_b^W of (1) with the following three properties:*

- *it has a representation $R_b \circ (\mathrm{id} + P)$ on an upper half-plane;*
- *it has a representation $S_b \circ (\mathrm{id} + Q)$ on a lower half-plane;*
- *P and Q are bounded and 1-periodic, and the seam lies half a period below the real axis.*

The precise statement is Definition 8.1. Its coefficients satisfy

$$\left| \frac{c_n}{\Lambda^n} - \tau_n(b) \right| \leq C_n \Lambda$$

with explicit constants C_n independent of b.

Corollary 1.1. *For K_b^W, $\hat{c}_n := c_n/\Lambda^n \to B_n e^{2\pi i n a}$ for every $n \geq 1$. In particular*

$$\lim_{b \to \eta^-} \frac{|c_1|}{\Lambda} = |B_1|.$$

The limiting phase is $\arg B_1 + 2\pi a$, and $c_n/c_1^n \to B_n/B_1^n$ for every $n \geq 1$.

In the coordinate $u = (w - e)/e$ the map η^w becomes $e^u - 1$, the germ used by Trappmann and Kouznetsov [30] at $b = \eta$. Their Abel function α has the normalisation of Φ_{att}, so $a = \alpha(\zeta(1))$ with $\zeta(z) = (z - e)/e$; the two agree to 40 digits. Paulsen [26, Prop. 1] shows that $\mathcal{K}_b(z)$ tends to their regular superexponential through 1 as $b \to \eta$ off $(-\infty, \eta]$. Corollary 1.1 describes the non-regular part that is lost in this limit: it is of size Λ, with the constants $B_n e^{2\pi i n a}$.

Theorem C (First-order term). *Let $s = 1 - e \log b$ and $p = |\log \lambda| \log \lambda_2 = 2s + O(s^2)$. For every $n \geq 1$ with $B_n \neq 0$ (in particular $n = 1$ by Proposition 2.4),*

$$\log \frac{\tau_n(b)}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^2) \qquad (b \to \eta^-),$$

where $\kappa^{(n)}$ is given by an absolutely convergent series over orbits of the parabolic germ $e^u - 1$ (Theorem 5.6; the remainder $O(p^2)$ is Theorem 5.8 with $m = 1$). More generally, $\log(\tau_n/B_n e^{2\pi i n a})$ has an asymptotic expansion $\sum_{j \geq 1} \kappa_j^{(n)} p^j$ to all orders, and every coefficient is built from convergent parabolic series of the same kind (Theorem 5.8).

The sewn solution is defined for $b < \eta$ without reference to Kneser's construction. The next theorem shows that it is exactly what Kneser's solution becomes when continued in the base around the cusp.

Theorem D (Continuation around the cusp). *Kneser's real-normalised solution for real $b > \eta$ continues analytically in the base through the upper half-disc about η, across the Shell–Thron boundary arc there (whatever the arithmetic of the neutral multiplier), and single-valued along every path in the upper half of the Shell–Thron region. Its boundary values on $(1, \eta)$ are $K_b = K_b^W$, so Corollary 1.1 holds for K_b: $c_n/\Lambda^n \to B_n e^{2\pi i n a}$ for every $n \geq 1$, and $|c_1|/\Lambda \to |B_1| > 0$. Continuation through the lower half-disc gives the complex-conjugate branch.*

Theorem E (The whole upper half-plane; computer-assisted). *The continuation of Theorem D is a single-valued holomorphic family on the whole upper half b-plane. It crosses every point of the open upper Shell–Thron boundary arc holomorphically, it is holomorphic at the neutral endpoint $b = e^{-e}$, and it extends holomorphically across every point of $(0, \eta) \setminus \{1\}$; $b = 0$ and $b = 1$ are genuine singularities. In particular the continuation is path independent, and it is*

the base-continued family of [25]*, read as the continuation from* (η,∞) *along paths in the upper half-plane (Corollary 10.41).*

The precise statements are Theorems 10.14 and 10.22 for Theorem D, and Theorems 10.33, 10.34, 10.40, 10.43, 10.45 and Proposition 10.46 for Theorem E.

- *Around the cusp* (Section 10), Kneser's initial region (the chord between the fixed points and its image) is tilted continuously along the path. In the coordinate $\log\frac{u-u_1}{u-u_2}$ it is a thin strip whose quotient is uniformly quasiconformal to a flat cylinder, so the neutral boundary is no obstacle. At real $b<\eta$ it is a fundamental domain of the sewn surface, and the seam condition of Definition 8.1 identifies the limit with K_b^W. A quasiconformal transport of the dynamics extends the family over the whole upper Shell–Thron region (Section 10.9).
- *Away from the cusp* (Sections 10.10–10.13), Kneser's own straight chord remains a crescent with a canonical marked point on the whole exterior part of the upper half-plane and across the boundary arc. This is verified by interval arithmetic, in a parametrisation by the multiplier of the continued fixed point.
- The family of [25] is defined by analytic continuation in the base, but there the crossing of the Shell–Thron boundary is argued only numerically, and the uniqueness statement needs an additional root-selection condition (Remark 1.2). Theorem E settles the first point. The nine-digit agreement at $b=\sqrt{2}$ is a numerical check.

By Lemma 2.1 and Proposition 2.2, Λ has the saddle-node form

$$\Lambda=\exp\bigl(-C_\eta(\eta-b)^{-1/2}(1+o(1))\bigr),\qquad C_\eta=\frac{4\pi^2}{\sqrt{2\mathrm{e}^{1-1/\mathrm{e}}}}=20.3508\ldots.$$

Thus the coefficients of the separation have the stated exponentially small upper bounds. Proposition 2.4 makes the first-mode asymptotic sharp with a strictly positive constant, and Proposition 2.6 gives a certified enclosure of it.

Numerical results. Theorems A and B are confirmed and extended numerically in Section 11.

(1) Ladder computations of K_b reproduce τ_1 and t_2/t_1^2 to their full precision, arguments included; for example, eight digits at $b=1.1$.
(2) At $b=\sqrt{2}$, far from η, the sewing equation gives $\operatorname{Im}K^W_{\sqrt{2}}(\frac12)=-1.18899697185\cdot10^{-48}$, which agrees with Paulsen's 120-digit value $-1.18899697\cdot10^{-48}$ for his base-continued family [25, §6] in all nine quoted digits, sign included, as Theorem E predicts.
(3) τ_1, t_2/t_1^2 and t_3/t_1^3 have been followed up to $\eta-b=3\cdot10^{-5}$.
(4) The parabolic series of Theorem C gives

$$\log\frac{\tau_1}{B_1\mathrm{e}^{2\pi\mathrm{i}a}}=(-0.010199006345897402441-0.013250956182506712250\,\mathrm{i})\,p+O(p^2),$$

in agreement with fits from the side $b<\eta$ to their eleven digits. The second coefficient, $\kappa_2^{(1)}=-7.88992074991511334\cdot10^{-5}+3.6554727325799459\cdot10^{-4}\,\mathrm{i}$, agrees with the fits to within one unit in their sixth digit.

Method. Four observations drive the proofs.

(1) The separation is a property of the transition map T_b between two classical Koenigs coordinates. A welding identity expresses c_n through the Fourier modes of T_b (Theorem 3.2). The scale Λ appears because the seam sits half a period below the real axis, and the gate carrying the positive modes a further half period below it.

(2) The limit of T_b is a confluence problem for the saddle-node unfolding in the direction where both fixed points are real (the "Glutsyuk" direction [14, 21, 6]). In the complementary Lavaurs direction, the corresponding statements are the classical results of parabolic implosion [19, 29, 23]. For vector fields, see [28]. We prove the needed statement directly (Section 4), using the model time $A_1 \log(u - u_1) + A_2 \log(u - u_2)$, whose one-step error vanishes at both fixed points.

(3) The Kneser-type solution is the conformal sewing of two half-cylinders along T_b. A contraction in a weighted Wiener algebra gives it with explicit bounds (Theorem 6.3). A proper-map argument proves uniqueness in the class of Definition 8.1 (Theorem 8.2).

(4) Kneser's initial region can be tilted continuously around the cusp. In the coordinate $\log \frac{u-u_1}{u-u_2}$ it is a thin strip on which the map is a translation up to relative error $O(|\varepsilon|)$, so its quotient is uniformly quasiconformal to a flat cylinder, whatever the multiplier. The Ahlfors–Bers theorem gives holomorphic dependence on the base, and at $b < \eta$ the tilted region is a fundamental domain of the sewn surface (Section 10).

Organisation. Section 2 collects the parametrisation and the parabolic data. Section 3 introduces T_b, the welding identity and the sewing construction. Section 4 proves Theorem A, and Section 5 proves Theorem C. Sections 6–8 prove Theorem B (existence with Wiener bounds, uniqueness) and its base dependence, and Section 10 proves Theorem D, including the crossing away from the cusp (Section 10.10). Section 9, placed before it, states Theorems A, B and C for general real saddle-node unfoldings and proves that the moduli $|\tau_n|$ do not depend on the base point. Section 11 reports the numerics, and Section 12 lists open problems.

Remark 1.2 (On the definition and uniqueness in [25])**.** In [25], $\mathcal{K}_b$ is defined by analytic continuation in b of Kneser's solution $\mathcal{K}_b(z_0)$, $\operatorname{Re} z_0 > -2$, from (η, ∞) (Prop. 3 there). That proposition asserts real-analytic dependence of Kneser's conformal uniformisation on the real base without a separate parameter-dependence proof; Proposition 10.11 supplies such a proof locally along the real exterior ray. Near the cusp, Theorem D carries the family through the Shell–Thron boundary by a different route. Away from the cusp, Theorems 10.33 and 10.34 cross the boundary arcs with Kneser's untilted chord. Off that boundary, Proposition 1 of that paper asserts two regular representations $\psi_1^{-1}(\rho_1)$ and $\psi_2^{-1}(\rho_2)$ from the limits at $\pm i\infty$. Its proof starts from $\psi_1(F(z+1)) = \psi_1(F(z)) + 1$; since $\psi_1 = \log \sigma_1 / \log s_1$ is multivalued, continuity on $\operatorname{Im} z > M$ only gives this up to a constant $2\pi i m / \log s_1$ with $m \in \mathbb{Z}$. We impose their degree-one form explicitly in Definition 8.1. A fixed-point limit alone leaves an integer end label open: near an end one only gets $\sigma(F(z)) = e^{\ell z} q^m \mathsf{g}(q)$ with $q = e^{\pm 2\pi i z}$, $\mathsf{g}(0) \neq 0$ and some integer $m \geq 0$, and only $m = 0$ is a time change id + (bounded periodic). The continuation across the Shell–Thron boundary is argued in [25, §5] by the continuity of a numerical scheme; it is not proved there.

The uniqueness statement [25, Prop. 2] requires only analyticity on $\operatorname{Re} z > -2$, $F(0) = 1$ and the two limits. As stated it is false for every real $b > \eta$. Let $K = K_b$ be Kneser's solution, $\mathcal{L} = \log b > 1/e$, and $u_n = K(n) \to \infty$.

- *Large image discs.* Choose N with $a u_{N+1} > 1$, and $\delta > 0$ with $B(u_N, \delta) \subset K(B(N, \frac{1}{2}))$ and $u_{N+1}(1 - e^{-\mathcal{L}\delta}) > \delta$. The inclusion $E_b(B(t, \delta)) \supset B(e^{\mathcal{L}t}, e^{\mathcal{L}t}(1 - e^{-\mathcal{L}\delta}))$ (apply the principal logarithm and $|\log(1+v)| \leq -\log(1 - |v|)$) propagates these discs. So $K(B(n+1, \frac{1}{2}))$ contains a disc about u_{n+1} of radius $u_{n+1}(1 - e^{-\mathcal{L}\delta}) \to \infty$.
- *A shifted solution.* For large n and $k = \lceil \mathcal{L} \exp(\mathcal{L} u_{n+1})/(2\pi) \rceil$, the point $w_k = \mathcal{L}^{-1}(\log(2\pi k/\mathcal{L}) + i\pi/2)$ lies in that disc, and $E_b(w_k) = 2\pi i k/\mathcal{L}$, $E_b^2(w_k) = 1$. Choose $z_k \in B(n+1, \frac{1}{2})$ with $K(z_k) = w_k$ and put $F_k(z) = K(z + z_k + 2)$. Then F_k is analytic on $\operatorname{Re} z > -2$, has $F_k(0) = 1$ and the same vertical limits as K, but $F_k(-1) = 2\pi i k/\mathcal{L} \neq 0 = K(-1)$.

Since k is unbounded, this gives infinitely many solutions. The gap in [25, Prop. 2] is the last step of its proof, which infers $C = 0$ from $F_1(C) = 1$. For real $b > \eta$ the uniqueness theorem of [24] is not affected: it also assumes analyticity on $\mathbb{C} \setminus (-\infty, -2]$ and $F(\bar{z}) = \overline{F(z)}$, both of which F_k violates. Indeed z_k is nonreal, because K is real on the real axis and w_k is not. And K cannot extend holomorphically through -2, since $K(-1) = 0$ whereas an exponential never vanishes. Hence F_k is singular at $-4 - z_k$, off the standard cut. For complex bases, some root-selection condition must be added to [25, Prop. 2]. The seam-height condition of Definition 8.1 is such a condition, and Theorem 8.2 shows that it suffices near $(1, \eta)$.

Replacing $F(0) = 1$ by $F(-1) = 0$ excludes the solutions F_k, and part of the argument then goes through. Suppose $F_2 = F_1(\cdot + C)$ on $\operatorname{Re} z > -2$, both with $F(-1) = 0$; exchanging F_1 and F_2 if necessary, $\operatorname{Re} C \geq 0$. A solution on $\operatorname{Re} z > -2$ can vanish only where $\operatorname{Re} z \leq -1$, since otherwise $F(z) = b^{F(z-1)} \neq 0$. Hence $F_1(C - 1) = F_2(-1) = 0$ forces $\operatorname{Re} C = 0$. To conclude $C = 0$ one still needs $F_1(-1 + \mathrm{i}t) \neq 0$ for real $t \neq 0$. This holds if F_1 continues to $\mathbb{C} \setminus (-\infty, -2]$ with the functional equation, as Kneser's solution does for real $b > \eta$: then $F_1(-1 + \mathrm{i}t) = b^{F_1(-2+\mathrm{i}t)}$. For complex bases this continuation is not available in general. For $b = \mathrm{i}$, [25] reports zeros of $\mathcal{K}_\mathrm{i}$ near $-3.810 + 0.146\mathrm{i}$ and $-6.591 + 0.273\mathrm{i}$, hence further branch points off $(-\infty, -2]$. So with $F(-1) = 0$ the extra condition needed is $F(-1 + \mathrm{i}t) \neq 0$ for $t \neq 0$, together with the degree-one form above.

2. Setup and parabolic data

Throughout, $1 < b < \eta$ and $E(w) = b^w$.

Lemma 2.1 (Parametrisation)**.** *Let L be a fixed point of E with multiplier $\lambda = L \log b$. Then $L = \mathrm{e}^\lambda$ and $\log b = \lambda \mathrm{e}^{-\lambda}$. The map $\lambda \mapsto \lambda \mathrm{e}^{-\lambda}$ increases on $(0, 1)$ and decreases on $(1, \infty)$, with maximum $\mathrm{e}^{-1} = \log \eta$. Hence $\lambda \in (0, 1)$ and $\lambda_2 \in (1, \infty)$ parametrise $(1, \eta)$ through the attracting and the repelling fixed point, and the two branches meet at $b = \eta$.*

Proof. $L = b^L$ gives $\log L = L \log b = \lambda$. □

Proposition 2.2. *With $\lambda = 1 - \varepsilon$ one has $\varepsilon = \sqrt{2\mathrm{e}(\log \eta - \log b)}\,(1 + O(\varepsilon))$, and as $b \to \eta^-$*

$$\Lambda = \exp\Big(-\frac{C_\eta}{\sqrt{\eta - b}}(1 + o(1))\Big), \qquad C_\eta = \frac{4\pi^2}{\sqrt{2\mathrm{e}^{1-1/\mathrm{e}}}}.$$

Proof. Expand $\lambda \mathrm{e}^{-\lambda} = \mathrm{e}^{-1}(1 - \varepsilon^2/2 - \varepsilon^3/3 + O(\varepsilon^4))$ and use $|\log \lambda| = \varepsilon + O(\varepsilon^2)$ and $\log \eta - \log b \sim (\eta - b)/\eta$. □

Remark 2.3 (Translation invariance)**.** A simultaneous shift $K(\cdot + \delta)$, $R(\cdot + \delta)$ sends $c_n \mapsto c_n \mathrm{e}^{2\pi \mathrm{i} n \delta}$. Thus $\Xi_n = c_n / c_1^n$, when defined, is invariant under complex translations, while $|c_1|$ is invariant only under real ones. The normalisation $F(0) = 1$ fixes the imaginary origin. This is why normalised first-mode limits require a specified base point, and why the constant a enters Corollary 1.1.

2.1. The parabolic germ.

At $b = \eta$, put $w = \mathrm{e}(1 + u)$. Then E becomes $f(u) = \mathrm{e}^u - 1$, with attracting petal on $\operatorname{Re} u < 0$ and formal Abel function

$$\alpha(u) = -\frac{2}{u} + \frac{\log(-u)}{3} + \sum_{k \geq 1} c_k^\alpha u^k, \qquad \alpha(f(u)) = \alpha(u) + 1. \tag{3}$$

The Fatou coordinates are $\Phi_{\mathrm{att}}(u) = \lim[\alpha(f^n u) - n]$ and $\Phi_{\mathrm{rep}}(u) = \lim[\alpha(f^{-n} u) + n]$, with $\log(-u)$ continued along orbits. On the upper gate, the horn map $\Phi_{\mathrm{rep}} \circ \Phi_{\mathrm{att}}^{-1}(z) = z +$

$\sum_{n\geq 1} A_n e^{2\pi i n z}$ and its inverse $\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}(z) = z + \sum_{n\geq 1} B_n e^{2\pi i n z}$ have no constant mode (Section 4, item (d)). Formal inversion gives

$$B_1 = -A_1, \qquad B_2 = -A_2 + 2\pi i A_1^2. \tag{4}$$

Proposition 2.4. *The first coefficient of the inverse upper horn map satisfies $B_1 \neq 0$.*

Proof. Let A be the immediate parabolic basin of $f(u) = e^u - 1$. The negative real ray is contained in A, since $u < f(u) < 0$ for $u < 0$. The entire map f has no critical points and exactly one finite singular value, -1, its asymptotic value along that ray. By [5, Lemma 87], the singular values of $f|_A : A \to A$ are contained in $A \cap \{-1\}$; the negative ray shows that -1 does occur. Hence f belongs to the one-petal unisingular class S_∞ of [5, Definition 8]. The parabolic-renormalisation theorem [5, Theorem 9 and Corollary 11] gives a renormalisation $R[f]$ with exactly one attracting petal at its parabolic fixed point. The upper renormalisation is the exponential projection of $\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}$, up to nonzero linear scalings of source and target [5, Appendix A]. In our normalisation, writing $\zeta = e^{2\pi i z}$, this projection is $H(\zeta) = \zeta + 2\pi i B_1 \zeta^2 + O(\zeta^3)$. Since $H'(0) = 1$, nonzero linear scalings that again give multiplier one preserve whether the quadratic coefficient vanishes. One attracting petal means that this coefficient is nonzero. Therefore $B_1 \neq 0$. □

Remark 2.5 (The mechanism)**.** The underlying reason is the classical one. The map f is of finite type in the sense of Epstein: it has no critical points and a single finite asymptotic value. Its horn maps are again of finite type, with singular values in $\mathbb{C}^*$ given by the projections of the single singular orbit [9]. So the upper horn map has exactly one singular value in $\mathbb{C}^*$. By the Fatou-type theorem for finite-type maps [9], the immediate basin of each attracting petal of a parabolic fixed point contains a singular value other than the fixed point. If $B_1 = 0$, then $H(\zeta) - \zeta = O(\zeta^3)$, and H would have at least two attracting petals at 0, each needing its own singular value.

Computing both maps directly on horizontal lines gives

$$\begin{aligned} |B_1| &= 0.089058436412213331563, \qquad |B_2| = 0.012487384111564700811, \\ \Xi_2^\infty &:= \frac{B_2}{B_1^2} = -0.02532049309975755123 + 1.57421906523195162 67\,\mathrm{i}, \\ \frac{B_3}{B_1^3} &= -2.254799124523803111 - 0.024408714296883171\,\mathrm{i}. \end{aligned} \tag{5}$$

Also $a = \Phi_{\mathrm{att}}(1/e - 1) = 3.0292972144180360989\ldots$, so the limiting phase in Corollary 1.1 is $\arg B_1 + 2\pi a = 1.5865330757\ldots \pmod{2\pi}$. These values are stable to all digits shown under changes of working precision (40–60 digits), transform size ($N = 16, 24, 32$) and sampling height ($\mathrm{Im}\, z = 1, 1.5, 2$). Digits beyond the intervals below remain numerical. The first three horn coefficients have independent certified enclosures of Proposition 2.6; nonvanishing of B_1 also follows from Proposition 2.4. The base-point coordinate a has the certified enclosure

$$3.029297214418036098924993 8 < a < 3.029297214418036098924994 0.$$

(Appendix A.5).

Proposition 2.6 (Certified first horn coefficients)**.** *The Fourier coefficients of the inverse upper horn map satisfy*

$$0.089058436412213331563 1 < |B_1| < 0.089058436412213331563 3, \qquad 0.012487384111564 69 < |B_2| < 0.012487 38$$

The ratios used in the numerical comparisons also satisfy

$$-0.025320493100 < \mathrm{Re}(B_2/B_1^2) < -0.025320493099, \quad 1.574219065231 < \mathrm{Im}(B_2/B_1^2) < 1.574219065233,$$
$$-2.25479914 < \mathrm{Re}(B_3/B_1^3) < -2.25479910, \qquad -0.02440873 < \mathrm{Im}(B_3/B_1^3) < -0.02440869.$$

Proof. Appendix A.5. □

Remark 2.7 (Formal invariants)**.** Put $\zeta = \mathrm{e}^{2\pi \mathrm{i} z}$. The inverse horn map becomes the cylinder germ $F(\zeta) = \zeta \exp(2\pi\mathrm{i} \sum B_n \zeta^n) = \zeta + a_2\zeta^2 + a_3\zeta^3 + \cdots$. Since $B_1 \neq 0$, its formal invariants under tangent-to-identity conjugacy include $a_2 = 2\pi\mathrm{i}B_1$ and the iterative residue $\rho = 1 - a_3/a_2^2$ [4]. One has

$$\Xi_2^{\infty} = 2\pi\mathrm{i}(\tfrac{1}{2} - \rho), \qquad \rho = 0.249455254258832 - 0.004029881638351\,\mathrm{i}. \tag{6}$$

When defined, the higher ratios B_n/B_1^n are not formal invariants of F. They are, however, invariants under the translations that preserve the normalisation of Fatou coordinates, and Theorem A compares them for every n.

3. The separation is a transition-map invariant

The separation problem compares a Kneser-type solution K_b with R_b. This section removes K_b from the problem; Section 10 identifies it with the sewn solution constructed below. The separation coefficients turn out to be determined, up to a relative error of order Λ, by the transition map between the two *regular* solutions at the two real fixed points — an object built from two convergent Schröder series in a fraction of a second, at any $b < \eta$.

3.1. **The transition map.** Let σ_{rep} be the Koenigs coordinate at the repelling point L_2, $\sigma_{\mathrm{rep}}(E(w)) = \lambda_2 \sigma_{\mathrm{rep}}(w)$, $\sigma'_{\mathrm{rep}}(L_2) = 1$, and put

$$S(z) = \sigma_{\mathrm{rep}}^{-1}(-\lambda_2^z),$$

continued to an entire function by $S(z+1) = E(S(z))$. It maps $\mathbb{R}$ decreasingly onto the segment (L, L_2) and has period $2\pi\mathrm{i}/\log\lambda_2$. Let $R^{-1}(w) = \log(\sigma(w)/\sigma(1))/\log\lambda$ be the attracting Abel coordinate, $R^{-1}(1) = 0$, which is multivalued with period $\mathrm{i}h = 2\pi\mathrm{i}/|\log\lambda|$. On the horizontal strip where S takes values in the immediate basin of L and a continuous branch is chosen, put

$$T(z) := R^{-1}(S(z)), \qquad T(z+1) = T(z) + 1, \qquad T(z) - z = \sum_{n\in\mathbb{Z}} t_n \mathrm{e}^{2\pi\mathrm{i}nz}.$$

Lemma 3.1 (Real structure)**.** *For real $b \in (1, \eta)$, $\mathrm{Im}\, T(x) \equiv h/2 \pmod h$ for every real x. Consequently, on the branch with $\mathrm{Im}\, t_0 = h/2$, $T(x) - x - \mathrm{i}h/2$ is real, $t_{-n} = \overline{t_n}$, and the line $\mathrm{Im}\, w = -h/2$ of attracting time is the image of $\mathbb{R}$ under the branch $T - \mathrm{i}h$.*

Proof. $S(x) \in (L, L_2)$, where $\sigma > 0$, while $1 < L$ gives $\sigma(1) < 0$. Hence $\sigma(S(x))/\sigma(1)$ is negative and $\mathrm{Im}\log(\sigma(S(x))/\sigma(1)) \in \pi + 2\pi\mathbb{Z}$; divide by $\log\lambda = -2\pi/h$. □

Translations of S ($S \mapsto S(\cdot + \delta)$, $\delta \in \mathbb{C}$) send $t_0 \mapsto t_0 + \delta$, $t_n \mapsto t_n \mathrm{e}^{2\pi\mathrm{i}n\delta}$, so

$$\tau_n(b) := t_n\, \mathrm{e}^{-2\pi\mathrm{i}nt_0} \qquad (\mathrm{Im}\, t_0 = h/2) \tag{7}$$

does not depend on the normalisation of S; it depends on that of R only through the real translation fixed by $R(0) = 1$. The condition $\mathrm{Im}\, t_0 = h/2$ refers to the real normalisation of S and fixes the branch of R^{-1}; a branch shift by $\mathrm{i}kh$ would multiply τ_n by Λ^{-nk}. The branch is fixed once in the real normalisation and then carried along.

3.2. **The welding identity.**

Theorem 3.2. *Let K be a solution of (1), holomorphic on the sets below where it is represented, and suppose there are 1-periodic holomorphic maps $\varphi_\pm = \mathrm{id} + (\text{periodic})$ with*

1. *$K = R \circ \varphi_+$ on an upper half-plane U_+, where $P = \varphi_+ - \mathrm{id} = \sum_{n\geq 0} c_n e^{2\pi i n z}$ converges on U_+;*
2. *$K = S \circ \varphi_-$ on a lower half-plane U_-, where φ_- is univalent, $G := \varphi_-^{-1} - \mathrm{id}$ is bounded on $V_- = \varphi_-(U_-)$, which contains a lower half-plane, and $G(w) \to g_0$ as $\operatorname{Im} w \to -\infty$;*
3. *$U_+ \cap U_-$ contains a horizontal line ℓ with $\varphi_-(\ell)$ inside the strip of T.*

Then, with the branch of T fixed by continuity from $\varphi_+ \circ \varphi_-^{-1}$, for every $n \geq 1$

$$t_n = \sum_{m\geq n} c_m \big[e^{2\pi i m G}\big]_{n-m}, \qquad t_0 = g_0 + \sum_{m\geq 0} c_m \big[e^{2\pi i m G}\big]_{-m}, \tag{8}$$

where $[F]_k$ is the k-th Fourier coefficient; in particular $[e^{2\pi i m G}]_0 = e^{2\pi i m g_0}$ and $[e^{2\pi i m G}]_k = 0$ for $k > 0$.

Proof. On ℓ, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$, so $\varphi_+ - T \circ \varphi_-$ takes values in the discrete period lattice of R and is constant; absorb it into the branch of T. Hence $T = \varphi_+ \circ \varphi_-^{-1}$ on the curve $\varphi_-(\ell)$, i.e. $T(w) - w = G(w) + P(w + G(w))$ there. The Fourier coefficients are computed by Cauchy's theorem on this curve. The region below it lies in V_-, and $\sum c_m e^{2\pi i m \varphi_-^{-1}(w)}$ converges uniformly on the curve $\varphi_-(\ell)$ itself, because $\ell \subset U_+$. Since G is bounded and holomorphic on V_-, with limit g_0 at $-i\infty$, so is $e^{2\pi i m G}$, whose Fourier series therefore contains only modes $k \leq 0$ and has constant term $e^{2\pi i m g_0}$. Expanding $P(w + G) = \sum_m c_m e^{2\pi i m w} e^{2\pi i m G(w)}$ and collecting modes gives (8); the Fourier coefficients of $T - \mathrm{id}$ do not depend on the line inside its strip of holomorphy. □

Corollary 3.3. *Under the hypotheses of Theorem 3.2, for every $n \geq 1$,*

$$t_n\, e^{-2\pi i n g_0} = c_n + e^{-2\pi i n g_0} \sum_{m>n} c_m [e^{2\pi i m G}]_{n-m}.$$

If $c_n \neq 0$, this may be written as

$$t_n\, e^{-2\pi i n g_0} = c_n(1 + \rho'_n), \qquad |\rho'_n| \leq \sum_{m>n} \frac{|c_m|}{|c_n|} \big|[e^{2\pi i m G}]_{n-m}\big|\, \big|e^{-2\pi i n g_0}\big|,$$

and $g_0 = t_0 - c_0 - \sum_{m\geq 1} c_m [e^{2\pi i m G}]_{-m}$. Heuristically, if the c_m decay like Λ^m and G is bounded on a line at height comparable to h above the gluing line, both corrections are $O(\Lambda)$. The precise statement for the sewing solution is Theorem 6.3(4). On the branch of Lemma 3.1, $g_0 = t_0 - ih + O(\Lambda)$, so that for the sewing solution, by Theorem 6.3(4),

$$\hat{c}_n := \frac{c_n}{\Lambda^n} = \tau_n(b) + O(\Lambda). \tag{9}$$

Since $B_1 \neq 0$, also $c_1 \neq 0$ for b sufficiently close to η and $\Xi_n := c_n / c_1^n = t_n / t_1^n + O(\Lambda)$ for each fixed n.

The identity $e^{-2\pi i n(t_0 - ih)} = \Lambda^n e^{-2\pi i n t_0}$ is where the scale Λ enters: *the Λ law is the statement that the gluing line sits at attracting time $-h/2$, half a period below the real axis, while the gate carrying the positive modes of T sits a further half period below it.* Nothing in the argument uses Kneser's construction.

3.3. **Welding construction.**

Proposition 3.4. *For $b \in (1,\eta)$ glue the half-cylinders $C_- = \{\operatorname{Im} w' \le 0\}/\mathbb{Z}$ (repelling time) and $C_+ = \{\operatorname{Im} w \ge -h/2\}/\mathbb{Z}$ (attracting time) along their boundaries by $w = T(w') - \mathrm{i}h$. The result is conformally $\mathbb{C}^*$; its universal-cover coordinate z defines a solution*

$$K^W(z) = R(w(z)) \text{ on } C_+, \qquad K^W(z) = S(w'(z)) \text{ on } C_-,$$

unique up to a translation in z, which satisfies the hypotheses of Theorem 3.2. It is holomorphic except where $w(z)$ meets the singular set $\bigcup_k((-\infty,-2]+\mathrm{i}kh)$ of R.

Unlike S, which is entire (the Poincaré function of a repelling point), R is not: its inverse Koenigs map extends only through inverse branches of E. So K^W carries, besides the cut near $(-\infty,-2]$, the copies near $(-\infty,-2]+\mathrm{i}kh$, $k \ge 1$, lying in its R-represented region. The same holds for the Kneser-type solutions at complex bases inside the Shell–Thron region, whose upper representation is through the attracting regular superfunction.

Proof. By Lemma 3.1 the gluing map is a real-analytic diffeomorphism between the boundary circles that extends holomorphically to a neighbourhood, so the glued surface X is a Riemann surface; each end is a punctured disc, hence $X \cup \{\text{two points}\}$ is a simply connected compact Riemann surface and $X \cong \mathbb{C}^*$. The definitions agree on the seam because $R \circ (T - \mathrm{i}h) = S$. The generator of $\pi_1(X)$ lifts to $z \mapsto z+1$ and acts as $w \mapsto w+1$, $w' \mapsto w'+1$, whence $K^W(z+1) = E(K^W(z))$. Near each puncture the change of coordinate is a holomorphic map of punctured discs fixing the puncture, which is the statement that $w - z$ (resp. $w' - z$) has only non-negative (resp. non-positive) Fourier modes. Uniqueness is the uniqueness of the uniformisation. □

Proposition 3.5 (Local parameter dependence of the sewn family)**.** *For every $b_0 \in (1,\eta)$, the normalised geometric sewing of Proposition 3.4 extends to a holomorphic family over a complex disc Δ_{b_0} about b_0. On every common regular domain of the universal-cover coordinate, $K_b^W(z)$ is jointly holomorphic in (b,z). These local families agree on overlaps and therefore define the inner sewn family on an open complex neighbourhood N_W of the entire real interval $(1,\eta)$, the union of the discs Δ_{b_0}. No uniform width of that neighbourhood is asserted.*

Proof. Fix b_0. Both fixed points and their Koenigs coordinates vary holomorphically near b_0. The compact set $S_{b_0}([0,1]) \subset (L, L_2)$ lies in the attracting basin and avoids its fixed point. Thus the chosen branch of $T_b = R_b^{-1} \circ S_b$ is jointly holomorphic on a parameter disc times a narrow strip about $[0,1]$; the relation $T_b(w'+1) = T_b(w')+1$ extends it to the whole strip. The real circle map is orientation preserving: $\sigma'_{\mathrm{rep},b}(S_b(x)) > 0$ by inverse iteration to L_2, $\sigma'_b(S_b(x)) > 0$ by forward iteration to L, and hence $S'_b(x) < 0$ and $(R_b^{-1})'(S_b(x)) = \sigma'_b(S_b(x))/(\sigma_b(S_b(x))\log\lambda_b) < 0$. Consequently $T'_{b_0}(x) > 0$. Compactness of the circle and local injectivity give a smaller strip on which T_{b_0} is injective modulo 1; shrinking the parameter disc preserves this property and makes the image of the real circle an analytic Jordan curve in the cylinder.

Put $U_b = T_b + \varpi_b$, where $\varpi_b = 2\pi\mathrm{i}/\log\lambda_b$; at b_0 this is $T_{b_0} - \mathrm{i}h$. Use U_b to identify a thin open annular collar in the repelling cylinder with its image in the attracting cylinder. More explicitly, for a sufficiently small $\delta > 0$ take the repelling chart $\{\operatorname{Im} w' < \delta/2\}/\mathbb{Z}$ and the attracting chart above the curve $U_b(\{\operatorname{Im} w' = -\delta/2\})$; identify them on $|\operatorname{Im} w'| < \delta/2$. After shrinking the parameter disc, these charts and the transition $(b,w') \mapsto (b, U_b(w'))$ form a Hausdorff holomorphic family of cylinders. At $b = b_0$ it is the analytic-collar version of the boundary welding in Proposition 3.4. Adjoin the two ends using the parameter independent local coordinates $\mathrm{e}^{2\pi\mathrm{i}w}$ and $\mathrm{e}^{-2\pi\mathrm{i}w'}$. This yields a proper holomorphic submersion whose fibres are compact Riemann surfaces of genus zero. The two added points and the upper-chart point $w = 0$ are three disjoint holomorphic sections (the last remains above the seam after shrinking the disc).

Local holomorphic triviality of a proper family with mutually biholomorphic compact fibres [10], followed by the unique fibrewise Möbius map sending these three sections to $0, \infty, 1$, gives a jointly holomorphic coordinate q_b on the compactified fibres. Choose the local lift $z = (2\pi\mathrm{i})^{-1} \log q_b$ for which the marked upper point $w = 0$ has $z = 0$. On the lift of the overlap, $R_b(U_b(w')) = S_b(w')$ because ϖ_b is a period of R_b. The two regular superfunctions therefore define K_b^W, jointly holomorphic wherever their represented branches are regular, and $K_b^W(0) = R_b(0) = 1$. On the real parameter slice this is exactly the geometric welding. Its marked coordinate makes the local families unique, so the identity theorem patches them on overlapping discs. □

For non-real b sufficiently close to any fixed real $b_0 \in (1, \eta)$, the sewn solution thus depends holomorphically on b. Theorem 7.1 gives the stronger quantitative Wiener bounds near η. For $\operatorname{Im} b > 0$ near (b_1, η) it tends (Theorem 7.1(2)) to L and L_2 as $\operatorname{Im} z \to \pm\infty$, respectively. Those limits alone do not characterise it: Proposition 2 of [25] has the counterexample in Remark 1.2. The base-continued Kneser family is identified with K^W in Theorem D; the seam condition of Definition 8.1 is what selects the right root there. Table 1 gives numerical evidence: τ_1 and t_2/t_1^2 reproduce the ladder values of $\hat{c}_1$ and Ξ_2 — moduli *and* arguments — to the precision of the ladder extrapolation, e.g. eight digits at $b = 1.10$.

TABLE 1. Kneser ladders (Sections 11.1, 11.2, extrapolated to $\operatorname{Im} b = 0$) against the Kneser-free invariants $\tau_1 = t_1 e^{-2\pi i t_0}$ and t_2/t_1^2 (script `transition_map.py`).

b	$\hat{c}_1$ (ladder)	τ_1	Ξ_2 (ladder)	t_2/t_1^2
1.05	$0.0850235\, e^{1.534136i}$	$0.0850226\, e^{1.534165i}$	$1.374049\, e^{1.413420i}$	$1.374336\, e^{1.413645i}$
1.10	$0.08642720\, e^{1.55109214i}$	$0.08642720\, e^{1.55109214i}$	$1.440851\, e^{1.477690i}$	$1.440851\, e^{1.477690i}$
1.15	$0.0872053\, e^{1.561012i}$	$0.0872052\, e^{1.561013i}$	$1.479111\, e^{1.511255i}$	$1.479107\, e^{1.511258i}$
1.30	0.088434	$0.0884321\, e^{1.577580i}$	$1.541516\, e^{1.561918i}$	$1.541515\, e^{1.561919i}$

3.4. **Reduction to confluence.** Write $a := \Phi_{\mathrm{att}}(1/e - 1) = 3.0292972144180360989\ldots$ for the α-normalised attracting Fatou coordinate of the base point $w = 1$ at $b = \eta$ (real, since the orbit of $u = 1/e - 1$ stays on the negative axis).

Theorem 3.6. *Assume the following confluence statement* (C)*: there are constants $s_*(b) \in \mathbb{C}$ such that, as $b \to \eta^-$,*

$$R_b^{-1}(e(1+u)) \to \Phi_{\mathrm{att}}(u) - a, \qquad S_b^{-1}(e(1+u)) + s_*(b) \to \Phi_{\mathrm{rep}}(u),$$

uniformly on compact subsets of the upper gate, with the branches continued as in Section 2.1. Then for every $n \geq 1$

$$\tau_n(b) \longrightarrow B_n\, e^{2\pi i n a}.$$

Consequently, also $t_n/t_1^n \to B_n/B_1^n$ for every $n \geq 1$. Together with Corollary 3.3 and the identification of K_b with K_b^W in Theorem 10.14, this gives $\hat{c}_n \to B_n e^{2\pi i n a}$ for all n. The limiting phase $\arg \hat{c}_1 \to \arg B_1 + 2\pi a = 1.5865330757\ldots \pmod{2\pi}$.

Proof. Replace S_b by $S_b(\cdot - s_*(b))$, which leaves τ_n unchanged. By (C), $T_b \to \Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1} - a = \mathfrak{h}^{-1} - a$ uniformly on $[0,1] + \mathrm{i}Y_g$ for fixed Y_g in the gate, so the Fourier coefficients on that line converge: $t_n \to B_n$ $(n \geq 1)$ and $t_0 \to -a$, the latter because $B_0 = 0$ in the α normalisation. Hence $\tau_n = t_n e^{-2\pi i n t_0} \to B_n e^{2\pi i n a}$. □

Statement (C) is a confluence theorem for the saddle-node unfolding $u \mapsto \mathrm{e}^{\lambda \mathrm{e}^{-\lambda}\mathrm{e}(1+u)}/\mathrm{e} - 1$ in the direction where the two fixed points are real (the "Glutsyuk" direction): the Koenigs coordinates at the two hyperbolic points converge, on the gate, to the sectorial Fatou coordinates of the parabolic germ. Results of this type are due to Glutsyuk [14] and, for the full unfolded modulus, Mardešić–Roussarie–Rousseau [21] and Christopher–Rousseau [6]. Rather than matching their normalisations, we prove (C) directly for this family in Section 4, so Theorem 3.6 holds unconditionally.

4. Proof of Theorem A: confluence

We prove (C) directly for the family at hand; no normalisation from the literature is needed. In u-coordinates ($w = \mathrm{e}(1+u)$, $\mu = \mathrm{e}\log b$)

$$f_b(u) = \mathrm{e}^{\mu - 1 + \mu u} - 1, \qquad f_\eta(u) = \mathrm{e}^u - 1.$$

For $b < \eta$ close to η the map has, in a fixed disc D_r, exactly the two fixed points $u_1 = \mathrm{e}^{\lambda - 1} - 1 < 0 < u_2 = \mathrm{e}^{\lambda_2 - 1} - 1$ (Rouché), and

$$f_b(u) - u = (u - u_1)(u - u_2)\, k_b(u), \qquad k_b \to k_\eta, \quad k_\eta(u) = \tfrac{1}{2} + \tfrac{u}{6} + \cdots, \tag{10}$$

uniformly on D_r, with k_b zero-free there. Put $A_j = 1/\log \lambda_j$, so $A_1 < 0 < A_2$.

Lemma 4.1 (Residues)**.** *As $b \to \eta^-$: $A_1(u_1 - u_2) \to 2$, $A_1 + A_2 \to \nu := 1 - 4k'_\eta(0) = \frac{1}{3}$, hence $A_1 u_1 + A_2 u_2 \to 2$ and $A_j u_j^2 \to 0$.*

Proof. $\lambda_1 = 1 + (u_1 - u_2)k_b(u_1)$ and $\lambda_2 = 1 + (u_2 - u_1)k_b(u_2)$. With $d = u_1 - u_2 \to 0$ and $1/\log(1+x) = 1/x + \frac{1}{2} + O(x)$, $A_1 + A_2 = \frac{k_b(u_2) - k_b(u_1)}{d\, k_b(u_1) k_b(u_2)} + 1 + O(d) \to 1 - 4k'_\eta(0)$, and $A_1 d \to 1/k_\eta(0) = 2$. □

Numerically $A_1 + A_2 = 0.3333331469$ and $A_1(u_1 - u_2) = 1.99676$ at $\lambda = 0.99$. Define the *model time*

$$\Psi_b(u) = A_1 \log(u - u_1) + A_2 \log(u - u_2),$$

which is exactly the time of the vector field $(u - u_1)(u - u_2)/(\dots)$ whose time-one map has multipliers λ_1, λ_2, and let $\alpha_2(u) = -2/u + \frac{1}{3}\log(-u)$. By Lemma 4.1, for u, p in a simply connected region at positive distance from 0, with logarithms continued along the same path,

$$\Psi_b(u) - \Psi_b(p) \longrightarrow \alpha_2(u) - \alpha_2(p). \tag{11}$$

Lemma 4.2 (Uniform one-step error)**.** *$F_b := \Psi_b \circ f_b - \Psi_b - 1$ is single-valued and holomorphic on a fixed disc $D_{r'}$, $F_b \to F_\eta := \alpha_2 \circ f_\eta - \alpha_2 - 1$ uniformly there, and*

$$|F_b(u)| \le M\, |u - u_1|\, |u - u_2| \qquad (u \in D_{r'})$$

with M independent of b.

Proof. By (10), $(f(u) - u_1)/(u - u_1) = 1 + (u - u_2)k(u)$ and $(f(u) - u_2)/(u - u_2) = 1 + (u - u_1)k(u)$, so

$$F = A_1 \log \frac{1 + (u - u_2)k}{1 + (u - u_1)k} + (A_1 + A_2)\log(1 + (u - u_1)k) - 1,$$

holomorphic for $|u| < r'$ small. The first logarithm is $O(|u_1 - u_2|)$ uniformly, and $A_1(u_1 - u_2)$, $A_1 + A_2$ are bounded (Lemma 4.1); so $|F| \le M_0$ on $|u| = r'$ uniformly, and the displayed formula converges. At $u = u_1$ the first term is $A_1 \log \lambda_1 = 1$ and the second vanishes, so $F(u_1) = 0$; likewise $F(u_2) = 0$. The maximum principle applied to $F/((u - u_1)(u - u_2))$ on $D_{r'}$ gives the bound, and letting $b \to \eta$ gives $F_\eta = O(u^2)$. □

Lemma 4.3 (Uniform petals and summability)**.** *Put* $Z_b(u)=A_1\log\frac{u-u_1}{u-u_2}$ *and* $\Omega_{R,b}=\{u:\operatorname{Re}Z_b(u)>R\}$ *(a disc of Apollonius about* u_1*). There is* R_0 *such that for* $R\ge R_0$ *and all* b *near* η*:* $\Omega_{R,b}\subset D_{r'/2}$*,* $f_b(\Omega_{R,b})\subset\Omega_{R,b}$*,* $\operatorname{Re}Z_b(f_b^k u)\ge\operatorname{Re}Z_b(u)+\frac34 k$*, and*

$$\sum_{k\ge N}|f_b^k u-u_1|\,|f_b^k u-u_2|\le\frac{C}{R+\frac34 N}\qquad(u\in\Omega_{R,b}),$$

with C *independent of* b *and* u*. As* $b\to\eta$*,* $\Omega_{R,b}$ *converges on compacta to the petal* $\{\operatorname{Re}(-2/u)>R\}$*.*

Proof. $Z(f(u))-Z(u)=A_1\log\frac{1+(u-u_2)k}{1+(u-u_1)k}=1+O(|u|+|u_1|+|u_2|)$ uniformly, by the expansion in the proof of Lemma 4.2; on $\Omega_{R,b}\subset D_{C/R+|u_1|}$ this is within $\frac14$ of 1 once R is large, which gives invariance and the growth of $\operatorname{Re}Z$. With $q=(u-u_1)/(u-u_2)$, $|q|=\mathrm{e}^{-\sigma}$, $\sigma=|\log\lambda_1|\operatorname{Re}Z$, one has $u-u_2=(u_2-u_1)/(q-1)$, whence

$$|u-u_1||u-u_2|=\frac{|q|\,|u_2-u_1|^2}{|1-q|^2}\le\Big(\frac{|u_2-u_1|}{|\log\lambda_1|}\Big)^2\frac{(1+\sigma)^2\mathrm{e}^{-\sigma}}{(\operatorname{Re}Z)^2},$$

using $1-\mathrm{e}^{-\sigma}\ge\sigma/(1+\sigma)$. The first factor is bounded (Lemma 4.1) and $(1+\sigma)^2\mathrm{e}^{-\sigma}\le4$; summing $(R+\frac34 k)^{-2}$ over $k\ge N$ gives the bound. The last assertion follows from $\operatorname{Re}Z_b(u)\to\operatorname{Re}(-2/u)$, i.e. $A_1(u_2-u_1)\to-2$. □

Theorem 4.4 (Confluence)**.** *Statement* (C) *holds. More precisely, with branches continued along paths in the respective parabolic basins:*

1. $R_b^{-1}(\mathrm{e}(1+u))\to\Phi_{\mathrm{att}}(u)-a$ *uniformly on compact subsets of the parabolic basin of* f_η*;*
2. *for a fixed* p' *in the gate (below), putting* $s_*(b)=\Phi_{\mathrm{rep}}(p')-\mathcal{S}_b(p')$*,* $\mathcal{S}_b(u)+s_*(b)\to\Phi_{\mathrm{rep}}(u)$ *uniformly on compact subsets of the repelling basin (backward orbits tending to* 0*). Here* $\mathcal{S}_b$ *is the real-normalised repelling time, i.e. the branch of* S_b^{-1} *that is real on the segment* (u_1,u_2) *and is continued through the band of Lemma 6.1, then along backward-invariant paths.*

The gate *of height* Y_* *is* $\mathcal{G}(Y_*)=\Phi_{\mathrm{rep}}^{-1}(\{Y_*\le\operatorname{Im}z\le Y_*+1,\ -1\le\operatorname{Re}z\le2\})$*. For* Y_* *large it is a compact subset of both parabolic basins of* f_η*. For* b *near* η *it lies in the band* $\mathcal{B}_b(Y_*-2)$*, because* $Z_b\to-2/u$ *uniformly on it (Lemma 4.1) and* $\operatorname{Im}(-2/u)=\operatorname{Im}\alpha(u)+O(1)=\operatorname{Im}\Phi_{\mathrm{rep}}(u)+O(1)$ *there.*

Proof. (1) On $\Omega_{R,b}$ put $\mathcal{A}_b(u)=\lim_n[\Psi_b(f_b^n u)-n]=\Psi_b(u)+\sum_{k\ge0}F_b(f_b^k u)$; the series converges absolutely by Lemmas 4.2 and 4.3. Since $f^n u\to u_1$ and the Koenigs map satisfies $\sigma(f^n u)=\lambda_1^n\sigma(u)$, $\Psi_b(f^n u)-n=A_1\log\sigma(u)+A_2\log(u_1-u_2)+o(1)$, because $A_1\log\lambda_1=1$. Thus $\mathcal{A}_b$ is the Koenigs Abel coordinate up to an additive constant, and $R_b^{-1}(\mathrm{e}(1+u))=\mathcal{A}_b(u)-\mathcal{A}_b(u^*)$, $u^*=1/\mathrm{e}-1$. Let K be a compact subset of the petal $P_R=\{\operatorname{Re}(-2/u)>R\}$ and $U\supset K\cup\{f_\eta^{N_0}u^*\}$ a simply connected region in P_R at positive distance from 0. For b near η, $U\subset\Omega_{R,b}$ and $u_1,u_2\notin U$. For $u,p\in U$,

$$\mathcal{A}_b(u)-\mathcal{A}_b(p)=\Psi_b(u)-\Psi_b(p)+\sum_{k\ge0}\big[F_b(f_b^k u)-F_b(f_b^k p)\big].$$

The first term converges by (11). In the sum, each term converges because $f_b^k\to f_\eta^k$ and $F_b\to F_\eta$ uniformly. The tails beyond N are $\le 2MC/(R+\frac34 N)$ uniformly in b (Lemma 4.3). Hence the difference converges uniformly to $\alpha_2(u)-\alpha_2(p)+\sum_k[F_\eta(f_\eta^k u)-F_\eta(f_\eta^k p)]=\Phi_{\mathrm{att}}(u)-\Phi_{\mathrm{att}}(p)$. The last identity holds because $\Phi_{\mathrm{att}}=\lim[\alpha(f^n u)-n]$ and $\alpha-\alpha_2=O(u)\to0$ along orbits. A general compact K of the basin satisfies $f_\eta^N(K)\Subset P_R$ for some N. Then $f_b^N(K)\Subset P_R$ for b near η, and $\mathcal{A}_b=\mathcal{A}_b\circ f_b^N-N$. Taking $p=f_b^{N_0}u^*$, which tends to $f_\eta^{N_0}u^*\in U$, and using

uniform convergence on a neighbourhood gives (1). (On $\Omega_{R,b}$ the function $\mathcal{A}_b$ is multivalued, since this punctured Apollonius disc surrounds u_1. It is single-valued on U, which avoids u_1. On K, $\sigma_b \neq 0$ because $f_b^N(K)$ avoids u_1. On non-immediate components of the basin the branch is defined by $\mathcal{A}_b \circ f_b^N - N$. The Koenigs map is normalised by $\sigma_b'(u_1) = 1$.)

(2) Apply (1)'s argument to $\check{f}_b(v) = -f_b^{-1}(-v)$, the principal inverse branch reflected. Then $\check{f}_\eta(v) = -\log(1-v) = v + \frac{v^2}{2} + \frac{v^3}{3} + \cdots$, which is of the same prepared form with $k_\eta'(0) = \frac{1}{3}$ and residue $-\frac{1}{3}$. The attracting fixed point of $\check{f}_b$ is $-u_2$, with multiplier $1/\lambda_2$. Its Koenigs Abel coordinate is $-S_b^{-1}(\mathrm{e}(1-v))$ up to a constant, and its parabolic Fatou coordinate is $-\Phi_{\mathrm{rep}}(-v)$ up to a constant (since $\alpha_f(-v) = -\alpha_{\check{f}}(v) + \mathrm{const}$). The free constant is fixed by p'. □

With Theorem 4.4, Theorem 3.6 becomes unconditional. Four points need checking; each is settled here. The proof of Lemma 6.1 uses only Lemmas 4.1–4.3, so there is no circularity.

(a) Inversion on the gate line. Fix $Y_* \geq Y_B + 7$ and the line $\ell = [0,1] + \mathrm{i}Y_*$. By Rouché, the solution $v_b(z)$ of $\mathcal{S}_b(v) + s_*(b) = z$ near $\Phi_{\mathrm{rep}}^{-1}(z)$ exists and converges uniformly on ℓ. Moreover $S_b(z - s_*(b)) = \mathrm{e}(1 + v_b(z))$, because $\mathcal{S}_b$ is a branch of S_b^{-1}.

(b) The gate line lies in the strip of holomorphy of T. The points $v_b(z)$, $z \in \ell$, lie within $o(1)$ of the gate. The gate lies in $\mathcal{B}_b(Y_* - 2) \subset \mathcal{B}_b(Y_B + 5)$ (Theorem 4.4), so these points lie in the band $\mathcal{B}_b(Y_B + 4)$. By Lemma 6.1(ii) their S-times $z - s_*(b)$ satisfy $|\operatorname{Im}(z - s_*(b))| \leq h/2 - Y_B - 4 + \pi/3 + C/Y_B + C\varepsilon < h/2 - Y_B - 2$, since $\pi/3 < 2$. So the shifted line $\ell - s_*(b)$ lies in the strip Σ of Lemma 6.1(iii) with $Y = Y_B$, where T is holomorphic. Therefore the Fourier coefficients of $\tilde{T} = T(\cdot - s_*)$ on ℓ are $t_n \mathrm{e}^{-2\pi i n s_*}$ and $\tilde{t}_0 = t_0 - s_*$, with the t_n those of T on $\mathbb{R}$. No prior knowledge of $\operatorname{Im} s_*$ is used.

(c) The branch of R_b^{-1}. Continue R_b^{-1} from u^* along the negative axis to $f_\eta^{N_0} u^*$, where it is real. Then go inside $P_R \setminus D_\delta(0)$ through the upper half-plane to the gate, and then inside the band down to the segment (u_1, u_2). This path passes above u_1 and does not wind around u_2. Along it $R_b^{-1} = \Psi_b + \sum_k F_b \circ f_b^k + c_R$, so the change of $\operatorname{Im} R_b^{-1}$ is $A_1 \Delta \arg(u - u_1) + A_2 \Delta \arg(u - u_2) + O(C/Y + \varepsilon) = h/2 + 0 + \text{small}$. Lemma 3.1 forces the exact value, so on the segment $\operatorname{Im} R_b^{-1} = +h/2$. The branch in Theorem 4.4(1) is therefore the one with $\operatorname{Im} t_0 = h/2$ used in (7).

(d) $B_0 = 0$. At $b = \eta$, on the gate, $\Phi_{\mathrm{att}} - \alpha_2 = \sum_{k \geq 0} F_\eta(f^k u)$ and $\Phi_{\mathrm{rep}} - \alpha_2 = -\sum_{k \geq 1} F_\eta(f^{-k} u)$, with the same principal $\log(-u)$ in α_2. That logarithm stays continuous along both orbits, since $-f^{\pm k} u$ remains in the lower half-plane. By Lemma 4.2 and $f^{\pm k} u \approx -2/(x \pm k + \mathrm{i}Y)$, both sums are $O(1/Y)$. Hence $\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}(z) - z \to 0$ as $\operatorname{Im} z \to +\infty$, so its constant Fourier mode vanishes: $B_0 = 0$. Numerically $|B_0| < 10^{-59}$.

By (a)–(d) and Theorem 4.4, $\tilde{T} \to \mathfrak{h}^{-1} - a$ uniformly on ℓ, where $\mathfrak{h}^{-1}$ denotes the inverse horn map. So $\tilde{t}_n \to B_n$ and $\tilde{t}_0 \to -a$, and $\tau_n = \tilde{t}_n \mathrm{e}^{-2\pi i n \tilde{t}_0}$ gives:

Corollary 4.5. *For every $n \geq 1$, $\tau_n(b) \to B_n \mathrm{e}^{2\pi i n a}$ as $b \to \eta^-$. Since $B_1 \neq 0$, also $t_n / t_1^n \to B_n / B_1^n$ for every $n \geq 1$.*

A direct pointwise check of Theorem 4.4 on two gate points (`confluence_check.py`) gives errors 0.086, 0.021, 0.0050, 0.00079 (attracting side) at $\lambda = 0.8, 0.9, 0.95, 0.98$, decreasing like ε^2.

5. The first-order term: a parabolic series

Theorem A gives the limit of τ_n. Here we prove that τ_n is differentiable at $b = \eta$ from the side $b < \eta$. The derivative is given by an absolutely convergent series computed entirely at the parabolic germ. Put

$$s = 1 - \mathrm{e} \log b = 1 - \mu, \qquad f_s(u) = \mathrm{e}^{-s + (1-s)u} - 1,$$

so $s \in (0, s_0]$ corresponds to $b < \eta$ near η and $f_0(u) = e^u - 1$. The map $(s, u) \mapsto f_s(u)$ is entire.

5.1. **Why termwise differentiation fails, and the remedy.** With the model time Ψ_b of Section 4, the one-step error is only $O(|u - u_1||u - u_2|)$ (Lemma 4.2). Along a parabolic orbit $u_k \approx -2/k$ one has $\partial_s u_k \asymp k$. Hence both $\partial_s F(u_k)$ and $F'(u_k)\partial_s u_k$ are of order one, and the differentiated series diverges. The remedy is the idea behind the prepared form of Mardešić–Roussarie–Rousseau [21]. We add to the model time a polynomial correction, holomorphic in s, that makes the one-step error vanish to order N at both fixed points simultaneously.

By the Weierstrass preparation theorem,

$$f_s(u) - u = q_s(u)\, k_s(u), \qquad q_s(u) = u^2 - \sigma_1(s)u + \sigma_2(s), \tag{12}$$

on a polydisc $|s| < s_0$, $|u| < r$. Here σ_1, σ_2 are holomorphic, k_s is holomorphic and zero-free, and $k_0(0) = \frac{1}{2}$. From $f_s(0) = -s + O(s^2)$ and $f_s'(0) - 1 = -2s + O(s^2)$ we get $\sigma_2 = -2s + O(s^2)$, $\sigma_1 = O(s)$, and the discriminant

$$D(s) = \sigma_1^2 - 4\sigma_2 = 8s + O(s^2)$$

is biholomorphic near 0. For real $s > 0$ the roots $u_1 < 0 < u_2$ are the fixed points of Section 4. Put $t = \sqrt{D(s)}$. Then $u_{1,2} = \frac{1}{2}(\sigma_1 \mp t)$ are holomorphic functions of t, interchanged by $t \mapsto -t$. Since $f_s'(u_1) = 1 + (u_1 - u_2)k_s(u_1)$, the residues are $A_1 = A_s(u_1, u_2)$ and $A_2 = A_s(u_2, u_1)$, where

$$A_s(x, y) = \frac{1}{\log(1 + (x - y)k_s(x))} = \frac{1}{(x - y)k_s(x)} + \frac{1}{2} + (x - y)\, \mathsf{q}_s(x, y)$$

with q_s holomorphic.

Lemma 5.1 (Uniform model)**.** *Fix $N \geq 1$. After shrinking s_0 and r, there are holomorphic functions $e_1, \dots, e_{2N-2}$ on $\{|s| < s_0\}$ with the following property. Put $P_s(u) = \sum_i e_i(s)u^i$ and, for $s \neq 0$,*

$$\Psi_s(u) = A_1 \log(u - u_1) + A_2 \log(u - u_2) + P_s(u).$$

Then:

1. *$\nu(s) = A_1 + A_2$ extends holomorphically to $|s| < s_0$. For every $\delta \in (0, r)$ there is $s_\delta > 0$ such that $\Psi_s(u) - \nu(s) \log u$ extends to a holomorphic function of $(s, u) \in \{|s| < s_\delta\} \times \{\delta < |u| < r\}$. Here, for $|u| > \delta$, the branches are fixed by $\log(u - u_j) := \log u + \operatorname{Log}(1 - u_j/u)$ with one common branch of $\log u$. Individually $A_j \sim \pm 1/\sqrt{2s}$, so any other convention would introduce $2\pi\mathrm{i}A_j$-jumps that are not holomorphic in s.*
2. *The one-step error*

$$F_s(u) = A_1 \log \frac{f_s(u) - u_1}{u - u_1} + A_2 \log \frac{f_s(u) - u_2}{u - u_2} + P_s(f_s(u)) - P_s(u) - 1$$

(principal logarithms) extends to a holomorphic function on $\{|s| < s_0\} \times \{|u| < r\}$, and $F_s = q_s^N\, \Gamma_s$ with Γ holomorphic and $|\Gamma| \leq M$ there.

Proof. Removability. Let $\Theta(s, x, y, u) = A_s(x, y)\ell(y) + A_s(y, x)\ell(x)$, where ℓ is holomorphic in (s, x, y, u). By the expansion of A_s, its singular part is

$$\frac{\ell(y)/k_s(x) - \ell(x)/k_s(y)}{x - y}.$$

The numerator vanishes on $x = y$, so Θ extends holomorphically across the diagonal. Evaluate Θ at $(x, y) = (u_1, u_2)$. The result is a holomorphic function of t, even in t, hence a holomorphic function of $t^2 = D(s)$, hence of s. The same holds for the other pairing $\Theta'(s, x, y, u) = A_s(x, y)\ell(x) + A_s(y, x)\ell(y)$, whose singular part $[\ell(x)/k_s(x) - \ell(y)/k_s(y)]/(x - y)$ is again removable and symmetric. These arguments apply to the following choices of ℓ:

- $\ell \equiv 1$ in Θ', which gives ν;
- $\ell(x) = \log(1 - x/u)$ in Θ', which gives $\Psi_s - \nu \log u - P_s$ for $|u| > \delta$ once $|u_j| < \delta/2$;
- $\ell(x) = \log(1 + (u - x)k_s(u))$ in Θ, which gives $F_s^{\log} := A_1 \log \frac{f-u_1}{u-u_1} + A_2 \log \frac{f-u_2}{u-u_2} - 1$, because $(f_s(u) - u_1)/(u - u_1) = 1 + (u - u_2)k_s(u)$.

This proves (1) and the holomorphy of $F^{\log}$.

Divisibility. For $s \neq 0$, $F_s^{\log}(u_j) = A_j \log f_s'(u_j) - 1 = 0$ (as in Lemma 4.2). Weierstrass division by q_s has a remainder $r_0(s) + r_1(s)u$ that is holomorphic in s and vanishes at two distinct points for $s \neq 0$. So the remainder is 0, and $F^{\log} = q_s\varphi$ with φ holomorphic. Likewise $D_i := (f_s(u))^i - u^i = q_s d_i$ (powers, not iterates). At $s = 0$, $D_i = u^i((1 + \frac{u}{2} + \cdots)^i - 1)$, so $d_i = \frac{i}{2}u^{i-1} + O(u^i)$.

The linear system. Divide $\varphi + \sum e_i d_i$ by the Weierstrass polynomial q_s^{N-1} of degree $2N - 2$. The $2N - 2$ coefficients of the remainder depend holomorphically on s and affinely on e, with matrix $M(s)$. At $s = 0$ the remainder is the Taylor polynomial modulo u^{2N-2}, so $M(0)$ is lower triangular with diagonal entries $i/2 \neq 0$. Hence $e(s) = -M(s)^{-1} \operatorname{rem}(\varphi)$ is holomorphic near 0. Then $F_s = F_s^{\log} + \sum e_i D_i = q_s(\varphi + \sum e_i d_i)$ is divisible by $q_s \cdot q_s^{N-1}$. The bound on Γ follows after shrinking the polydisc. □

Only the one-step error enters the estimates below, and it is single-valued. The logarithms in Ψ_s are continued along orbits exactly as in the proof of Theorem 4.4. Since P_s is single-valued, replacing Ψ_b by Ψ_s changes no branch. The path of item (c) in Section 4 runs outside $D_\delta(0)$ and does not pass between u_1 and u_2. So along it the convention of Lemma 5.1(1) agrees with the continuation used there.

5.2. **Orbits for complex parameters.** Let $P_R = \{\operatorname{Re}(-2/u) > R\}$ and put $\zeta = -2/u$.

Lemma 5.2 (Finite orbits)**.** *Let $K \subset P_R$ be compact, with R large, and let $Z = \max_K |2/u|$. There is $c > 0$ with the following property. For every integer $J \geq 1$, every $\sigma \in \mathbb{C}$ with $|\sigma| \leq c/J^2$, every $u \in K$ and every $0 \leq k \leq J$, the iterate $u_k = f_\sigma^k(u)$ is defined and $\zeta_k = -2/u_k$ satisfies*

$$\operatorname{Re}\zeta_k \geq \operatorname{Re}\zeta_0 + k/2, \qquad |\zeta_k| \leq |\zeta_0| + 2k.$$

In particular $|u_k| \leq 2/(R + k/2)$. For $\sigma = 0$ this holds for all k.

Proof. For $|u| \leq r$ we have $-2/f_0(u) = -2/u + 1 + \mathsf{e}(u)$ with $|\mathsf{e}(u)| \leq C_0|u|$, and $|f_0(u)| \geq |u|/2$. Write $f_\sigma = f_0 + \sigma g$ with $|g| \leq G$. Suppose the claim holds up to $k < J$. Then $|u_k| \leq 2/R \leq r$ and $w := \sigma g(u_k)/f_0(u_k)$ satisfies

$$|w| \leq G|\sigma||\zeta_k| \leq Gc(Z + 2) \leq \tfrac{1}{2}.$$

Hence

$$\zeta_{k+1} = \frac{-2}{f_0(u_k)}(1 + w)^{-1} = \zeta_k + 1 + e_k, \qquad |e_k| \leq \frac{2C_0}{R} + 4G|\sigma||\zeta_k|^2 \leq \frac{2C_0}{R} + 4Gc(Z + 2)^2.$$

This is at most $\frac{1}{2}$ for $R \geq 8C_0$ and c small. So $\operatorname{Re}\zeta_{k+1} \geq \operatorname{Re}\zeta_k + \frac{1}{2}$ and $|\zeta_{k+1}| \leq |\zeta_k| + \frac{3}{2}$. □

Lemma 5.3. *Let $u \in K$, $u_k = f_0^k(u)$ and $\dot{u}_k = \partial_\sigma f_\sigma^k(u)|_{\sigma=0}$. Then*

$$|\dot{u}_k| \leq C(k + 1), \qquad |\partial_\sigma F_\sigma(v)|_{\sigma=0}| \leq M'|v|^{2N-2}, \qquad |F_0'(v)| \leq M'|v|^{2N-1} \quad (|v| \leq r/2).$$

Proof. For the first bound, apply Cauchy's estimate on $|\sigma| = c/(k+1)^2$ together with Lemma 5.2 (with $J = k + 1$). This gives $|\dot{u}_k| \leq \frac{2}{R+k/2} \cdot \frac{(k+1)^2}{c}$. For the other two, write $F_\sigma = q_\sigma^N \Gamma_\sigma$ with $q_0(v) = v^2$ and $\partial_\sigma q_\sigma$ bounded, and bound $\partial_\sigma\Gamma$ and Γ_0' by Cauchy's estimates. □

5.3. **One-sided differentiability.** For u in a compact $K \subset P_R$ and real $s \in [0, s_0]$ put

$$\mathcal{A}_s(u) = \Psi_s(u) + \sum_{k\geq 0} F_s(f_s^k u),$$

with a fixed branch of $\log u$ on a simply connected neighbourhood $U \subset P_R$ of K. For $s > 0$ this is the attracting Koenigs Abel coordinate up to an additive constant, by the argument of Theorem 4.4(1). For $s = 0$, $\mathcal{A}_0 = \Phi_{\mathrm{att}} + \mathrm{const}$, because $\Psi_0 - \alpha = \mathrm{const} + O(u)$.

Proposition 5.4. *Let $N \geq 3$. Uniformly for $u \in K$, as $s \to 0^+$,*

$$\mathcal{A}_s(u) = \mathcal{A}_0(u) + s\,\dot{\mathcal{A}}(u) + O(s^{10/9}),$$

where

$$\dot{\mathcal{A}}(u) = \partial_s \Psi_s(u)\big|_{s=0} + \sum_{k\geq 0}\Big[\partial_\sigma F_\sigma(u_k)\big|_{\sigma=0} + F_0'(u_k)\,\dot{u}_k\Big], \qquad u_k = f_0^k(u). \tag{13}$$

The series converges absolutely and uniformly on K, with terms $O(k^{2-2N})$. Hence $\dot{\mathcal{A}}$ is holomorphic on the interior of K.

Proof. The Ψ-part is holomorphic in s by Lemma 5.1(1), so it contributes $O(s^2)$. Fix $J \geq 1$ and split the sum at J.

Head. $H_J(\sigma) = \sum_{k<J} F_\sigma(f_\sigma^k u)$ is holomorphic on $|\sigma| < c/J^2$. There $|u_j(\sigma)| \leq C\sqrt{|\sigma|} \leq C\sqrt{c}/J$. Since $k < J$ gives $1/J \leq (R+1)/(R+k/2)$, Lemma 5.2 yields $|q_\sigma(u_k)| \leq C_3(R+k/2)^{-2}$. Therefore

$$|H_J(\sigma)| \leq MC_3^N \sum_k (R + k/2)^{-2N} =: M_1,$$

uniformly in J. Cauchy's estimate gives $|H_J''| \leq 8M_1J^4/c^2$ on $[0, c/(2J^2)]$, so

$$|H_J(s) - H_J(0) - sH_J'(0)| \leq 4M_1c^{-2}s^2J^4.$$

Tails. We may assume $K \subset P_{R+1}$. Since $Z_s \to -2/u$ uniformly on K (Lemma 4.1), $K \subset \Omega_{R,s}$ for all small $s > 0$. If r was shrunk in Lemma 5.1, increase R so that $\Omega_{R,s} \subset D_{r/2}$, and reach a general K by f^m as below. For real $s \in (0, s_0]$ the proof of Lemma 4.3 gives $|q_s(f_s^k u)| \leq C_1(R+\frac{3}{4}k)^{-2}$. For $s = 0$, Lemma 5.2 gives the analogous bound $|u_k|^2 \leq 4(R+k/2)^{-2}$. Hence $\big|\sum_{k\geq J} F_s(f_s^k u)\big| \leq C_4J^{1-2N}$ for all $s \in [0, s_0]$. By Lemma 5.3, the terms of (13) are $O((R + k/2)^{2-2N})$. So $H_J'(0)$ differs from the series by at most C_5J^{3-2N}.

Balance. For $s \leq c/(2J^2)$,

$$|\mathcal{A}_s - \mathcal{A}_0 - s\dot{\mathcal{A}}| \leq Cs^2 + 4M_1c^{-2}s^2J^4 + 2C_4J^{1-2N} + C_5sJ^{3-2N}.$$

Take $J = \lceil s^{-2/9}\rceil$, which is admissible for small s. With $N \geq 3$ every term is $O(s^{10/9})$. □

Remark 5.5. Taking $J = s^{-2/(2N+3)}$ instead gives the error $O(s^{2-8/(2N+3)})$, so $O(s^{2-\delta})$ for any $\delta > 0$ with N large. The proof needs $N \geq 3$ only because of the crude bound on the head ($N = 2$ would give $s^{6/7}$). The series converges for $N \geq 2$.

The derivative does not depend on N. The condition $F_0 = O(u^{2N})$ forces Ψ_0 to agree with the formal Abel function α up to a constant and $O(u^{2N-1})$, so the $e_i(0)$ are the coefficients of α and do not depend on N. Hence $\mathcal{A}_s^{(N)} - \mathcal{A}_s^{(N')}$, which equals the difference of the two polynomials evaluated at $u_1(s) = O(\sqrt{s})$, is a constant plus $O(s^{3/2})$. In any case additive constants cancel in $\dot{\mathcal{A}}(v) - \dot{\mathcal{A}}(u^*)$. Numerically, $N = 3, 4, 6, 8$ give the same $\kappa^{(1)}$ to 16–20 digits.

5.4. **The derivative of τ_n.** Compact subsets of the basin are handled by $\mathcal{A}_s = \mathcal{A}_s \circ f_s^m - m$, where m is chosen so that $f_0^m(K) \Subset P_R$. The iterate f_s^m is jointly holomorphic, so the expansion of Proposition 5.4 persists, with $\dot{\mathcal{A}} = \dot{\mathcal{A}} \circ f_0^m + \mathcal{A}_0' \circ f_0^m \cdot \partial_s f_s^m$.

For the repelling side, apply everything to the reflected family $\check{f}_s(v) = -f_s^{-1}(-v) = -(\log(1-v)+s)/(1-s)$, exactly as in Theorem 4.4(2). This family is again of the form (12), with fixed points $-u_2, -u_1$. Let $\mathcal{S}_s(u) = -\mathcal{A}_s^{\check{f}}(-u)$. By Remark 2.3 and (7), τ_n does not depend on the additive normalisation of the repelling coordinate. We use $\mathcal{S}_s^\sharp = \mathcal{S}_s - \mathcal{S}_s(p') + \Phi_{\mathrm{rep}}(p')$. This differs from the real normalisation by the constant called $s_*(b)$ in Theorem 4.4(2), and it is right-differentiable with derivative $\dot{\mathcal{S}}^\sharp$. On the attracting side, $R_b^{-1}(\mathrm{e}(1+u)) = \mathcal{A}_s(u) - \mathcal{A}_s(u^*)$, with the branch fixed in item (c) of Section 4.

Theorem 5.6. *Let $\ell = [0,1] + \mathrm{i}Y_*$ be the gate line of Section 4, and put $v(z) = \Phi_{\mathrm{rep}}^{-1}(z)$ and $\tilde{T}_0 = \mathfrak{h}^{-1} - a$. Define*

$$\delta T(z) = \dot{\mathcal{A}}(v(z)) - \dot{\mathcal{A}}(u^*) - \tilde{T}_0'(z)\,\dot{\mathcal{S}}^\sharp(v(z)), \qquad \delta t_m = \int_\ell \delta T(z)\,\mathrm{e}^{-2\pi\mathrm{i}mz}\,dz.$$

Then for every $n \geq 1$, as $s \to 0^+$,

$$\tau_n = B_n \mathrm{e}^{2\pi\mathrm{i}na} + s\,\dot{\tau}_n + O(s^{10/9}), \qquad \dot{\tau}_n = \mathrm{e}^{2\pi\mathrm{i}na}\big(\delta t_n - 2\pi\mathrm{i}n\,B_n\,\delta t_0\big).$$

Moreover $p = -\log\lambda_1 \log\lambda_2 = 2s + \frac{10}{9}s^2 + O(s^3)$ is holomorphic in s. Consequently, if $B_n \neq 0$,

$$\log\frac{\tau_n}{B_n \mathrm{e}^{2\pi\mathrm{i}na}} = \kappa^{(n)} p + O(p^{10/9}), \qquad \kappa^{(n)} = \frac{\delta t_n}{2B_n} - \pi\mathrm{i}n\,\delta t_0.$$

Proof. By Proposition 5.4 and Cauchy's estimates, both $\mathcal{A}_s$ and $\mathcal{S}_s^\sharp$ have expansions $X_0 + s\dot{X} + O(s^{10/9})$ in the C^1 topology on a compact neighbourhood of $v(\ell)$. By Rouché, the inverse $v_s = (\mathcal{S}_s^\sharp)^{-1}$ exists near $v(\ell)$ and satisfies

$$v_s = v - s\,\dot{\mathcal{S}}^\sharp(v)/\Phi_{\mathrm{rep}}'(v) + O(s^{10/9}).$$

Hence $\tilde{T}_s := R_b^{-1} \circ \mathrm{e}(1+v_s)$ satisfies $\tilde{T}_s = \tilde{T}_0 + s\,\delta T + O(s^{10/9})$ uniformly on ℓ, where we used $\mathcal{A}_0'(v)/\Phi_{\mathrm{rep}}'(v) = \tilde{T}_0'$. By items (a)–(c) of Section 4, the Fourier coefficients $\tilde{t}_m(s)$ of $\tilde{T}_s - \mathrm{id}$ on ℓ satisfy $\tau_n = \tilde{t}_n \mathrm{e}^{-2\pi\mathrm{i}n\tilde{t}_0}$. Differentiate, using $\tilde{t}_n(0) = B_n$ and $\tilde{t}_0(0) = -a$. The expansion of p follows from the removability argument of Lemma 5.1 applied to $\log\lambda_1 \log\lambda_2$, together with a direct Taylor expansion. Finally, $\log(\tau_n/\tau_n(0)) = s\,\dot{\tau}_n/\tau_n(0) + O(s^{10/9})$ and $s = p/2 + O(p^2)$. □

5.5. **Higher orders.** The argument of Proposition 5.4 extends to every order, at the price of a larger N.

Proposition 5.7. *Let $m \geq 1$ and $N \geq m^2 + m + 1$. For $u \in K$ and $j \geq 0$ put*

$$\mathcal{A}^{(j)}(u) = \partial_s^j \Psi_s(u)\big|_{s=0} + \sum_{k\geq 0} \partial_\sigma^j \big[F_\sigma(f_\sigma^k u)\big]_{\sigma=0}. \tag{14}$$

For $j \leq m$ the series converges absolutely and uniformly on K, and its terms are $O(k^{2j-2N})$. Uniformly for $u \in K$, as $s \to 0^+$,

$$\mathcal{A}_s(u) = \sum_{j=0}^{m} \frac{s^j}{j!}\mathcal{A}^{(j)}(u) + O(s^{m+\gamma}), \qquad \gamma = 1 - \frac{2(m+1)^2}{2N+2m+1} > 0.$$

Proof. Term bounds. Fix $k \geq 0$ and apply Lemma 5.2 with $J = k+1$. For $|\sigma| \leq c/(k+1)^2$ the head estimate in the proof of Proposition 5.4 gives $|q_\sigma(f_\sigma^k u)| \leq C_3(R + k/2)^{-2}$, hence $|F_\sigma(f_\sigma^k u)| \leq M C_3^N (R + k/2)^{-2N}$. By Cauchy's estimate on this disc,

$$\big|\partial_\sigma^j [F_\sigma(f_\sigma^k u)]_{\sigma=0}\big| \leq j!\, M C_3^N\, c^{-j} (k+1)^{2j} (R + k/2)^{-2N} = O(k^{2j-2N}).$$

This is summable for $j \le m < N - \frac{1}{2}$. For $m = 1$ this recovers the bounds of Lemma 5.3, without computing $\dot{u}_k$.

Head. $H_J(\sigma) = \sum_{k<J} F_\sigma(f_\sigma^k u)$ is holomorphic and bounded by M_1 on $|\sigma| < \rho := c/J^2$. For real $0 \le s \le \rho/2$, the Taylor remainder of order m satisfies

$$\Big|H_J(s) - \sum_{j\le m} \frac{s^j}{j!} H_J^{(j)}(0)\Big| \le M_1 \sum_{j>m} (s/\rho)^j \le 2M_1 c^{-m-1}\, s^{m+1} J^{2m+2}.$$

Tails. As before, $\big|\sum_{k\ge J} F_s(f_s^k u)\big| \le C_4 J^{1-2N}$ for $s \in [0, s_0]$. By the term bounds, the tails of the j-th derivative series from J on are $O(J^{2j+1-2N})$. Since $sJ^2 \le c/2$, the tails contribute $\sum_{j\le m} s^j J^{2j+1-2N} = O(J^{1-2N})$.

Balance. Write $\mathcal{T}_J^{(j)}$ for the tail from $k = J$ on of the j-th derivative series. The identity

$$\mathcal{A}_s - \sum_{j\le m} \frac{s^j}{j!}\mathcal{A}^{(j)} = \Big[\Psi_s - \sum_{j\le m} \frac{s^j}{j!}\partial_s^j \Psi\Big] + \Big[H_J(s) - \sum_{j\le m} \frac{s^j}{j!} H_J^{(j)}(0)\Big] + \sum_{k\ge J} F_s(f_s^k u) - \sum_{j\le m} \frac{s^j}{j!}\mathcal{T}_J^{(j)}$$

holds exactly. The Ψ-part is holomorphic in s. Altogether the error is $O(s^{m+1}J^{2m+2} + J^{1-2N})$. Take $J = \lceil s^{-\beta} \rceil$ with $\beta = (m+1)/(2N+2m+1) < \frac{1}{2}$, which is admissible. Both terms are then $O(s^{m+1-2(m+1)\beta}) = O(s^{m+\gamma})$, and $\gamma > 0$ exactly when $N \ge m^2 + m + 1$. □

For $m = 1$ this is Proposition 5.4; the error there is $O(s^{10/9})$ for $N = 3$.

Theorem 5.8. *For every $n \ge 1$ with $B_n \ne 0$, the function $\log(\tau_n/\tau_n(0))$ has an asymptotic expansion to all orders in p as $b \to \eta^-$:*

$$\log \frac{\tau_n}{B_n e^{2\pi i n a}} \sim \sum_{j\ge 1} \kappa_j^{(n)} p^j, \qquad \kappa_1^{(n)} = \kappa^{(n)}.$$

For each order m, the remainder after the p^m term is $O(p^{m+1})$. Each coefficient $\kappa_j^{(n)}$ is a polynomial in the following data:

- *B_n^{-1} and a;*
- *the Taylor coefficients of $s(p)$;*
- *finitely many Fourier coefficients on ℓ of explicit polynomial expressions in $\mathcal{A}^{(i)} \circ v$, $\mathcal{S}^{\sharp(i)} \circ v$ $(i \le j)$, their u-derivatives, and $1/\Phi'_{\mathrm{rep}} \circ v$.*

Here $\mathcal{A}^{(i)}$ and $\mathcal{S}^{\sharp(i)}$ are given by the convergent series (14) over parabolic orbits of f_0 and G_0. The coefficients do not depend on N.

Proof. Fix m and $N \ge m^2 + m + 1$. By Proposition 5.7 and the reduction $\mathcal{A}_s = \mathcal{A}_s \circ f_s^M - M$ (with f_s^M jointly holomorphic), $\mathcal{A}_s$ and $\mathcal{S}_s^\sharp$ have expansions $\sum_{j\le m} s^j X_j/j! + O(s^{m+\gamma})$. These hold uniformly on a compact neighbourhood of $v(\ell) \cup \{u^*\}$ and are holomorphic in u, so by Cauchy's estimates they also hold in the C^r topology on a smaller neighbourhood, for every r.

Inverse. Formal inversion of $\sum_j s^j \mathcal{S}_j^\sharp/j!$ gives a polynomial approximant $\tilde{v}_s = v + \sum_{1\le j\le m} s^j V_j$ with $\mathcal{S}_s^\sharp(\tilde{v}_s) = z + O(s^{m+\gamma})$. Since $\mathcal{S}_s^\sharp \to \Phi_{\mathrm{rep}}$ in C^1 and $\Phi'_{\mathrm{rep}} \ne 0$ near $v(\ell)$, the maps $\mathcal{S}_s^\sharp$ are uniformly bi-Lipschitz there, and the true inverse satisfies $v_s = \tilde{v}_s + O(s^{m+\gamma})$.

Composition. Taylor-expand $\mathcal{A}_s$ about v, using the bounded u-derivatives. This gives $\tilde{T}_s = \sum_{j\le m} s^j T_j/j! + O(s^{m+\gamma})$ uniformly on ℓ. Hence the Fourier coefficients $\tilde{t}_i(s)$ and $\tau_n = \tilde{t}_n e^{-2\pi i n \tilde{t}_0}$ have expansions to order m. Since $\tau_n(0) \ne 0$, so does $\log(\tau_n/\tau_n(0))$.

Change of parameter. Finally, p is holomorphic in s with $p'(0) = 2$, so s is a holomorphic function of p.

Independence of N. An asymptotic expansion of a fixed function is unique, so the coefficients do not depend on the choice of N.

Remainder. Applying the argument at order $m+1$ and absorbing the p^{m+1} term gives the remainder $O(p^{m+1})$. □

Remark 5.9. In particular, Theorem 5.8 with $m=2$ improves the remainder in Theorem 5.6 from $O(p^{10/9})$ to $O(p^2)$. Theorem 5.8 gives explicit asymptotic coefficients in the normalisation of τ_n, but says nothing about summability. The $\frac{1}{2}$-summability of [6], in their normalisation, suggests Gevrey-2 growth $|\kappa_j^{(n)}| \lesssim C^j (j!)^2$ in $p \propto \epsilon_{\mathrm{CR}}$ (Gevrey-1 in $\sqrt{p}$). This matches the scale $\Lambda = \exp(-C/\sqrt{p})$ of the beyond-all-orders terms. The constants in our proof grow faster, because N must grow like m^2.

Remark 5.10 (Relation to the literature)**.** Christopher and Rousseau show that the unfolded modulus, in a particular normalisation, is $\frac{1}{2}$-summable in the canonical parameter $\epsilon_{\mathrm{CR}} = (A_2 - A_1)^{-2}$, with the Glutsyuk direction as the singular direction [6, Cor. 4.9]. One checks the exact identity $p = 4\epsilon_{\mathrm{CR}}/(1-\nu^2\epsilon_{\mathrm{CR}})$. Their result gives the existence of an asymptotic expansion. Mardešić–Roussarie–Rousseau proved continuity at $\epsilon_{\mathrm{CR}} = 0$ and conjectured 1-summability in $\sqrt{\epsilon}_{\mathrm{CR}}$ [21, §11]; this is the result of [6], and Ribón proved multisummability of Fatou coordinates and Écalle–Voronin invariants for unfoldings of any finite codimension [27]. These results give existence of the expansion but not its coefficients. Theorems 5.6 and 5.8 give the coefficients, in the normalisation of τ_n, as explicit convergent series over parabolic orbits, with a direct proof that does not use summability theory. This parallels the expression of the Écalle–Voronin invariants themselves by convergent numerical series [7]. On the Lavaurs side, normalised perturbed Fatou coordinates and horn maps depend holomorphically on the perturbation [29] (see also [15, Thms. 1.3, 2.1]), so there too an expansion exists, but we have not found its first-order coefficients in the literature. The closest analogue is the splitting of separatrices of area-preserving maps near a saddle-centre bifurcation: Gelfreich and Brännström prove that the homoclinic invariant is $\mathrm{e}^{-2\pi^2/\log\lambda}$ times an asymptotic series $\sum_k \mathsf{c}_k \delta^{2k}$, with c_0 the Stokes constant of the parabolic limit [11, 12], and Gelfreich and Simó extract the higher c_k numerically to high precision [13]. That setting concerns one Fourier mode of a two-dimensional map, and its coefficients for $k \geq 1$ are defined through auxiliary problems rather than convergent orbit sums. Within a parabolic family (no splitting of the fixed point), orbit-sum expansions of the horn map in a parameter appear in [3, Prop. 9.1]. We know of no closed form. An integer-relation search on $\kappa^{(1)}$ against $\pi, \pi^2, \log 2, \gamma$ returns only spurious relations.

5.6. **Numerics.** The series (13) is evaluated with $N = 3, \dots, 8$ on 24 points of the line $\operatorname{Im} z = 1$ or 1.5, with orbits of length 300–3000. The s-derivative of each (finite, holomorphic) truncation is taken by a central difference at $s = \pm 10^{-20}$ to $\pm 10^{-30}$ in 60–90 digit arithmetic (`mrr_derivative.py`). The results are stable to all printed digits:

$$\kappa^{(1)} = -0.010199006345897402441 - 0.013250956182506712250\,\mathrm{i},$$
$$\kappa^{(2)} = -0.05105655620132933 - 0.06231578053849410\,\mathrm{i},$$
$$\kappa^{(3)} = -0.140044348129 - 0.169224463926\,\mathrm{i}.$$

These agree with the fits (18) from the side $b < \eta$ to all of their eleven digits. At $s = 0$ the same computation reproduces $\tau_n(0) = B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ to the precision of the stored B_n. The second coefficients $\kappa_2^{(n)}$ of Theorem 5.8 are computed by a five-point stencil ($s = \pm 10^{-12}, \pm 2\cdot 10^{-12}$, 90 digits) applied to the same finite truncations (`mrr_second.py`), with $N = 7$ ($\operatorname{Im} z = 1$, orbits of length 2000) and $N = 10$ ($\operatorname{Im} z = 1.5$, length 3000). The series converge for $N \geq 3$; the condition $N \geq m^2 + m + 1 = 7$ is needed only for the proof. The two computations agree to 18,

16 and 12 digits respectively:

$$\kappa_2^{(1)} = -7.88992074991511334 \cdot 10^{-5} + 3.6554727325799459 \cdot 10^{-4}\,\mathrm{i},$$
$$\kappa_2^{(2)} = -1.895481905818775 \cdot 10^{-4} + 1.106655243691997 \cdot 10^{-4}\,\mathrm{i},$$
$$\kappa_2^{(3)} = 4.07137984831 \cdot 10^{-4} - 8.49506527544 \cdot 10^{-3}\,\mathrm{i}.$$

The fits from the side $b < \eta$ (`deficit_modes.py`) give $-7.88991\cdot10^{-5}+3.65547\cdot10^{-4}$i, $-1.89548\cdot 10^{-4} + 1.10666 \cdot 10^{-4}$i and $4.07137 \cdot 10^{-4} - 8.49507 \cdot 10^{-3}$i, in agreement to within one unit in the sixth digit.

6. The sewing solution: decay estimate

Throughout, $b < \eta$ is real and close to η, and S has the real normalisation of Section 3, so $T(z) - z = \sum t_n e^{2\pi i n z}$ with $\operatorname{Im} t_0 = h/2$ and $t_{-n} = \overline{t_n}$ (Lemma 3.1).

Lemma 6.1 (The band)**.** *Let $Z_b(u) = A_1 \log \frac{u-u_1}{u-u_2}$ with the branch $\operatorname{Im} Z_b \in (0, h)$ on $\mathbb{C} \setminus ((-\infty, u_1] \cup [u_2, \infty))$. This branch has $\operatorname{Im} Z_b = h/2$ on (u_1, u_2) and $\operatorname{Im} Z_b < h/2$ on $\operatorname{Im} u > 0$. For $Y > 0$ let*

$$\mathcal{B}_b(Y) = \{u :\ Y < \operatorname{Im} Z_b(u) < h - Y\},$$

a lens bounded by two circular arcs through u_1, u_2. There is an absolute Y_B such that for $Y \geq Y_B$ and all b near η:

(i) *every $u \in \mathcal{B}_b(Y)$ has its forward orbit in $D_{r'}$ tending to u_1 and its principal backward orbit in $D_{r'}$ tending to u_2, with $\sum_{k\in\mathbb{Z}} |F_b(f_b^k u)| \leq C/Y + C|\log\lambda|$;*

(ii) *the real-normalised times $\mathcal{R}_b = \Psi_b + \sum_{k\geq 0} F_b \circ f_b^k + c_R$ and $\mathcal{S}_b = \Psi_b - \sum_{k\geq 1} F_b \circ f_b^{-k} + c_S$ are holomorphic and injective on $\mathcal{B}_b(Y)$. They are branches of R_b^{-1} and S_b^{-1}, with $\mathcal{S}_b$ real on (u_1, u_2) and $\mathcal{R}_b$ the branch with $\operatorname{Im} \mathcal{R}_b = h/2$ on (u_1, u_2) (Lemma 3.1), and*

$$\left|\operatorname{Im} \mathcal{S}_b(u) - (\operatorname{Im} Z_b(u) - h/2)\right| \leq \tfrac{\pi}{3} + C/Y + C|\log\lambda|;$$

(iii) *S_b maps the strip $\Sigma = \{|\operatorname{Im} w'| < h/2 - Y - 2\}$ into $\mathcal{B}_b(Y)$ with $\mathcal{S}_b \circ S_b = \mathrm{id}$. Consequently $T = \mathcal{R}_b \circ S_b$ is holomorphic on Σ, and*

$$T(w') - w' = (c_R - c_S) + \sum_{k\in\mathbb{Z}} F_b(f_b^k(S_b(w'))).$$

Proof. (i) Put $q = e^{Z/A_1} = (u - u_1)/(u - u_2)$ and $\zeta = Z/A_1 = -|\log\lambda| Z$, so $\operatorname{Im}\zeta \in (-2\pi + |\log\lambda| Y, -|\log\lambda| Y)$ on the band. On $-2\pi \leq \operatorname{Im}\zeta \leq 0$ one has $|e^\zeta - 1| \geq c \min(1, \delta(\zeta)) \max(1, |e^\zeta|)$ with $\delta(\zeta) = \min(|\zeta|, |\zeta + 2\pi i|)$ and an absolute $c > 0$; the only zeros are 0 and $-2\pi i$. On the band $\delta(\zeta) = |\log\lambda| \min(|Z|, |Z - ih|) > |\log\lambda| Y$. Since $u - u_2 = (u_2 - u_1)/(q - 1)$ and $|u_2 - u_1|/|\log\lambda| \to 2$ (Lemma 4.1), this gives

(15)

$$|u - u_2| \leq C(|\log\lambda| + Y^{-1}), \qquad |u - u_1||u - u_2| = \frac{|q|\,|u_2 - u_1|^2}{|q-1|^2} \leq \begin{cases} C \min(|Z|, |Z - ih|)^{-2}, & \delta(\zeta) \leq 1, \\ C|\log\lambda|^2 e^{-|\log\lambda|\,|\operatorname{Re} Z|}, & \delta(\zeta) > 1, \end{cases}$$

using $|q|/\max(1, |q|)^2 = e^{-|\operatorname{Re}\zeta|}$. So $\mathcal{B}_b(Y) \subset D_{C/Y + |u_1| + |u_2|} \subset D_{r'/2}$ for Y large. By the proof of Lemma 4.3, $Z \circ f - Z = 1 + O(|u| + |\log\lambda|)$, so $\operatorname{Re} Z$ increases by at least $\frac{3}{4}$ per step while the orbit is in $D_{r'/2}$.

The imaginary part is controlled through $\Psi_b = Z_b + (A_1 + A_2)\log(u - u_2)$, with step error $|F_b| \leq M|u - u_1||u - u_2|$ (Lemma 4.2). Summing (15) along $\operatorname{Re} Z_k \geq \operatorname{Re} Z_0 + \frac{3}{4}k$ bounds $\sum_k |F_b(u_k)|$ by $C'/Y + C'|\log\lambda|$. Points with $\delta(\zeta_k) \leq 1$ contribute $\sum_k (x_k^2 + Y^2)^{-1} \leq C/Y$, and the others $|\log\lambda|^2 \sum e^{-|\log\lambda|\,|x_k|} \leq C|\log\lambda|$. Now $\operatorname{Im} Z - \operatorname{Im}\Psi = -(A_1 + A_2)\arg(u - u_2)$ with

$\arg(u-u_2)\in(0,2\pi)$ on the lens. So $\operatorname{Im} Z$ drifts by at most $C'/Y+C'|\log\lambda|+\frac{2\pi}{3}+o(1)$ along the orbit, which is $<Y/2$ for $Y\ge Y_B\ge 8$. By induction the orbit stays in $\mathcal{B}_b(Y/2)\subset D_{r'/2}$, and $\operatorname{Re} Z_k\to+\infty$, i.e. $u_k\to u_1$. The continued values of Z along the orbit have $\operatorname{Im}\in(Y/2,h-Y/2)\subset(0,h)$, so they agree with the fixed slit-plane branch. The backward orbit is the forward orbit of the reflected inverse family $\check f_b(v)=-f_b^{-1}(-v)$, to which the same argument applies (its prepared form is checked in the proof of Theorem 4.4(2)). Its period is $h_2=h+2\pi/3+o(1)$, so $\mathcal{B}_b(Y)$ is contained in the corresponding band of $\check f_b$ at height $Y(1+O(|\log\lambda|))-1$. This is harmless for $Y\ge Y_B$.

(ii) As in the proof of Theorem 4.4(1), $\Psi_b(f^n u)-n\to A_1\log\sigma_b(u)+\text{const}$ along forward orbits. So $\mathcal{R}_b$ is a branch of the attracting Abel coordinate, and $\mathcal{S}_b$ likewise of the repelling one. ($\mathcal{B}_b(Y)$ is simply connected and σ_b has no zero on it, since f_b is injective on $D_{r'}$ and $u_1\notin\mathcal{B}_b(Y)$.) Both F_b and f_b are real, so the sums are real on the segment. By Lemma 3.1, $\operatorname{Im} R_b^{-1}\equiv h/2 \pmod h$ on $(u_1,u_2)=S_b(\mathbb{R})$; choose c_R so that $\mathcal{R}_b$ is the branch of R_b^{-1} with $\operatorname{Im}\mathcal{R}_b=h/2$ there, and c_S so that $\mathcal{S}_b$ is real on the segment. On the segment $\operatorname{Im} Z=h/2$ and $\arg(u-u_2)=\pi$, which fixes $\operatorname{Im} c_S$. The displayed estimate then follows from $\operatorname{Im}\Psi=\operatorname{Im} Z+(A_1+A_2)\arg(u-u_2)$, $A_1+A_2\to\frac13$ and (i). Injectivity: if $\mathcal{S}_b(u)=\mathcal{S}_b(u')$ then the principal backward orbits satisfy $\sigma_{\mathrm{rep},b}(u_{-n})=\sigma_{\mathrm{rep},b}(u'_{-n})$ near u_2, where $\sigma_{\mathrm{rep},b}$ is injective, so $u_{-n}=u'_{-n}$. Moreover f_b is injective on $D_{r'}$ (its non-injectivity has period $2\pi\mathrm{i}/\log b$ in w, far larger than the disc), so $u=u'$. The same argument with σ_b and forward orbits proves injectivity of $\mathcal{R}_b$.

(iii) Fix $x\in\mathbb{R}$. $S_b(x)\in(u_1,u_2)\subset\mathcal{B}_b(Y)$ and $\mathcal{S}_b(S_b(x))=x$, since both sides are the real-normalised repelling time. Move y' from 0 towards $\pm(h/2-Y-2)$. As long as $S_b(x+\mathrm{i}y')$ stays in the lens, continuation gives $\mathcal{S}_b(S_b(x+\mathrm{i}y'))=x+\mathrm{i}y'$. It cannot leave through the corners u_1,u_2, where $|\operatorname{Re}\mathcal{S}_b|\to\infty$. It cannot leave through the arcs either: there $\operatorname{Im} Z\in\{Y,h-Y\}$, so by (ii) $|\operatorname{Im}\mathcal{S}_b|\ge h/2-Y-\pi/3-o(1)>|y'|$. Hence $S_b(\Sigma)\subset\mathcal{B}_b(Y)$, $T=\mathcal{R}_b\circ S_b$ on Σ, and $T(w')-w'=\mathcal{R}_b(u)-\mathcal{S}_b(u)$ with $u=S_b(w')$ is the displayed bi-infinite sum. □

Lemma 6.2 (Uniform mode bound)**.** *Put $Y_0:=Y_B+3$. There is M, which can be taken ≤ 1, and $b_1<\eta$ such that for $b\in(b_1,\eta)$ and all $n\ne 0$,*

$$|t_n|\le M\mathrm{e}^{-2\pi|n|d},\qquad d:=\tfrac{h}{2}-Y_0-1.$$

Proof. Take $Y=Y_B+1$ in Lemma 6.1(iii), so that the estimate (ii) is available on the closure of the lens. Σ has half-width $h/2-Y_B-3=d+1$, so the lines $\operatorname{Im} w'=\pm d$ lie in Σ. On them, by (iii) and (i), $T-\mathrm{id}-t_0=\sum_k F_b(f_b^k S_b)-\text{mean}$ has modulus at most $2(C/Y_B+C|\log\lambda|)$. Let M be the supremum of this over $b\in(b_1,\eta)$. Cauchy's estimate on these lines gives the bound for $n>0$ and $n<0$. This does not use Theorem 4.4. □

For $H>0$ let $\mathfrak{A}_H$ be the space of 1-periodic $g=\sum g_n\mathrm{e}^{2\pi\mathrm{i}nz}$ with $\|g\|_H=\sum|g_n|\mathrm{e}^{2\pi|n|H}<\infty$. It is a Banach algebra (the weight is submultiplicative), its elements are holomorphic and bounded by $\|g\|_H$ on $|\operatorname{Im} z|\le H$, and the projections $\Pi_{<0},\Pi_{\ge0}$ onto negative and non-negative modes have norm 1. Also $\|\mathrm{e}^g\|_H\le\mathrm{e}^{\|g\|_H}$ and $\|\mathrm{e}^{g_1}-\mathrm{e}^{g_2}\|_H\le\|g_1-g_2\|_H\,\mathrm{e}^{\max\|g_j\|_H}$.

Theorem 6.3 (W2)**.** *Let M,d be as in Lemma 6.2, $\beta=\frac14$, and choose $D\ge1$ with $4\pi M\mathrm{e}^{-2\pi D}\le\beta(1-\mathrm{e}^{-2\pi D})^2$. Put $H=d-D-\beta$ and $T^\circ=T-\mathrm{id}-t_0$. Assume $H>1$, which holds for b near η since $d\sim\pi/|\log\lambda|$, and that b is so close to η that $C_1\Lambda\mathrm{e}^{4\pi(Y_0+D+1)}<\frac12$, where $C_1=4M+2\beta$. Then:*

(1) *The map $\mathcal{T}(Q)=-\Pi_{<0}[T^\circ\circ(\mathrm{id}+Q)]$ is a contraction, with constant $\le\frac12$, of the ball $\{Q\in\Pi_{<0}\mathfrak{A}_H:\|Q\|_H\le\beta\}$ into itself. Its fixed point Q^* and $P':=\Pi_{\ge0}[T^\circ\circ(\mathrm{id}+Q^*)]$ satisfy*

$$T\circ(\mathrm{id}+Q^*)=\mathrm{id}+t_0+P'\qquad\text{on } |\operatorname{Im} z|<H.$$

(2) *$\varphi_- = \mathrm{id}+Q^*$ is univalent on $\{\operatorname{Im} z < H-1\}$ and $\varphi_+ = \mathrm{id}+t_0+P'$ on $\{\operatorname{Im} z > -(H-1)\}$. The function equal to $R\circ(\varphi_+ - \mathrm{i}h)$ above and $S\circ\varphi_-$ below is a solution of (1); it is the sewing solution K_b^W of Proposition 3.4.*

(3) *The equation $\varphi_+(z_0) = \mathrm{i}h$ has a unique solution with $|z_0 - (\mathrm{i}h - t_0)| < 1$, and $|z_0 - (\mathrm{i}h - t_0)| \le C_1\Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$. After the translation $z \mapsto z + z_0$ one has $K_b^W(0) = 1$ and $P = R^{-1}\circ K_b^W - \mathrm{id}$ with $P(0) = 0$.*

(4) *For every $n \ge 1$ the Fourier coefficients of this P satisfy*

$$\Big|\frac{c_n}{\Lambda^n} - \tau_n(b)\Big| \le C_n\Lambda, \qquad C_n = C\,M(M+1)\,\mathrm{e}^{2\pi(n+2)(Y_0+D+2)},$$

with an absolute constant C.

Proof. (1) Write $T^\circ\circ(\mathrm{id}+Q) = \sum_{m\neq 0} t_m \mathrm{e}^{2\pi\mathrm{i}mz}\mathrm{e}^{2\pi\mathrm{i}mQ}$. Then

$$\|T^\circ\circ(\mathrm{id}+Q)\|_H \le \sum_{m\neq 0}|t_m|\mathrm{e}^{2\pi|m|(H+\beta)} \le 2M\sum_{m\ge 1}\mathrm{e}^{-2\pi mD} \le \beta$$

by Lemma 6.2 and the choice of D. Similarly

$$\|T^\circ\circ(\mathrm{id}+Q_1) - T^\circ\circ(\mathrm{id}+Q_2)\|_H \le \sum_{m\neq 0}|t_m|\mathrm{e}^{2\pi|m|(H+\beta)}2\pi|m|\,\|Q_1-Q_2\|_H \le \frac{4\pi M\mathrm{e}^{-2\pi D}}{(1-\mathrm{e}^{-2\pi D})^2}\|Q_1-Q_2\|_H,$$

which is at most $\frac12\|Q_1-Q_2\|_H$. Since $|\operatorname{Im}(z+Q)| \le H+\beta < d$ on the strip, the Fourier series of T converges at $z+Q(z)$ and equals T there (Lemma 6.1). The identity follows by splitting $T^\circ\circ(\mathrm{id}+Q^*) = P' + \Pi_{<0}[\ldots] = P' - Q^*$.

(2) The series of Q^* has only negative modes and converges on $\operatorname{Im} z \le H$, so it converges on the whole lower half-plane $\operatorname{Im} z < H$. There $|Q^{*\prime}| \le 2\pi\mathrm{e}^{-2\pi}\beta/(1-\mathrm{e}^{-2\pi})^2 < 1$ on $\operatorname{Im} z \le H-1$, so $\operatorname{Re}\varphi_-' > 0$ and φ_- is univalent (Noshiro–Warschawski). The same holds for φ_+ above. On the strip, $R\circ(\varphi_+ - \mathrm{i}h) = R\circ(T\circ\varphi_- - \mathrm{i}h) = R\circ T\circ\varphi_- = S\circ\varphi_-$, because $\mathrm{i}h$ is a period of R and $R\circ T = S$. The functional equation holds because $\varphi_\pm$ commute with $z\mapsto z+1$. The resulting function realises the sewing of Proposition 3.4: the images of $\varphi_\pm$ are the two half-cylinders and the seam in the z-plane is $\Gamma = \varphi_-^{-1}(\mathbb{R})$, a curve in $|\operatorname{Im} z| \le \beta$. Since $\varphi_\pm - \mathrm{id}$ is bounded, the argument principle shows that the restrictions of $\varphi_\pm$ to the two sides of Γ map onto the two half-cylinders. By uniqueness of the uniformisation it is K_b^W up to translation. (The identity $R\circ T = S$ is used on the band, where orbits stay in $D_{r'}$ and $\mathcal{R}_b$ inverts R; Lemma 6.1.)

(3) For $\operatorname{Im} z \ge h/2-1$ we have $|P'(z)-p_0'| \le \sum_{n\ge1}|p_n'|\mathrm{e}^{-2\pi n(h/2-1)}$ and $|p_n'| \le \beta\,\mathrm{e}^{-2\pi nH}$. Moreover $p_0' = \sum_{m\ge1} t_m[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_{-m}$, and $|[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_{-m}| \le \mathrm{e}^{2\pi m\beta}\mathrm{e}^{-2\pi mH}$, so $|p_0'| \le 2M\mathrm{e}^{-2\pi(d+H-\beta)} \le 2M\Lambda\mathrm{e}^{2\pi(2Y_0+D+2+2\beta)}$, and $\sum_{n\ge1}|p_n'|\mathrm{e}^{-2\pi n(h/2-1)} \le 2\beta\Lambda\mathrm{e}^{2\pi(Y_0+D+2+\beta)}$. Both are bounded by (using $D\ge1$ and $Y_0\ge1$)

$$\tfrac12 C_1\Lambda\mathrm{e}^{4\pi(Y_0+D+1)}.$$

Since $\operatorname{Im}(\mathrm{i}h - t_0) = h/2$ and $|P''| \le \beta$ there, Rouché's theorem on the disc of radius 1 gives the unique root and the bound, using the smallness hypothesis. Then $K_b^W(z_0) = R(\varphi_+(z_0) - \mathrm{i}h) = R(0) = 1$, and after translation $P(z) = \varphi_+(z+z_0) - \mathrm{i}h - z$, so $P(0) = 0$.

(4) For $n \ge 1$ the n-th mode of P is $c_n = p_n'\mathrm{e}^{2\pi\mathrm{i}nz_0}$. Because Q^* has no constant or positive modes, $[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_0 = 1$ and $[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_k = 0$ for $k>0$. Hence $p_n' = t_n + \sum_{m>n} t_m[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_{n-m}$, and

$$|p_n' - t_n| \le \sum_{m>n} M\mathrm{e}^{-2\pi md}\mathrm{e}^{2\pi m\beta}\mathrm{e}^{-2\pi(m-n)H} \le 2M\mathrm{e}^{-2\pi n(d-\beta)}\mathrm{e}^{-2\pi(d+H-\beta)}.$$

With $z_0 = \mathrm{i}h - t_0 + \zeta$, $|\zeta| \le C_1\Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$ by (3), we have $\mathrm{e}^{2\pi\mathrm{i}nz_0} = \Lambda^n \mathrm{e}^{-2\pi\mathrm{i}nt_0}\mathrm{e}^{2\pi\mathrm{i}n\zeta}$ and $|t_n \mathrm{e}^{-2\pi\mathrm{i}nt_0}| = |\tau_n|$, so $|t_n| = |\tau_n|\mathrm{e}^{-\pi nh}$. Dividing,

$$\frac{c_n}{\Lambda^n} - \tau_n = \tau_n(\mathrm{e}^{2\pi\mathrm{i}n\zeta} - 1) + (p_n' - t_n)\mathrm{e}^{-2\pi\mathrm{i}nt_0}\mathrm{e}^{2\pi\mathrm{i}n\zeta}.$$

In the first term $|\tau_n| = |t_n|\mathrm{e}^{\pi nh} \le M\mathrm{e}^{2\pi n(Y_0+1)}$ and $|\mathrm{e}^{2\pi\mathrm{i}n\zeta} - 1| \le 4\pi n|\zeta|$ when $2\pi n|\zeta| \le 1$ (since $|\mathrm{e}^w - 1| \le (\mathrm{e}-1)|w|$ for $|w| \le 1$); in the second, $|\mathrm{e}^{2\pi\mathrm{i}n\zeta}| \le \mathrm{e}$ in that case. If instead $2\pi n|\zeta| > 1$, then $\Lambda \ge (2\pi nC_1)^{-1}\mathrm{e}^{-4\pi(Y_0+D+1)}$, and we use a trivial bound. The display above gives $|p_n'| \le 3M\mathrm{e}^{-2\pi n(d-\beta)}$, and $|\zeta| < 1$, so $|c_n|\Lambda^{-n} = |p_n'|\mathrm{e}^{2\pi n(h/2-\operatorname{Im}\zeta)} \le 3M\mathrm{e}^{2\pi n(Y_0+2+\beta)}$. Hence $|c_n|\Lambda^{-n} + |\tau_n| \le 4M\mathrm{e}^{2\pi n(Y_0+2+\beta)}$, which is already $\le C_n\Lambda$ because $\mathrm{e}^{2\pi n(D-\beta)} \ge 2\pi n$ and $C_1 \le 5$ (this uses $D \ge 1$). In the second term, $|(p_n' - t_n)\mathrm{e}^{-2\pi\mathrm{i}nt_0}| \le 2M\mathrm{e}^{2\pi n(Y_0+1+\beta)}\Lambda\mathrm{e}^{2\pi(2Y_0+2+D+2\beta)}$, by $d = h/2 - Y_0 - 1$ and $H = d - D - \beta$. Adding the two gives the stated bound, since $n(Y_0+1+\beta) + 2Y_0 + 2D + 2.5 \le (n+2)(Y_0+D+2)$. □

Corollary 6.4. *For the sewing solution, $\hat{c}_n(K_b^W) \to B_n\mathrm{e}^{2\pi\mathrm{i}na}$ for every $n \ge 1$ as $b \to \eta^-$. Moreover, $\Xi_n(K_b^W) \to B_n/B_1^n$ and $\lim \arg \hat{c}_1 = \arg B_1 + 2\pi a$.*

Proof. Theorem 6.3(4) and Corollary 4.5: C_n does not depend on b and $\Lambda \to 0$. Proposition 2.4 makes c_1 nonzero for b near η, justifying division and the limiting phase. □

Corollary 6.5 (Local parabolic limit of the sewn solution)**.** *Put $u^* = 1/\mathrm{e} - 1$ and $a = \Phi_{\mathrm{att}}(u^*)$. On some fixed disk about $z = 0$, the function*

$$F_\eta^{\mathrm{att}}(z) = \mathrm{e}(1 + \Phi_{\mathrm{att}}^{-1}(z + a))$$

is holomorphic, satisfies $F_\eta^{\mathrm{att}}(0) = 1$, and has $(F_\eta^{\mathrm{att}})'(0) \ne 0$. As $b \to \eta^-$ through real bases, $K_b^W \to F_\eta^{\mathrm{att}}$ locally uniformly on that disk, with all z-derivatives. Consequently there are $r > 0$ and $b_2 < \eta$ such that, for $b_2 < b < \eta$, the only preimage of 1 under K_b^W in $B(0,r)$ is 0; both K_b^W and $1/K_b^W$ are uniformly bounded there.

Proof. Theorem 4.4(1) gives locally uniform convergence of $A_b(u) := R_b^{-1}(\mathrm{e}(1+u))$ to $A_\eta(u) := \Phi_{\mathrm{att}}(u) - a$ near u^*, with $A_b(u^*) = 0$. The derivative $A_\eta'(u^*)$ is nonzero: the series in the proof of Theorem 4.4, Lemmas 4.2 and 4.3, and Cauchy's estimate give $\Phi_{\mathrm{att}}'(u) = 2/u^2 + O(1/u)$ on a narrower attracting petal. The Abel identity and $f_\eta'(u) \ne 0$ propagate nonvanishing back along the orbit of u^*. Uniform convergence of these local inverses therefore gives $R_b \to F_\eta^{\mathrm{att}}$ on a fixed disk about 0.

In Theorem 6.3, write $P_b(z) = \zeta_b + P_b'(z + z_0(b))$, where $\zeta_b = z_0(b) + t_0(b) - \mathrm{i}h(b) \to 0$. The proof of that theorem gives $p_0'(b) \to 0$ and $|p_n'(b)| \le \beta\mathrm{e}^{-2\pi nH(b)}$ for $n \ge 1$, while $H(b) \to \infty$ and $\operatorname{Im} z_0(b) = h(b)/2 + o(1)$. Hence $P_b \to 0$ uniformly on every fixed bounded z-disk on which the upper representation is defined. That representation contains a fixed disk about 0 for all sufficiently large b, and $K_b^W(z) = R_b(z + P_b(z))$ there. This proves the asserted convergence; Cauchy's formula gives convergence of derivatives. Choose a small closed disk on which F_η^{att} is nonzero and $F_\eta^{\mathrm{att}} - 1$ has only its simple zero at 0. Rouché's theorem and $K_b^W(0) = 1$ give the final assertions. □

Proposition 6.6 (First-order correction)**.** *In the setting of Theorem 6.3, put $K_0 = Y_0 + D + 2$ and assume also $\Lambda\mathrm{e}^{4\pi(K_0+1)} \le \frac{1}{2}$. Then*

$$\hat{c}_1 = \tau_1 + \Lambda\big[-4\pi^2|\tau_1|^2\tau_1 - 2\pi\mathrm{i}\tau_1^2 - 4\pi\mathrm{i}\tau_2\bar{\tau}_1\big] + E, \qquad |E| \le C'\Lambda^2,$$

with C' depending only on M, Y_0, D. Consequently, as $b \to \eta^-$,

$$\frac{\hat{c}_1}{\tau_1} - 1 = \Lambda\Big[-4\pi^2|\tau_1|^2 - 2\pi\mathrm{i}\tau_1 - 4\pi\mathrm{i}\frac{\tau_2\bar{\tau}_1}{\tau_1}\Big] + O(\Lambda^2).$$

Proof. Let $x=\Lambda^{1/2}\mathrm{e}^{2\pi K_0}$ and let q_{-1} be the (-1)-mode of Q^*. Since $\mathrm{e}^{-2\pi d}=\Lambda^{1/2}\mathrm{e}^{2\pi(Y_0+1)}$ and $\mathrm{e}^{-2\pi H}=\Lambda^{1/2}\mathrm{e}^{2\pi(Y_0+1+D+\beta)}$, we have $|t_m|\le Mx^{|m|}$ and $|g_k|\le\|g\|_H x^{|k|}$ for $g\in\mathfrak{A}_H$; also $\Lambda\le x^2$ and $x^2\mathrm{e}^{\pi}\le\frac12$. Below C depends only on M,Y_0,D.

The mode q_{-1}. Take the (-1)-mode of $Q^*=-\Pi_{<0}\sum_{m\neq0}t_m\mathrm{e}^{2\pi\mathrm{i}mz}\mathrm{e}^{2\pi\mathrm{i}mQ^*}$. Since $\mathrm{e}^{2\pi\mathrm{i}mQ^*}$ has only non-positive modes and constant mode 1, the terms $m\le-2$ do not contribute, $m=-1$ contributes t_{-1}, and $m\ge1$ contributes $t_m[\mathrm{e}^{2\pi\mathrm{i}mQ^*}-1]_{-1-m}$. With $\|\mathrm{e}^{2\pi\mathrm{i}mQ^*}-1\|_H\le2\pi m\beta\mathrm{e}^{2\pi m\beta}$ this gives $q_{-1}=-t_{-1}+O(x^3)$.

The modes p_1',p_0'. By the formulas in the proof of Theorem 6.3(3),(4), and because $(Q^*)^j$ has only modes $\le-j$,

$$p_1'=t_1+4\pi\mathrm{i}t_2q_{-1}+\sum_{m\ge3}t_m[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_{1-m}=t_1-4\pi\mathrm{i}t_2t_{-1}+O(x^5),$$

$$p_0'=2\pi\mathrm{i}t_1q_{-1}+\sum_{m\ge2}t_m[\mathrm{e}^{2\pi\mathrm{i}mQ^*}]_{-m}=-2\pi\mathrm{i}t_1t_{-1}+O(x^4).$$

Real structure. By Lemma 3.1, $t_{-1}=\bar t_1$ and $\bar t_0=t_0-\mathrm{i}h$, so $t_{-1}=\bar\tau_1\Lambda\mathrm{e}^{-2\pi\mathrm{i}t_0}$, $t_1t_{-1}=\Lambda|\tau_1|^2$ and $t_2t_{-1}\mathrm{e}^{-2\pi\mathrm{i}t_0}=\Lambda\tau_2\bar\tau_1$. With $|\mathrm{e}^{-2\pi\mathrm{i}t_0}|=\Lambda^{-1/2}$ and $|\tau_n|\le M\mathrm{e}^{2\pi n(Y_0+1)}$,

$$p_1'\mathrm{e}^{-2\pi\mathrm{i}t_0}=\tau_1-4\pi\mathrm{i}\Lambda\tau_2\bar\tau_1+O(x^4),\qquad p_0'=-2\pi\mathrm{i}\Lambda|\tau_1|^2+O(x^4).$$

The root. Write $z_0=\mathrm{i}h-t_0+\zeta$ as in Theorem 6.3(3), so $\zeta+p_0'+\sum_{n\ge1}p_n'\mathrm{e}^{2\pi\mathrm{i}nz_0}=0$ and $\mathrm{e}^{2\pi\mathrm{i}z_0}=\Lambda\mathrm{e}^{-2\pi\mathrm{i}t_0}\mathrm{e}^{2\pi\mathrm{i}\zeta}$. Since $|\zeta|<\frac12$, $|\mathrm{e}^{2\pi\mathrm{i}nz_0}|\le x^n\mathrm{e}^{\pi n}$, and with $|p_n'|\le\beta x^n$ the terms $n\ge2$ sum to $O(x^4)$. Hence first $|\zeta|\le Cx^2$, and then $\zeta=-p_0'-\Lambda\tau_1+O(x^4)=\Lambda(2\pi\mathrm{i}|\tau_1|^2-\tau_1)+O(x^4)$.

Conclusion. $\hat c_1=c_1/\Lambda=p_1'\mathrm{e}^{-2\pi\mathrm{i}t_0}\mathrm{e}^{2\pi\mathrm{i}\zeta}$ and $\mathrm{e}^{2\pi\mathrm{i}\zeta}=1-4\pi^2\Lambda|\tau_1|^2-2\pi\mathrm{i}\Lambda\tau_1+O(x^4)$. Multiplying gives the first formula, with $|E|\le Cx^4=C\Lambda^2\mathrm{e}^{8\pi K_0}$. The second follows by dividing by τ_1, which tends to $B_1\mathrm{e}^{2\pi\mathrm{i}a}\neq0$ (Corollary 4.5). □

The construction is directly computable (`weld_solve.py`). At $b=1.10$ the iteration contracts by $\sim10^{-4}$ per step and gives $\hat c_1=0.0864272029\,\mathrm{e}^{1.5510921403\mathrm{i}}$, against $\tau_1=0.0864272022\,\mathrm{e}^{1.5510921408\mathrm{i}}$. The relative difference is $8.8\cdot10^{-9}=0.38\Lambda$, and at $b=1.30$ it is $3.9\cdot10^{-19}=0.40\Lambda$. At these bases h is too small for the hypotheses of Theorem 6.3, so the agreement is not an instance of the theorem or of Proposition 6.6. It does match the first-order term of Proposition 6.6, whose modulus is 0.3834Λ and 0.3984Λ respectively.

7. Dependence on the base

This section proves that K_b^W depends holomorphically on the base near (b_1,η), with quantitative bounds; the identification with the boundary value K_b of (2) is completed in Section 10.

Theorem 7.1 (W1, local)**.** *Let $b_0\in(b_1,\eta)$. There is a disc $\Delta\ni b_0$ in the base plane such that:*

1. *For $b'\in\Delta$, with $L(b'),L_2(b')$ the continued fixed points, $R_{b'},S_{b'}$ the regular superfunctions, $\varpi(b')=2\pi\mathrm{i}/\log\lambda(b')$ the period of $R_{b'}$, and $T_{b'}=R_{b'}^{-1}\circ S_{b'}$ with its branch continued from b_0, the function $T_{b'}-\mathrm{id}$ is holomorphic on $|\operatorname{Im}z|<d(b_0)+\frac34$, and the contraction of Theorem 6.3 works with β, D and $H=H(b_0)$ unchanged and $2M$ in place of M. The solution is $K_{b'}^W=R_{b'}\circ(\varphi_++\varpi)$ above, where $\varphi_+=T_{b'}\circ\varphi_-$, and $S_{b'}\circ\varphi_-$ below. It is normalised by the root of $\varphi_+(z_0)=-\varpi$ in the disc of radius 1 about $-\varpi-t_0$. (For real b', $\varpi=-\mathrm{i}h$.) Its fixed point, the resulting solution $K_{b'}^W$, and the normalising root $z_0(b')$ are holomorphic in b'.*
2. *For $b'\in\Delta$ with $\operatorname{Im}b'>0$ one has $\arg\lambda>0$, $\arg\lambda_2<0$, and $K_{b'}^W(z)\to L(b')$ as $\operatorname{Im}z\to+\infty$, $K_{b'}^W(z)\to L_2(b')$ as $\operatorname{Im}z\to-\infty$, uniformly for $\operatorname{Re}z$ in compacts. In the upper and lower half-planes (of the translated coordinate) it has the representations*

$R_{b'} \circ (\mathrm{id} + \theta_{\mathrm{up}})$ *and* $S_{b'} \circ (\mathrm{id} + \theta_{\mathrm{dn}})$. *These are the defining representations of the complex-base construction in* `_cbuild.py`.

(3) *(Local uniqueness.) If a solution* F *with* $F(0) = 1$ *admits such representations and, after the translation that puts its seam representation in the form* $T \circ (\mathrm{id} + Q) = \mathrm{id} + t_0 + P'$, *has* $\|Q\|_H \le \beta$ *and its normalising root in that disc, then* $F = K^W_{b'}$. *(Without the root condition the statement is false: the translates of* $K^W_{b'}$ *by the other roots* c_j, $j \ge 1$, *have the same* Q.*)*

Proof. (1) By Lemma 6.1(iii), at b_0 the map T is holomorphic on the strip of half-width $d(b_0)+1$, and by the proof of Lemma 6.2 (the bound there holds on all of Σ) $|T_{b_0} - \mathrm{id} - t_0| \le M$ on $|\operatorname{Im} z| \le d(b_0) + \frac{1}{2}$. The Koenigs maps at $L(b')$, $L_2(b')$ and the principal logarithms defining R^{-1} depend holomorphically on b'. To choose one parameter disc for the unbounded strip, first work on the compact fundamental rectangle $0 \le \operatorname{Re} z \le 1$, $|\operatorname{Im} z| \le d(b_0) + \frac{3}{4}$. Its image under S_{b_0} is compactly contained in the common domain of the two chosen Abel charts; their local parameter continuations therefore give a single disc Δ on this rectangle. The identity $T_{b'}(z+1) = T_{b'}(z)+1$ extends the result to the whole strip. Thus $T_{b'} - \mathrm{id}$ is jointly holomorphic on $\Delta \times \{|\operatorname{Im} z| < d(b_0) + \frac{3}{4}\}$, and so is its mean $t_0(b')$. Shrinking Δ again, uniform continuity on the compact rectangle $|\operatorname{Im} z| \le d(b_0) + \frac{1}{2}$ gives $|T_{b'} - \mathrm{id} - t_0(b')| \le 2M$ on the lines $\operatorname{Im} z = \pm(d(b_0) + \frac{1}{2})$. Cauchy's estimate on these lines gives $|t_n(b')| \le 2M\mathrm{e}^{-2\pi|n|(d(b_0)+1/2)} \le 2M\mathrm{e}^{-2\pi|n|d(b_0)}$, so the estimates of Theorem 6.3(1), with the same $H = d(b_0) - D - \beta$, hold with $2M$ in place of M: the original choice of D gives a Lipschitz constant at most $2\beta = \frac{1}{2}$ and a ball-image norm at most $\beta(1 - \mathrm{e}^{-2\pi D})/\pi < \beta$. The iterates $\mathcal{T}^k(0)$ are holomorphic in b' and converge uniformly, so Q^* is holomorphic in b'. So are P' and, by the implicit function theorem (Rouché), z_0.

(2) From $\lambda \mathrm{e}^{-\lambda} = \log b'$, $\partial\lambda/\partial b = \mathrm{e}^{\lambda}/(b(1-\lambda))$ is positive for $\lambda < 1$ and negative for $\lambda_2 > 1$. Hence $\operatorname{Im} b' > 0$ gives $\operatorname{Im}\lambda > 0 > \operatorname{Im}\lambda_2$ for Δ small. Above, $K^W = R(z + c + o(1))$, and $R(w) = \sigma^{-1}(c'\lambda^w) \to L$ when $\operatorname{Re}(w \log\lambda) \to -\infty$. This is the case as $\operatorname{Im} w \to +\infty$, since $\operatorname{Re}(w\log\lambda) = \operatorname{Re} w \log|\lambda| - \operatorname{Im} w \arg\lambda$. Below, the same argument with $\arg\lambda_2 < 0$ applies. The representations are those of Theorem 6.3(2), transported by $z \mapsto z + z_0$.

(3) The fixed point of a contraction is unique in its ball. P' and z_0 are then determined, the latter by $F(0) = 1$ together with the root condition. □

Theorem 7.1 gives quantitative control of the sewn family on complex discs about real bases near η. The base continuation from $b > \eta$ passes the Shell–Thron boundary, where $|\Lambda| = 1$ and the contraction does not apply; Section 10 reaches the sewn family by a different route, a tilted initial region that is insensitive to $|\Lambda|$.

8. Characterisation of the sewn solution

The sewn solution is characterised by its two regular representations and the position of its seam. This is the uniqueness statement in Theorem B; it is also what fixes the root in the identification of Section 10.

Definition 8.1. For real $b \in (1, \eta)$ and $Y > 0$ let $\mathcal{C}_b(Y)$ be the set of solutions F of (1) with $F(0) = 1$ such that ("solution" means holomorphic off a closed singular set and satisfying the equation where defined)

(a) $F = R \circ (\mathrm{id} + P)$ on $\{\operatorname{Im} z > -2Y\}$, with P holomorphic, 1-periodic and bounded there, and $P(0) = 0$ (the branch $R^{-1}(1) = 0$);

(b) $F = S \circ (\mathrm{id} + Q)$ on $\{\operatorname{Im} z < -Y\}$, with Q holomorphic, 1-periodic and bounded there, and $|\operatorname{Im} Q - h/2| \le 1$ on the strip $O = \{-2Y < \operatorname{Im} z < -Y\}$.

Condition (b) says that the real line of repelling time, i.e. the segment (L, L_2), sits half a period of attracting time below the real axis of F. Some such condition is necessary. In the coordinate of Theorem 6.3(1), for each integer $j \geq 0$ the equation $\varphi_+(c) = \mathrm{i}h(1+j)$ has a unique root $c_j = \mathrm{i}h(1+j) - t_0 + \zeta_j$ with $|\zeta_j| < 1$, and $c_0 = z_0$: the Rouché argument of Theorem 6.3(3) applies verbatim, since it only uses $\operatorname{Im} z \geq h/2 - 1$, and gives $|\zeta_j| \leq C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$. The translate $K_b^{(j)} = K_b^W(\cdot + c_j - c_0)$ satisfies $K_b^{(j)}(0) = R(\mathrm{i}hj) = 1$, and has the representations $R \circ (\mathrm{id} + P_j)$ above, with $P_j(z) = \varphi_+(z + c_j) - \mathrm{i}h(1+j) - z$, and $S \circ (\mathrm{id} + Q_j)$ on a lower half-plane, with $Q_j(z) = c_j + Q^*(z + c_j)$. The n-th mode $c_n^{(j)}$ of P_j is $p_n' \mathrm{e}^{2\pi \mathrm{i} n c_j}$, so

$$c_n^{(j)} = c_n \mathrm{e}^{2\pi \mathrm{i} n(c_j - c_0)} = c_n \Lambda^{nj} \mathrm{e}^{2\pi \mathrm{i} n(\zeta_j - \zeta_0)} = c_n \Lambda^{nj}(1 + O(n\Lambda)) \qquad (n\Lambda \leq 1).$$

By Corollary 6.4 and $B_1 \neq 0$, $|c_1^{(j)}| \asymp \Lambda^{1+j}$, so $K_b^{(j)} \neq K_b^W$ for $j \geq 1$ (the function P is determined by the solution, as $P(0) = 0$). On the other hand $|\operatorname{Im} Q_j - (h/2 + jh)| \leq \beta + O(\Lambda)$, so condition (b) fails for $j \geq 1$; the proof shows that it excludes every $j \neq 0$.

Theorem 8.2 (Characterisation)**.** *There are $b_1 < \eta$ and Y_B such that for $b \in (b_1, \eta)$ and $Y_0 + D + 3 < Y < (h - Y_0 - D - 3)/2$ (with Y_0, D as in Theorem 6.3),*

$$\mathcal{C}_b(Y) = \{K_b^W\},$$

with K_b^W normalised as in Theorem 6.3(3). In particular $\hat{c}_n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for every n holds for the unique element of $\mathcal{C}_b(Y)$.

Proof. $K_b^W \in \mathcal{C}_b(Y)$. In the normalised coordinate, $\varphi_-(z) = z + z_0 + Q^*(z + z_0)$ is defined for $\operatorname{Im}(z + z_0) < H$, i.e. for $\operatorname{Im} z < -(Y_0 + 1 + D + \beta) + O(\Lambda)$, which contains $\{\operatorname{Im} z < -Y\}$. There $|Q - z_0| \leq \beta$, so $|\operatorname{Im} Q - h/2| \leq \beta + O(\Lambda) < 1$. The R-representation holds for $\operatorname{Im}(z + z_0) > -(H-1)$, i.e. above $-(h - Y_0 - D - 2 - \beta)$, which contains $\{\operatorname{Im} z > -2Y\}$. P and Q are bounded there.

Uniqueness. Let $F \in \mathcal{C}_b(Y)$, $\varphi_+ = \mathrm{id} + P$ and $\varphi_- = \mathrm{id} + Q$. Work with the strip $\Sigma_q = \{|\operatorname{Im} w'| < d - D\}$ of S-time. By Lemma 6.2 and the choice of D, on $\overline{\Sigma_q}$

$$|T - \mathrm{id} - t_0| \leq \frac{2M\mathrm{e}^{-2\pi D}}{1 - \mathrm{e}^{-2\pi D}} \leq \frac{\beta}{2\pi}, \qquad |T' - 1| \leq \beta < 1,$$

so T is univalent on the convex set Σ_q (Noshiro–Warschawski), continuous up to the boundary, and within $\beta/2\pi$ of a translation.

(i) *The overlap.* Let $O = \{-2Y < \operatorname{Im} z < -Y\}$. For $z \in O$, $\operatorname{Im} \varphi_-(z) \in (h/2 - 2Y - 1,\ h/2 - Y + 1) \subset (-(d-D), d-D)$, because $2Y < h - Y_0 - D - 2$ and $Y > Y_0 + D + 2$. Since $\sigma \circ R(w) = \sigma(1)\lambda^w$ wherever R is defined, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$ gives $\lambda^{\varphi_+} = \lambda^{T \circ \varphi_-}$. Hence $\varphi_+ = T \circ \varphi_- - \mathrm{i}h + k\mathrm{i}h$ on O for a constant $k \in \mathbb{Z}$. No injectivity is needed for this step.

(ii) *The lifted sewn surface.* Let $V_- = \{\operatorname{Im} w' < d - D\}$ (repelling time) and let V_+ be the region above the curve $T(\{\operatorname{Im} w' = -(d-D)\}) - \mathrm{i}h$ (attracting time). Let $\tilde{X}$ be $V_- \sqcup V_+$ modulo $w = T(w') - \mathrm{i}h$ for $w' \in \Sigma_q$. Since T is a homeomorphism of $\overline{\Sigma_q}$ onto its image, the identification graph is closed and $\tilde{X}$ is a Hausdorff Riemann surface. Translation by 1 acts freely on it, and $X = \tilde{X}/\mathbb{Z} \cong \mathbb{C}^*$, both ends being punctured discs. The function $\tilde{K}$, equal to R on V_+ and S on V_-, is well defined, because $R(T(w') - \mathrm{i}h) = S(w')$ on the band. The sewing solution is $\tilde{K} \circ \tilde{\Phi}^W$, where $\tilde{\Phi}^W$ is the lift of its uniformisation; sewing along the real line or along any other essential curve in Σ_q gives the same $\tilde{X}$.

Define $\tilde{\Phi} : \mathbb{C} \to \tilde{X}$ by $\varphi_+ - k\mathrm{i}h$ on $\{\operatorname{Im} z > -2Y\}$ and φ_- on $\{\operatorname{Im} z < -Y\}$. These agree on O by (i). They land in $V_\mp$ as follows. $\operatorname{Im} Q$ is bounded and harmonic on a half-cylinder, hence extends harmonically over the puncture. So on $\{\operatorname{Im} z \leq -Y - \delta\}$ it is bounded by its values on the line $\operatorname{Im} z = -Y - \delta \subset O$. This gives $\operatorname{Im} \varphi_- \leq -Y + h/2 + 1 < d - D$. Likewise, on a line in O

near $\operatorname{Im} z = -2Y$, (i) gives $\operatorname{Im}(\varphi_+ - kih) \geq \operatorname{Im} z - 1 - \beta/2\pi$. The minimum principle then gives $\operatorname{Im}(\varphi_+ - kih) \geq -2Y - 1 - \beta/2\pi$ above that line. This exceeds the height $-h/2 - d + D + \beta/2\pi$ of the lower boundary of V_+, because $2Y < h - Y_0 - D - 3$.

(iii) $\tilde{\Phi}$ *is a translate of* $\tilde{\Phi}^W$. $\tilde{\Phi}$ commutes with $z \mapsto z+1$ and descends to a holomorphic $\Phi : \mathbb{C}/\mathbb{Z} \to X \cong \mathbb{C}^*$. Φ is proper, since $\varphi_\pm - \mathrm{id}$ are bounded, and has local degree one at each end, since $P \to p_0$ and $Q \to q_0$. A proper holomorphic map of $\mathbb{C}^*$ that fixes the ends with local degree one is linear. Hence $\tilde{\Phi} = \tilde{\Phi}^W(\cdot + c)$ for some $c \in \mathbb{C}$, and $F = \tilde{K} \circ \tilde{\Phi} = K_b^W(\cdot + c)$ in the coordinate of Theorem 6.3(1)–(2). Univalence of $\varphi_\pm$ is a conclusion, not an assumption.

(iv) *The root.* Take a line $\ell \subset O$ with $\operatorname{Im} z = -Y''$ and Y'' close to $2Y$. There $\operatorname{Im} \varphi_- \leq -2Y + h/2 + 1 + o(1) < H - \beta$, so $\varphi_-(\ell)$ lies in $\varphi_-^W(\{\operatorname{Im} < H\})$. The inclusion $\varphi_-^W(\{\operatorname{Im} < H\}) \supset \{\operatorname{Im} < H - \beta\}$ holds because $|Q^*| \leq \beta$: for w in the smaller set, $\varphi_-^W - w$ has winding number 1 around 0 on the boundary of a large fundamental rectangle. By injectivity of $\tilde{\Phi}^W$, the points $z + c$ with $z \in \ell$ lie in $\{\operatorname{Im} < H\}$. There $Q(z) = c + Q^*(z + c)$, so $|\operatorname{Im} c - h/2| \leq 1 + \beta$. On the upper chart, $\varphi_+(z) - kih = \varphi_+^W(z + c) - ih$, and $P(0) = 0$ gives $\varphi_+^W(c) = ih(1 - k)$. Since $|P'| \leq \beta$ where φ_+^W is defined, $|c - (ih(1-k) - t_0)| \leq \beta$. Comparing imaginary parts with the bound on $\operatorname{Im} c$ gives $|k|h \leq 1 + 2\beta$, so $k = 0$. Then c lies in the Rouché disc of Theorem 6.3(3), so $c = z_0$ and $F = K_b^W$. The last sentence of the theorem is Corollary 6.4. □

9. General real saddle-node unfoldings

Sections 3–8 use the exponential only through a few constants. This section states their results for a general real saddle-node unfolding, lists the changes in the proofs, and adds one exact invariance statement. Tetration is the case $a_2 = \frac{1}{2}$, $\gamma = 1$, $\rho = \frac{1}{3}$ below, in the coordinate $w = \mathrm{e}(1 + u)$ (so $c = \mathrm{e}$ below) and with the parameter $\mu = \mathrm{e} \log b$, $\mu_0 = 1$.

9.1. **Setting.** Fix an affine coordinate u centred at the saddle-node, $w = w^* + c\,u$ with a constant $c > 0$, and write maps in this coordinate as $f_\mu(u) = (f_\mu(w^* + cu) - w^*)/c$. Let $f_\mu(w)$ be a family with the following properties.

- (H1) $f_\mu(w)$ is entire in w, holomorphic in μ near μ_0, and real for real (μ, w).
- (H2) $f_{\mu_0}(w^*) = w^*$, $f'_{\mu_0}(w^*) = 1$ and $a_2 := c\, f''_{\mu_0}(w^*)/2 > 0$ (replace w by $-w$ if necessary).
- (H3) $\gamma := c^{-1} \partial_\mu f_\mu(w^*)|_{\mu_0} > 0$ (reverse μ if necessary). For $s := \mu_0 - \mu > 0$ small there are two real fixed points $L < L_2$ near w^*, L attracting with multiplier λ and L_2 repelling with multiplier λ_2.
- (H4) The base point $w_0 < w^*$ is real, and f_{μ_0} is increasing with $f_{\mu_0}(w) > w$ on $[w_0, w^*)$. Then for s small, $w_0 < L$ lies in the immediate basin of L, and the Koenigs map satisfies $\sigma(w_0) < 0$ (Lemma 9.2).
- (H5) $B_1 \neq 0$, where B_n are the Fourier coefficients of the inverse upper horn map of the germ at $s = 0$ (used only where indicated).

Write the germ as $g(u) = (f_{\mu_0}(w^* + cu) - w^*)/c = u + a_2 u^2 + a_3 u^3 + \cdots$, with iterative residue $\rho = 1 - a_3/a_2^2$. Its formal Abel function is $\alpha(u) = -1/(a_2 u) + \rho \log(-u) + \sum_{k \geq 1} d_k u^k$, and Φ_{att}, Φ_{rep}, B_n are defined as in Section 2.1. The normalisation $F(0) = 1$ becomes $F(0) = w_0$: $R^{-1}(w_0) = 0$, $u^* := (w_0 - w^*)/c$ and $a := \Phi_{\mathrm{att}}(u^*) \in \mathbb{R}$.

The choice of c does not matter. Replacing c by c' conjugates the germ by the linear map $\psi(v) = (c'/c)v$; then ρ, $a_2\gamma$ and the multipliers are unchanged, while the Fatou coordinates become $\Phi \circ \psi - \kappa$ with the real constant $\kappa = \rho \log(c'/c)$. Hence $B_n \mapsto B_n \mathrm{e}^{2\pi i n \kappa}$ and $a \mapsto a - \kappa$, so $B_n \mathrm{e}^{2\pi i n a}$ and $|B_n|$ do not depend on c. More generally, for a real germ $\psi(v) = c_\psi v + O(v^2)$ with $c_\psi > 0$, the Fatou coordinates of $\psi^{-1} \circ g \circ \psi$ are $\Phi_{\mathrm{att}} \circ \psi - \kappa$ and $\Phi_{\mathrm{rep}} \circ \psi - \kappa$ with $\kappa = \rho \log c_\psi + \psi''(0)/(2a_2 c_\psi^2) \in \mathbb{R}$: both functions satisfy the Abel equation of the conjugated

germ, and their difference from its formal Abel function tends to κ along orbits. So $B_n \mathrm{e}^{2\pi i n a}$ is unchanged under real-analytic conjugacy of the germ, with the base point transported.

The definitions of S, T, t_n, τ_n, h, Λ, K^W and $\mathcal{C}(Y)$ are those of Sections 3 and 8, with μ in place of b.

No global hypothesis enters. All estimates take place in a fixed disc about w^*; (H1) is used only to make S entire.

Proposition 9.1 (Scales)**.** *As $s \to 0^+$,*

$$|\log\lambda|,\ \log\lambda_2 = 2\sqrt{a_2\gamma s}\,(1+O(\sqrt{s})), \qquad \Lambda = \exp\Big(-\frac{2\pi^2}{\sqrt{a_2\gamma s}}(1+o(1))\Big),$$

and $p := |\log\lambda|\log\lambda_2$ is holomorphic in s with $p = 4a_2\gamma s + O(s^2)$. Moreover, with $A_1 = 1/\log\lambda$, $A_2 = 1/\log\lambda_2$ and $u_1 = (L - w^*)/c$, $u_2 = (L_2 - w^*)/c$,

$$A_1 + A_2 \to \rho, \qquad A_1(u_1 - u_2) \to 1/a_2, \qquad h_2 - h \to 2\pi\rho,$$

where $h_2 = 2\pi/\log\lambda_2$.

Proof. Write the Weierstrass preparation (12) as $f(u) - u = (u^2 - \sigma_1 u + \sigma_2)k_s(u)$ with $k_0(0) = a_2$. Then $\sigma_2 = f_s(0)/a_2 + O(s^2) = -\gamma s/a_2 + O(s^2)$, so the discriminant is $D(s) = 4\gamma s/a_2 + O(s^2)$, and $\lambda_{1,2} = 1 \mp a_2\sqrt{D} + O(D)$. The limits of $A_1 + A_2$ and $A_1(u_1 - u_2)$ are those of Lemma 4.1, with $1 - k'(0)/k(0)^2 = 1 - a_3/a_2^2$ and $1/k(0) = 1/a_2$. Finally $h_2 - h = 2\pi(A_1 + A_2)$. For tetration this gives C_η of Proposition 2.2 and $p = 2s + \cdots$.

Holomorphy of p. Shrink r and s_0 so that $|f_s'(u) - 1| \le \frac{1}{4}$ on $\{|s| < s_0\} \times \overline{D_r}$ and both roots of q_s lie in $D_{r/2}$. For $\phi(s,u)$ jointly holomorphic there, the power sum

$$\phi(s,u_1(s)) + \phi(s,u_2(s)) = \frac{1}{2\pi\mathrm{i}} \oint_{|u|=r} \phi(s,u)\,\frac{q_s'(u)}{q_s(u)}\,du$$

(roots counted with multiplicity) is holomorphic in s, since q_s has no zeros on $|u| = r$. Apply this to $\ell = \operatorname{Log} f_s'$ and to ℓ^2. For real $s > 0$, $\log\lambda_j = \ell(s,u_j)$, so

$$p = -\ell(s,u_1)\,\ell(s,u_2) = \tfrac{1}{2}\Big[\big(\ell(s,u_1)^2 + \ell(s,u_2)^2\big) - \big(\ell(s,u_1) + \ell(s,u_2)\big)^2\Big]$$

extends holomorphically to $|s| < s_0$, with $p(0) = 0$. By the first display, $p(s)/s \to 4a_2\gamma$ as $s \to 0^+$, so $p'(0) = 4a_2\gamma$ and $p = 4a_2\gamma s + O(s^2)$. □

Lemma 9.2 (Real dynamics)**.** *Assume* (H1)–(H4). *For $s > 0$ small, f_μ is increasing on $[L, L_2]$ with $f_\mu(w) < w$ on (L, L_2), so (L, L_2) lies in the immediate basin of L. Moreover $w_0 < L$, f_μ maps $[w_0, L)$ into itself with $f_\mu(w) > w$ there, and w_0 lies in the immediate basin of L. The Koenigs map σ of L, $\sigma'(L) = 1$, is negative on $[w_0, L)$ and positive on (L, L_2).*

Proof. We work in the coordinate u; since $c > 0$, order and signs are the same as in w. Shrinking r and s_0, in (12) we may assume $|k_s - a_2| \le a_2/2$ and $|f_s' - 1| \le \frac{1}{4}$ on D_r. For real $s > 0$ the discriminant $D(s) = 4\gamma s/a_2 + O(s^2)$ is positive, so $u_1 < u_2$ are real and $|u_j| \le C\sqrt{s}$; on $D_r \cap \mathbb{R}$, f_s is increasing and $k_s > 0$.

On (u_1, u_2), $q_s < 0$, so $f_s(u) < u$, while $f_s(u) > f_s(u_1) = u_1$. Orbits in (u_1, u_2) therefore decrease to a fixed point in $[u_1, u_2)$, i.e. to u_1; the connected set $[u_1, u_2)$ lies in the basin and contains u_1.

Take $\delta \in (0, r)$ with $g(-\delta) < -\delta/2$. By (H4), $g(u) - u$ has a positive minimum m on the compact interval $I = [u^*, -\delta]$, and since g is increasing, $g(I) \subset [u^* + m, -\delta/2]$. By uniform convergence $f_s \to g$ on I, for s small $f_s(u) > u$ on I and $f_s(I) \subset (u^*, -\delta/4)$; and $-\delta/4 < u_1$ because $|u_1| \le C\sqrt{s}$. On $[-\delta, u_1)$, $q_s > 0$, so $f_s(u) > u$, and $f_s(u) < f_s(u_1) = u_1$ by monotonicity.

Hence f_s maps $[u^*, u_1)$ into itself with $f_s(u) > u$; orbits increase to a fixed point in $(u^*, u_1]$, which can only be u_1. As before $[u^*, u_1]$ lies in the immediate basin.

The map σ is real and increasing near L with $\sigma(L) = 0$. For $w \in [w_0, L)$, $f^n(w) \uparrow L$, so $\sigma(f^n w) < 0$ for large n and $\sigma(w) = \lambda^{-n}\sigma(f^n w) < 0$, as $\lambda \in (0, 1)$. For $w \in (L, L_2)$ the same argument gives $\sigma(w) > 0$. □

9.2. **The theorems.**

Theorem 9.3 (Confluence, general)**.** *Under* (H1)–(H4)*,* $\tau_n(\mu) \to B_n \mathrm{e}^{2\pi i n a}$ *as* $\mu \to \mu_0^-$ *for every* $n \geq 1$*, with* τ_n *taken on the branch* $\operatorname{Im} t_0 = h/2$*. Under* (H5) *also* $t_n/t_1^n \to B_n/B_1^n$*.*

Theorem 9.4 (Sewing solution, general)**.** *Under* (H1)–(H4) *there is* $\mu_1 < \mu_0$ *such that for* $\mu \in (\mu_1, \mu_0)$ *the following hold. The sewn solution* K_μ^W*, normalised by* $K_\mu^W(0) = w_0$*, is the unique element of* $\mathcal{C}_\mu(Y)$*. Its coefficients satisfy* $|c_n/\Lambda^n - \tau_n(\mu)| \leq C_n \Lambda$*, with* C_n *as in Theorem 6.3(4) and constants* Y_B, Y_0, D, M *that depend on the family (with* $Y_0 = Y_B + 2 + \pi|\rho|$*; see the proof below). The family is holomorphic in* μ *near* (μ_1, μ_0)*, and for* $\operatorname{Im}\mu > 0$ *small it tends to* L *and* L_2 *as* $\operatorname{Im} z \to +\infty$ *and* $-\infty$*, respectively.*

Theorem 9.5 (First and higher orders, general)**.** *Under* (H1)–(H4)*, and for* n *with* $B_n \neq 0$*,*

$$\log \frac{\tau_n}{B_n \mathrm{e}^{2\pi i n a}} \sim \sum_{j \geq 1} \kappa_j^{(n)} p^j, \qquad \kappa_1^{(n)} = \frac{\delta t_n}{4 a_2 \gamma\, B_n} - \frac{2\pi i n\, \delta t_0}{4 a_2 \gamma},$$

with δt_m *defined as in Theorem 5.6 from the orbits of the germ* g *and of its reflected local inverse. Each* $\kappa_j^{(n)}$ *is given by convergent parabolic series as in Theorem 5.8.*

Proof of Theorems 9.3–9.5. The proofs of Sections 3–8 apply verbatim, with the following replacements.

- In Lemmas 4.1–4.3, Theorem 4.4 and Lemma 5.2, replace $\alpha_2 = -2/u + \frac{1}{3}\log(-u)$ by $-1/(a_2 u) + \rho \log(-u)$ and $-2/u$ by $-1/(a_2 u)$. The limiting petal is $\{\operatorname{Re}(-1/(a_2 u)) > R\}$.
- The gate of Theorem 4.4 needs a margin depending on ρ. Put $c_\rho := \pi|\rho| + 1$. On the gate, $\operatorname{Im}\Phi_{\mathrm{rep}} - \operatorname{Im}(-1/(a_2 u)) = \rho \arg(-u) + O(1/Y_*)$, so $|\operatorname{Im} Z - \operatorname{Im}\Phi_{\mathrm{rep}}| < c_\rho$ for μ near μ_0, and the gate lies in the band $\mathcal{B}_\mu(Y')$, $Y' := Y_* - c_\rho - 2$, rather than $\mathcal{B}_\mu(Y_* - 2)$. In step (b) after the proof of Theorem 4.4, fix $Y_* \geq Y_B + 3c_\rho + 3$, so that $Y' \geq Y_B + 2c_\rho + 1$. For points within $o(1)$ of the gate, Lemma 6.1(ii) (with the constant of the next item) gives $|\operatorname{Im}\mathcal{S}_\mu| \leq h/2 - Y' + \pi|\rho| + \frac{1}{2} + o(1) < h/2 - Y' + c_\rho \leq h/2 - (Y_B + 1) - c_\rho$. So the shifted gate line lies in the strip Σ of Lemma 6.1(iii) taken with $Y = Y_B + 1$, where T is holomorphic, and the rest of step (b) is unchanged.
- In Theorem 4.4(2), the reflected inverse $G_\mu(v) = -f_\mu^{-1}(-v)$ is the local inverse near w^*. It is jointly holomorphic, its germ is $v + a_2 v^2 + (2a_2^2 - a_3)v^3 + \cdots$, and its iterative residue is $-\rho$.
- In Lemma 3.1, $1 < L$ is replaced by (H4): S maps $\mathbb{R}$ onto (L, L_2), where $\sigma > 0$, while $\sigma(w_0) < 0$ (Lemma 9.2).
- In Lemma 6.1, the bound $\pi/3$ in (ii) becomes $\pi|\rho|$ (and $C/Y + C|\log\lambda| \leq \frac{1}{2}$ for $Y \geq Y_B$ and μ near μ_0), and $h_2 = h + 2\pi/3 + o(1)$ becomes $h_2 = h + 2\pi\rho + o(1)$. Since $\pi|\rho|$ may exceed 2, the half-width $h/2 - Y - 2$ of the strip in (iii) is replaced by $h/2 - Y - c_\rho$, $\Sigma = \{|\operatorname{Im} w'| < h/2 - Y - c_\rho\}$; the proof of (iii) is unchanged, because on the arcs $|\operatorname{Im}\mathcal{S}_\mu| \geq h/2 - Y - \pi|\rho| - \frac{1}{2} > h/2 - Y - c_\rho$. In (i) the drift $2\pi/3$ becomes $2\pi|\rho|$, and we take $Y_B \geq 8c_\rho$. Accordingly, Lemma 6.2 holds with $Y_0 := Y_B + 1 + c_\rho$ in place of $Y_B + 3$: with $Y = Y_B + 1$, Σ has half-width $h/2 - Y_B - 1 - c_\rho = d + 1$, $d = h/2 - Y_0 - 1$. Injectivity of f_μ on $D_{r'}$ follows from $f'_{\mu_0}(w^*) = 1$ after shrinking r'.

- In Proposition 3.4, the singular set of R is the closed set Σ_R of inverse-branch images of the singular values of f.
- In Lemma 5.1, $d_i = \frac{i}{2}u^{i-1} + \cdots$ becomes $d_i = ia_2u^{i-1} + \cdots$, and $D(s) = 8s + \cdots$ becomes $4\gamma s/a_2 + \cdots$. In Theorem 5.6, $p'(0) = 2$ becomes $p'(0) = 4a_2\gamma$ (Proposition 9.1).
- In Theorem 7.1(2), $\lambda_{1,2} = 1 \mp a_2\sqrt{D(s)} + O(D)$ gives $\partial_\mu\lambda_1 > 0 > \partial_\mu\lambda_2$, hence $\operatorname{Im}\mu > 0 \Rightarrow \arg\lambda_1 > 0 > \arg\lambda_2$.
- Statements involving ratios or logarithms use (H5) in place of Proposition 2.4. Corollary 6.5 needs, in addition, that w_0 is not precritical for f_{μ_0}.

Everything else is literally unchanged. In particular the welding identity, the Wiener-algebra contraction and the characterisation use only the two Koenigs charts and the relation $\sigma \circ R(w) = \sigma(w_0)\lambda^w$. □

Remark 9.6 (On (H5))**.** The proof of Proposition 2.4 gives $B_1 \neq 0$ whenever f_{μ_0} restricted to its immediate parabolic basin has exactly one singular value. This covers $w^2 + c$, whose basin contains the single critical point. For $v - \sin v + c$ the critical values $2\pi k + 1$ are real and only $k = 0$ lies in the basin, but the map is not of finite type, so applying the argument requires a check. For $w + (w^2 - c)e^w$ we only know $B_1 \neq 0$ numerically.

9.3. **Independence of the base point.**

Proposition 9.7. *Let w_0 and w_0' both satisfy* (H4)*, and mark with a prime the quantities computed from w_0'. Then for every $\mu < \mu_0$ near μ_0 and every $n \geq 1$,*

$$\tau_n'(\mu) = \tau_n(\mu)\, e^{2\pi i n\Delta(\mu)}, \qquad B_n e^{2\pi i n a'} = B_n e^{2\pi i n a}\, e^{2\pi i n \Delta_0},$$

with $\Delta(\mu) = R_\mu^{-1}(w_0')$ and $\Delta_0 = \Phi_{\mathrm{att}}(u^{*\prime}) - \Phi_{\mathrm{att}}(u^*)$ both real, and $\Delta(\mu) \to \Delta_0$ as $\mu \to \mu_0^-$. Hence $|\tau_n(\mu)|$ does not depend on the base point, and neither does $\operatorname{Re}\kappa_j^{(n)}$ for any j. The moduli $|c_n|$ and $|\hat{c}_n|$ of the sewn solution change only at relative order $O(n\Lambda)$.

The second identity is the definition $\Delta_0 = a' - a$. Its content is that Δ_0 is real and is the limit of $\Delta(\mu)$, so that the first identity is compatible with Theorem 9.3. The independence of $B_n e^{2\pi i n a}$ from the choice of coordinates is a different statement; it is the conjugation invariance shown in the Setting.

Proof. By Lemma 9.2, $\sigma(w_0)$ and $\sigma(w_0')$ are negative, so $\Delta = \log(\sigma(w_0')/\sigma(w_0))/\log\lambda$ is real. Then $R'^{-1} = R^{-1} - \Delta$ and $T' = T - \Delta$, so $t_n' = t_n$ for $n \neq 0$ and $t_0' = t_0 - \Delta$. The branch condition $\operatorname{Im} t_0 = h/2$ is preserved, and $\tau_n' = \tau_n e^{2\pi i n\Delta}$. At μ_0, by (H4) the g-orbits of u^* and $u^{*\prime}$ increase along the negative axis to 0, where α is real, so $a = \lim_k[\alpha(g^k u^*) - k]$ and a' are real. By Theorem 4.4(1) in the general form, $\Delta(\mu) = R_\mu^{-1}(w_0') \to \Phi_{\mathrm{att}}(u^{*\prime}) - a = \Delta_0$. Taking real parts of the logarithm of the first identity divided by the second gives the statement on $\operatorname{Re}\kappa_j^{(n)}$. For the sewn solutions, the normalisation $K^W(0) = w_0'$ gives $K^W[w_0'] = K^W[w_0](\cdot + \delta)$ with $\delta + P(\delta) = \Delta$. Since K^W is not real on the real axis, $\operatorname{Im}\delta = -\operatorname{Im} P(\delta) = O(\Lambda)$ is in general not zero, so by Remark 2.3 c_n is multiplied by $e^{2\pi i n\delta} = e^{2\pi i n \operatorname{Re}\delta}(1 + O(n\Lambda))$. □

Corollary 9.8. *$|B_n|$ and the real parts $\operatorname{Re}\kappa_j^{(n)}$ are invariant under orientation-preserving real-analytic conjugacy of the unfolding near (μ_0, w^*) and under reparametrisation of μ.*

Proof. Let $(\tilde{f}_{\tilde{\mu}}, \tilde{w}_0)$ be a second unfolding satisfying (H1)–(H4), $\tilde{\mu} = \theta(\mu)$ a real-analytic reparametrisation with $\theta(\mu_0) = \tilde{\mu}_0$, $\theta' > 0$, and ϕ_μ univalent on a disc N about w^*, continuous in μ, real with $\phi_\mu' > 0$ on $N \cap \mathbb{R}$, such that $\phi_\mu \circ f_\mu = \tilde{f}_{\theta(\mu)} \circ \phi_\mu$ on $N \cap f_\mu^{-1}(N)$. Fix $\mu < \mu_0$ near μ_0 and write $\phi = \phi_\mu$, $\tilde{f} = \tilde{f}_{\theta(\mu)}$. Then $\phi(L) = \tilde{L}$ and $\phi(L_2) = \tilde{L}_2$ with the same multipliers, so h, Λ and p agree; in particular p depends only on the multipliers.

The base point w_0 need not lie in N. Fix an auxiliary base point $w_0' \in N$ with $w_0 < w_0' < w^*$ and $[w_0', w^*] \subset N$; it satisfies (H4), and for μ near μ_0, $[w_0', L_2] \subset N$.

Attracting side. By uniqueness of Koenigs coordinates, $\tilde\sigma = \phi'(L)\,\sigma \circ \phi^{-1}$ near $\tilde L$, and $\sigma = \lambda^{-n}\sigma \circ f^n$ extends this to $\tilde\sigma \circ \phi = \phi'(L)\,\sigma$ at every w whose forward orbit stays in N. This includes w_0' and a complex neighbourhood of (L, L_2). In particular $\tilde\sigma(\phi(w_0')) = \phi'(L)\sigma(w_0') < 0$ by Lemma 9.2, and the attracting Abel coordinates with base points w_0' and $\phi(w_0')$ satisfy $\tilde R'^{-1} \circ \phi = R'^{-1}$ there.

Repelling side. Likewise $\tilde\sigma_{\mathrm{rep}} = \phi'(L_2)\,\sigma_{\mathrm{rep}} \circ \phi^{-1}$ near $\tilde L_2$, i.e. $\tilde\sigma_{\mathrm{rep}}^{-1}(\zeta) = \phi(\sigma_{\mathrm{rep}}^{-1}(\zeta/\phi'(L_2)))$ for small ζ. Both sides continue along a neighbourhood of the negative axis, because $\sigma_{\mathrm{rep}}^{-1}((-\infty, 0)) = (L, L_2) \subset N$. Hence, for z near $\mathbb{R}$,

$$\tilde S(z) = \tilde\sigma_{\mathrm{rep}}^{-1}(-\lambda_2^z) = \phi(S(z - c')), \qquad c' = \frac{\log \phi'(L_2)}{\log \lambda_2} \in \mathbb{R}.$$

Combining, $\tilde T'(z) = \tilde R'^{-1}(\tilde S(z)) = T'(z - c')$ near $\mathbb{R}$, where T' and $\tilde T'$ are computed from w_0' and $\phi(w_0')$. Since c' is real, both sides lie on the branch $\operatorname{Im} t_0 = h/2$, and by the translation invariance (7), $\tilde\tau_n' = \tau_n'$. The proof of Proposition 9.7 uses only $\sigma < 0$ at the two base points at the given μ; applied to f with w_0, w_0' and to $\tilde f$ with $\tilde w_0, \phi(w_0')$, it gives $|\tau_n| = |\tau_n'| = |\tilde\tau_n'| = |\tilde\tau_n|$.

By Theorem 9.3 and $a \in \mathbb{R}$, $|B_n| = \lim |\tau_n|$, so $|\tilde B_n| = |B_n|$. Since a is real, $\operatorname{Re}\log(\tau_n/B_n \mathrm{e}^{2\pi i n a}) = \log|\tau_n| - \log|B_n|$, which is therefore the same function of p for both unfoldings. As p is increasing in s (Proposition 9.1), the coefficients of the asymptotic expansion in p are unique, and $\operatorname{Re}\tilde\kappa_j^{(n)} = \operatorname{Re}\kappa_j^{(n)}$. □

9.4. **The first-order term as a derivative on the parabolic stratum.** Theorem 9.5 gives $\kappa^{(n)}$ as a convergent series. Here we identify it. It is the one-sided derivative of a function that extends continuously to the parabolic maps, and on those maps the function is the horn coefficient. So $\kappa^{(n)}$ splits into one number attached to the germ and the derivative of the horn map along the unfolding.

Work in the affine coordinate u of the Setting, $w = w^* + cu$. Let $\mathcal{O}_r$ be the Banach space of bounded holomorphic functions on $D_r = \{|u| < r\}$ with the supremum norm. An unfolding in this subsection is a real-analytic curve $s \mapsto f_s \in \mathcal{O}_r$ with $f_0 = g$ and the sign condition of (H3). No global hypothesis such as (H1) is needed: τ_n denotes the invariant $t_n \mathrm{e}^{-2\pi i n t_0}$ of the local transition $\mathcal{A} \circ (\mathcal{S}^\sharp)^{-1}$ on the gate line, which is all that Theorems 3.6–5.8 use. We write

$$f_s = g - sX + O(s^2), \qquad X \in \mathcal{O}_r, \qquad X(0) = \gamma > 0.$$

Call X the *direction* of the unfolding. The radius r is fixed small enough that all compositions below ($f \circ g$, $\phi \circ g$ for $\phi \in \mathcal{O}_{2r}$) are defined on a neighbourhood of $\overline{D_{r/2}}$, where all estimates take place. Let $\mathcal{P}$ be the set of $f \in \mathcal{O}_r$ near g with a double fixed point in $D_{r/2}$ (the *parabolic stratum*). Near g, $\mathcal{P}$ is the zero set of the discriminant $D[f] = \sigma_1^2 - 4\sigma_2$ of the Weierstrass polynomial of $f - \mathrm{id}$. Since $\sigma_1[g] = \sigma_2[g] = 0$ and $\sigma_2[f] = f(0)/a_2 + O(\|f - g\|^2)$, D is a submersion at g. Hence $\mathcal{P}$ is a complex-analytic hypersurface with tangent space

$$T_g\mathcal{P} = \{Y \in \mathcal{O}_r : \ Y(0) = 0\},$$

where a tangent vector Y means the curve $g - tY + O(t^2)$. Every $Y \in T_g\mathcal{P}$ is tangent to a holomorphic curve in $\mathcal{P}$, for instance $g - tY + c(t)$ with $c(t) = O(t^2)$ given by the implicit function theorem; for real Y the curve is real.

For real f near g with two real fixed points (attracting one on the left), put $\mathcal{I}_n[f] = \tau_n[f]$. For $f \in \mathcal{P}$ put $\mathcal{I}_n[f] = B_n[f]\,\mathrm{e}^{2\pi i n a[f]}$, computed from the germ of f at its double fixed point with the same base point w_0. Both are the invariant $t_n \mathrm{e}^{-2\pi i n t_0}$ of $T = \mathcal{A}[f] \circ \mathcal{S}^\sharp[f]^{-1}$ on the gate line, with $\mathcal{A}[f]$ normalised by $\mathcal{A}[f](u^*) = 0$ (Theorems 3.6 and 5.6). This normalisation matters: an

additive constant on the attracting side changes $t_n \mathrm{e}^{-2\pi i n t_0}$, while one on the repelling side does not. By Theorem 9.3, $\mathcal{I}_n$ is continuous at g from the side of two real fixed points.

Lemma 9.9. *Fix $N \geq 3$. After shrinking r, the coefficient $\nu[f] = A_1 + A_2$ and the one-step error $F[f] = \Psi[f] \circ f - \Psi[f] - 1 = q[f]^N \Gamma[f]$ of Lemma 5.1 are holomorphic functions of f on a neighbourhood of g in $\mathcal{O}_r$, with values in bounded holomorphic functions on $D_{r/2}$. For each $\delta \in (0, r/2)$, the model time $\Psi[f] - \nu[f] \log u$ is holomorphic in f on a (smaller) neighbourhood of g, with values in bounded holomorphic functions on the annulus $\delta < |u| < r/2$. The same holds for the reflected inverse $G[f](v) = -f^{-1}(-v)$.*

Proof. The proof of Lemma 5.1 uses only Weierstrass preparation and division, the removability of the symmetric expressions Θ, Θ' across the diagonal, and the inversion of the triangular matrix $M(0)$. Each of these steps depends holomorphically on a Banach parameter. The inverse function theorem gives $G[f]$. □

Proposition 9.10. *There is a continuous complex-linear functional $\mathfrak{L}_n$ on $\mathcal{O}_r$ with the following two properties.*

1. *For every unfolding with direction X, $\mathcal{I}_n[f_s] = \mathcal{I}_n[g] + s\, \mathfrak{L}_n(X) + O(s^{10/9})$ as $s \to 0^+$.*
2. *For every holomorphic curve $t \mapsto g_t \in \mathcal{P}$ with $g_0 = g$, the function $t \mapsto \mathcal{I}_n[g_t]$ is holomorphic near 0, with derivative $\mathfrak{L}_n(-\partial_t g_t|_{t=0})$ at 0.*

Proof. Define $\mathfrak{L}_n(X)$ by the formula of Theorem 5.6 for $\dot\tau_n$, where every derivative in s is replaced by the derivative in the direction $-X$:

$$\dot{\mathcal{A}}(u) = d\Psi[g](-X)(u) + \sum_{k\geq 0} \Big[dF[g](-X)(u_k) + F[g]'(u_k)\, \dot u_k \Big], \qquad \dot u_{k+1} = g'(u_k)\dot u_k - X(u_k),\ \dot u_0 = 0,$$

on the compact set $K \subset P_R$ of Proposition 5.4; at the base point and on the rest of the basin it is continued by $\dot{\mathcal{A}} = \dot{\mathcal{A}} \circ g^m + \mathcal{A}_0' \circ g^m \cdot \dot u_m$, as in Section 5. Define $\dot{\mathcal{S}}^\sharp$ in the same way from G, with the constant $\Phi_{\mathrm{rep}}[g](p')$ of $\mathcal{S}^\sharp$ held fixed, and then δT, δt_m and $\mathfrak{L}_n(X) = \mathrm{e}^{-2\pi i n t_0}(\delta t_n - 2\pi i n\, t_n\, \delta t_0)$ at g. By Lemma 9.9 every term is linear in X. The bounds of Lemma 5.3 hold with $\|X\|$ as a factor, so the series converge and $\mathfrak{L}_n$ is continuous.

(1) For a single unfolding, $\partial_s \Psi_s|_{s=0} = d\Psi[g](-X)$ and $\partial_\sigma F_\sigma|_{\sigma=0} = dF[g](-X)$ by the chain rule, so $\dot{\mathcal{A}}$ above is the derivative of Proposition 5.4, and Theorem 5.6 (in the form of Theorem 9.5) is statement (1).

(2) Let $t \mapsto g_t \in \mathcal{P}$ be a holomorphic curve with $g_0 = g$, with double fixed point $u_0(t)$. The germ of g_t at $u_0(t)$ has $a_2(t) \to a_2$, so the proof of Lemma 5.2 with $\sigma = 0$, in the coordinate $-1/(a_2(t)(u - u_0(t)))$, gives the petal bounds for all k, uniformly for small complex t. Hence $\mathcal{A}[g_t](u) = \Psi[g_t](u) + \sum_k F[g_t](g_t^k u)$ converges uniformly and is holomorphic in t. By Weierstrass' theorem its t-derivative is the termwise derivative, which is the series for $\dot{\mathcal{A}}$ with $-X$ replaced by $\partial_t g_t|_{t=0}$. As at $s = 0$ in Section 5, $\mathcal{A}[g_t]$ is $\Phi_{\mathrm{att}}[g_t]$ plus a constant; this constant is removed by the normalisation $\mathcal{A}[g_t](u^*) = 0$, so that $\mathcal{A}[g_t] - \mathcal{A}[g_t](u^*) = \Phi_{\mathrm{att}}[g_t] - a[g_t]$. The repelling side is the same with G, and there additive constants do not affect $\mathcal{I}_n$. The inversion and the Fourier integrals are holomorphic in t, so $\mathcal{I}_n[g_t]$ is holomorphic with derivative $\mathfrak{L}_n(-\partial_t g_t|_{t=0})$. □

Theorem 9.11 (Variation formula)**.** *Assume $B_n \neq 0$. Put*

$$\kappa_{\mathrm{tr}}^{(n)}(g) = \frac{\mathfrak{L}_n(1)}{4a_2\, \mathcal{I}_n[g]}, \qquad \mathcal{D}_n(g)[Y] = \frac{d}{dt}\Big|_{t=0} \log \mathcal{I}_n[g_t] \quad (Y \in T_g\mathcal{P}),$$

where $g_t \in \mathcal{P}$ is any holomorphic curve with $g_0 = g$ and $\partial_t g_t|_{t=0} = -Y$. By Proposition 9.10(2), $\mathcal{D}_n(g)[Y] = \mathfrak{L}_n(Y)/\mathcal{I}_n[g]$, so $\mathcal{D}_n(g)$ is well defined and linear. Then for every unfolding of g

with direction X,

$$\kappa^{(n)} = \kappa^{(n)}_{\mathrm{tr}}(g) + \frac{\mathcal{D}_n(g)[X - X(0)]}{4a_2\, X(0)}.$$

In particular $\kappa^{(n)}_{\mathrm{tr}}(g)$ is the first-order coefficient of the translation unfolding $g - s$, and

$$\operatorname{Re}\kappa^{(n)} = \operatorname{Re}\kappa^{(n)}_{\mathrm{tr}}(g) + \frac{d\log|B_n|\,[X - X(0)]}{4a_2\, X(0)},$$

where $d\log|B_n|[Y] = \frac{d}{dt}\log|B_n[g_t]|$ along real curves in $\mathcal{P}$.

Proof. By Proposition 9.1, $p = 4a_2\gamma s + O(s^2)$ with $\gamma = X(0)$. By Proposition 9.10(1) and $B_n \neq 0$, $\kappa^{(n)} = \mathfrak{L}_n(X)/(4a_2\gamma\,\mathcal{I}_n[g])$. Write $X = \gamma\cdot 1 + (X - \gamma)$. Then $X - \gamma \in T_g\mathcal{P}$, and by definition $\mathfrak{L}_n(X - \gamma) = \mathcal{I}_n[g]\,\mathcal{D}_n(g)[X - \gamma]$. By linearity,

$$\kappa^{(n)} = \frac{\gamma\,\mathfrak{L}_n(1) + \mathcal{I}_n[g]\,\mathcal{D}_n(g)[X - \gamma]}{4a_2\gamma\,\mathcal{I}_n[g]}.$$

For the real part, let $f \in \mathcal{P}$ be real. Its double fixed point is real and the base point lies on the real attracting axis, so $a[f]$ is real and $|\mathcal{I}_n[f]| = |B_n[f]|$. For real X, take a real curve g_t in $\mathcal{P}$ tangent to $X - \gamma$. Along it $\operatorname{Re}\log\mathcal{I}_n[g_t] = \log|B_n[g_t]|$, so $\operatorname{Re}\mathcal{D}_n(g)[X - \gamma] = d\log|B_n|\,[X - \gamma]$. □

Corollary 9.12. (1) $\operatorname{Re}\kappa^{(n)}$ *depends on X only through its class modulo the infinitesimal conjugacies $L\phi = \phi\circ g - g'\phi$, ϕ real. Numerically, it is not a function of the germ alone (Remark 9.14).*

(2) *For tetration the direction is $X = (1 + u)\mathrm{e}^u$, and $X - 1 = L(-2 - u)$. Hence $\operatorname{Re}\kappa^{(n)} = \operatorname{Re}\kappa^{(n)}_{\mathrm{tr}}(\mathrm{e}^u - 1)$.*

Proof. (1) $L\phi(0) = 0$, so $X \mapsto X + L\phi$ keeps $X(0)$ and adds $L\phi \in T_g\mathcal{P}$ to $X - X(0)$. The curve $(\mathrm{id} + t\phi)^{-1}\circ g\circ(\mathrm{id} + t\phi)$ lies in $\mathcal{P}$ and equals $g - tL\phi + O(t^2)$, so its direction is $L\phi$. Along it $|B_n|$ is constant: the Fatou coordinates are transported by the real conjugacy, which changes their α-normalisation by a real constant C, and $B_n \mapsto B_n\mathrm{e}^{-2\pi\mathrm{i}nC}$ preserves $|B_n|$. So $d\log|B_n|\,[L\phi] = 0$. (2) $L(-2 - u) = -(\mathrm{e}^u - 1) - 2 + (2 + u)\mathrm{e}^u = (1 + u)\mathrm{e}^u - 1$. Apply (1). □

Remark 9.13 (The constant $\kappa^{(n)}_{\mathrm{tr}}$ is a cocycle)**.** Let ψ be real-analytic near 0 with $\psi(0) = 0$, $\psi'(0) = 1$, and put $\tilde g = \psi^{-1}\circ g\circ\psi$. Conjugating the translation unfolding $g - s$ by ψ gives $\psi^{-1}\circ(g - s)\circ\psi = \tilde g - s\tilde X + O(s^2)$ with $\tilde X = (1/\psi')\circ\tilde g$, and $\tilde X(0) = 1$. Take the base point in the domain of ψ; by Proposition 9.7 this does not affect real parts. Corollary 9.8 and Theorem 9.11 give

$$\operatorname{Re}\kappa^{(n)}_{\mathrm{tr}}(\tilde g) = \operatorname{Re}\kappa^{(n)}_{\mathrm{tr}}(g) - \frac{d\log|B_n|(\tilde g)\,[\tilde X - 1]}{4a_2}.$$

So $\kappa^{(n)}_{\mathrm{tr}}$ is not an invariant of the germ: it depends on the choice of the transverse direction 1, which is not preserved by changes of coordinates. The invariant object is the affine function $X \mapsto \operatorname{Re}\kappa^{(n)}[g; X]$ on directions normalised by $X(0) = 1$, and Theorem 9.11 describes it completely: its linear part is $d\log|B_n|$ on the parabolic stratum, and its value on one transverse direction is $\operatorname{Re}\kappa^{(n)}_{\mathrm{tr}}$.

Remark 9.14 (Numerics)**.** Two independent computations confirm Theorem 9.11. The left side is the first-order coefficient of the unfolding $g - sX$, computed by the uniform model at $s = \pm 10^{-15}$. The right side uses $\kappa^{(n)}_{\mathrm{tr}}$ and the horn variation of the parabolic germs $g \mp \delta X_1$, $X_1 = X - X(0) + \frac{X'(0)}{2a_2}(1 - g')$, together with the base-point term. In 45-digit arithmetic (`kappa_variation.py`) the two sides agree to $3\cdot 10^{-17}$ for $n = 1$, $2\cdot 10^{-13}$ for $n = 2$ and $3\cdot 10^{-9}$ for $n = 3$. The last column is $|\text{left} - \text{right}|$ for the complex identity.

germ	direction X	$\operatorname{Re}\kappa^{(1)}_{\mathrm{tr}}$	$\operatorname{Re}\kappa^{(1)}$	difference ($n = 1$)
$e^u - 1$	$(1+u)e^u$	-0.0101990063458974	-0.0101990063458974	$1.6 \cdot 10^{-17}$
$e^u - 1$	$1 + 0.6u^2$	-0.0101990063458974	39.5149228923293	$1.4 \cdot 10^{-17}$
$e^u - 1$	$1 + 0.3u + 0.5u^4$	-0.0101990063458974	395.094296950776	$2.1 \cdot 10^{-17}$
$u + u^2$	$1 + 0.7u^2$	-0.0150441028391897	-0.0150441028391897	$1.3 \cdot 10^{-20}$
$u + u^2$	$1 + 0.3u - 0.4u^3$	-0.0150441028391897	-0.0143510016842305	$1.1 \cdot 10^{-17}$

For $u + u^2$, $u^2 = -L(u)$, so the fourth row has no horn term. The base point $u^* = -\frac{1}{2}$ used for $u + u^2$ is its critical point, so $\Phi'_{\mathrm{att}}(u^*) = 0$ and the base-point term vanishes there; the exp rows test it. Under a change of coordinates $\psi(u) = u + 0.3u^2$ of $e^u - 1$ (`kappa_cocycle.py`), $\operatorname{Re}\kappa^{(1)}_{\mathrm{tr}}$ moves from -0.0101990 to -2.8764956, and the formula of Remark 9.13 predicts the new value to $3 \cdot 10^{-17}$. The horn variation itself is also evaluated by the regularised orbit series $\dot\Phi_{\mathrm{att}}(u) = \lim_K [\dot\alpha(u_K) - \sum_{k<K} \Phi'_{\mathrm{att}}(u_{k+1}) X_1(u_k)]$ and its repelling analogue, without finite differences in the germ; it agrees with the finite differences to 10^{-16} for $n = 1$.

Remark 9.15 (Numerical illustration)**.** The script `generality_check.py` implements the construction of this section for any family given by f and its Taylor coefficients at the fixed points. The table lists the values at $\lambda = 0.97$, from runs over $\lambda = 0.5, \dots, 0.97$; the fourth column tends to $\operatorname{Re}\kappa^{(1)}$ as $p \to 0$.

family	ρ	$\lvert B_1\rvert$	$\operatorname{Re}\log(\tau_1/B_1 e^{2\pi i a})/p$	$\lvert\tau_1\rvert/\lvert B_1\rvert - 1$
b^w	$1/3$	0.0890584364122	-0.0101990	$-9.4 \cdot 10^{-6}$
$w^2 + c$	1	0.0597942361690	-0.0150481	$-1.4 \cdot 10^{-5}$
$v - \sin v + c$	1	0.0723786416245	-0.0142133	$-1.3 \cdot 10^{-5}$
$w + (w^2 - c)e^w$	0	2744.09379131	$+1.08542$	$+1.0 \cdot 10^{-3}$

In each family τ_n, t_n/t_1^n and $\log(\tau_1/B_1 e^{2\pi i a})/p$ converge at the rate $O(p)$. The branch $\operatorname{Im} t_0 = h/2$ holds to 10^{-31} or better, and the sewing iteration of Theorem 6.3 gives $\hat c_n = \tau_n(1 + O(\Lambda))$. For $w^2 + c$ with the base points $w_0 = 0$, 0.2 and 0.35, the real part in the fourth column agrees to all ten digits shown at each λ, as Proposition 9.7 requires. The imaginary part ranges from -0.07 to 116. The germs of $w^2 + c$ and $v - \sin v + c$ have the same ρ but different $|B_1|$ and $\operatorname{Re}\kappa^{(1)}$.

Remark 9.16 (Numerical evidence for the analogue of Theorem D for $w^2 + c$)**.** For $f_c(w) = w^2 + c$ the parameter $c = 1/4$ is the cusp of the main cardioid, which plays the role of the Shell–Thron region. For real $c > 1/4$ the two fixed points are complex conjugate and repelling, and the real axis escapes. The two-fixed-point construction of Kneser type converges at the sampled parameters with the normalisation $F(0) = f_c^3(0)$, corresponding to the intended critical-point marking $F(-3) = 0$ (script `quad_kneser.py`). The separate proof document `docs/proofs/quadratic-continuation.tex` establishes its quadratic F and G using marked tilted domains, an explicit polynomial multiplier deformation with open-overlap comparison, and an independently replayed 4,013-box Arb exterior cover. Its conclusion concerns regular height germs with this normalisation; it does not imply convergence of the present numerical iteration. The following earlier observations provide numerical evidence only.

- Along the arc $c = \frac{1}{4} + 0.1e^{i\varphi}$ the sampled values are consistent with continuation across the cardioid boundary. Interpolation through points on both sides of the neutral band predicts held-out points to $2 \cdot 10^{-8}$, as accurately as interpolation within one side.
- On vertical ladders $c = c_r + iy$, extrapolation towards $y = 0$ supports agreement with the sewn coefficient. At $\lambda = 0.1$ the extrapolated value gives $(\hat c_1/\tau_1 - 1)/\Lambda = 0.2272 - 0.1120\,i$, against $0.2279 - 0.1115\,i$ from the first-order formula at the end of Section 6. So the data support agreement with K^W including its $O(\Lambda)$ correction; they do not prove existence or identification of the boundary limit. The ratio c_2/c_1^2 agrees with t_2/t_1^2 to $4 \cdot 10^{-8}$.

For the analogue of Theorem E the numerics are only partial. The construction converges at sampled points in a wedge to the lower right of the cusp. In the multiplier coordinate $c = \mu/2 - \mu^2/4$, at $\arg\mu = 0.25\pi$ and 0.4π, interpolation across the neutral band errs by $\le 10^{-7}$ against a variation of order 10^{-1}. Further left the iteration does not converge, and above it converges only after the θ lines are raised in some tested cases. These observations are consistent with local holomorphic continuation, but do not establish it. Moving the Taylor centre or fixing the constant Fourier mode of θ_{up} may enlarge the regular numerical region; these changes remain untested. They cannot alone give direct coverage of the whole upper half-plane while retaining both Schröder charts: at $c = 1/4 + i/2$, $\lambda_{\mathrm{up}} = i$, the fifth inverse-Schröder coefficient requires $0 = (3 - 5i)/2$. The separate quadratic proof document supplies the exact obstruction and a conditional local convergence theorem. A full numerical method must also replace the neutral-end representation. A pointwise algorithm that does not record paths does not by itself establish single-valued analytic continuation; parameter holomorphy and agreement of local branches must also be established.

10. Continuation of Kneser's solution around the cusp

This section proves Theorem D, for the continuation that passes near η. The continuation leaves the real exterior ray, crosses the upper Shell–Thron boundary arc close to the cusp, and returns to the real axis below η from the upper half-plane. Near the cusp the two fixed points are close, and Kneser's initial region (the chord between the fixed points together with its image) can be tilted continuously into an initial region for the attracting–repelling configuration below η. The multiplier of the upper fixed point crosses the unit circle on the way, but this plays no role. In the logarithmic coordinate of the pair of fixed points the tilted region is a thin strip on which the map is a translation up to a relative error $O(|\varepsilon|)$, so its quotient is quasiconformally a flat cylinder, whether or not the fixed points are linearisable. At the end of the path the tilted region is a fundamental domain for the sewn surface of Proposition 3.4, and a proper-map argument identifies the continued solution with K_b^W.

Conventions for this section. Here ε is the complex parameter $1 - \lambda_1$ defined below; for real $b < \eta$ it is the ε of the earlier sections, and we use results of those sections only in the form stated there. The letters α (a tilt angle), ς (a normal coordinate), $\mathsf{r} \in [0, 1]$ (an interpolation parameter), H, X, M, q and f have local meanings (in the subsection "Consequences at the cusp" $f = F_\eta^{\mathrm{att}}$). L_1 denotes the fixed point with multiplier $1 - \varepsilon$: it is the upper fixed point L_+ for real $b > \eta$ and the attracting fixed point L for real $b < \eta$. Constants C, c do not depend on ε; r_0 is decreased finitely many times.

10.1. **The cusp parameter.** For small complex ε put

$$\lambda_1 = 1 - \varepsilon, \qquad \log b(\varepsilon) = \lambda_1 \mathrm{e}^{-\lambda_1} = (1 - \varepsilon)\mathrm{e}^{\varepsilon - 1},$$

and let $\lambda_2(\varepsilon)$ be the other root of $\lambda \mathrm{e}^{-\lambda} = \log b(\varepsilon)$ near 1. It is holomorphic, $\lambda_2 = 1 + \varepsilon + \frac{2}{3}\varepsilon^2 + O(\varepsilon^3)$. By Lemma 2.1, $L_j = \mathrm{e}^{\lambda_j}$ are fixed points of E_b with multipliers λ_j. In the coordinate $w = \mathrm{e}(1 + u)$,

$$f_\varepsilon(u) = \mathrm{e}^{\mu(1+u)-1} - 1, \qquad \mu = \mathrm{e}\log b = (1 - \varepsilon)\mathrm{e}^{\varepsilon},$$

with fixed points $u_j = \mathrm{e}^{\lambda_j - 1} - 1$, so that $u_1 - u_2 = -2\varepsilon(1 + O(\varepsilon))$. For $0 < r \le \frac{1}{10}$ let

$$\mathcal{S}_r^{\mathrm{c}} = \Big\{\varepsilon :\ 0 < |\varepsilon| < r,\ -\tfrac{7\pi}{12} < \arg\varepsilon < \tfrac{\pi}{12}\Big\}, \qquad \mathcal{S}_r^{\mathrm{c}\prime} = \Big\{\varepsilon \in \mathcal{S}_r^{\mathrm{c}} :\ -\tfrac{7\pi}{12} + \tfrac{1}{10} < \arg\varepsilon < \tfrac{\pi}{12} - \tfrac{1}{10}\Big\}.$$

The lemmas are stated on $\mathcal{S}_r^{\mathrm{c}}$; the family is built on $\mathcal{S}_r^{\mathrm{c}\prime}$, so that the parameter discs of Proposition 10.9 stay in $\mathcal{S}_r^{\mathrm{c}}$. Put $u^* = \mathrm{e}^{-1} - 1$, the base point $w = 1$, and $u_n^* = f_\varepsilon^n(u^*)$.

Lemma 10.1 (Cusp parameter)**.** *There are $r_0, r_1 > 0$ with the following properties.*

(1) $\varepsilon\mapsto b(\varepsilon)$ *is injective on* $\mathcal{S}^{\mathrm{c}}_{r_0}$. *For every* $r\le r_0$ *there is* $r'>0$ *such that* $b(\mathcal{S}^{\mathrm{c}\prime}_r)$ *contains* $\{b:|b-\eta|<r',\ \operatorname{Im}b>0\}$ *and the segments* $(\eta-r',\eta)$ *and* $(\eta,\eta+r')$. *Real* $b\in(\eta-r_1,\eta)$ *corresponds to real* $\varepsilon\in(0,r_0)$, *and then* $L_1=L$ *is the attracting and* L_2 *the repelling real fixed point. Real* $b\in(\eta,\eta+r_1)$ *corresponds to* $\varepsilon(\theta)=1-\theta\cot\theta-\mathrm{i}\theta=\frac{1}{3}\theta^2-\mathrm{i}\theta+O(\theta^4)$, $0<\theta<\theta_1$; *then* $L_1=L_+$ *is the fixed point in the upper half-plane used in Proposition 10.11, and* $\lambda_2(\varepsilon(\theta))=\overline{\lambda_1}$, $u_2=\overline{u_1}$.

(2) *On* D_{ρ_0}, $\rho_0=\frac12$, $f_\varepsilon(u)-u=(u-u_1)(u-u_2)k_\varepsilon(u)$ *with* k_ε *jointly holomorphic in* (ε,u) *and* $|k_\varepsilon-\frac12|\le\frac14$. *Moreover* $f'_\varepsilon(u_j)=\lambda_j$.

(3) $g_\varepsilon(v)=(\operatorname{Log}(1+v)+1)/\mu-1$ *is jointly holomorphic on* $\{\operatorname{Re}(1+v)>0\}$, *and* $g_\varepsilon(f_\varepsilon(u))=u$ *whenever* $|\operatorname{Im}(\mu(1+u))|<\pi$, *in particular on* D_{ρ_0} *and near the real segment* $[u^*,0]$.

Proof. (1) $(1-\varepsilon)\mathrm{e}^\varepsilon-1=-\frac12\varepsilon^2(1+O(\varepsilon))$, so $\log b-\log\eta=-(2\mathrm{e})^{-1}\psi_0(\varepsilon)^2$ with ψ_0 holomorphic and $\psi_0(\varepsilon)=\varepsilon(1+O(\varepsilon))$. For small r_0, ψ_0 is injective and maps $\mathcal{S}^{\mathrm{c}}_{r_0}$ into a sector of opening $2\pi/3+o(1)<\pi$, on which squaring is injective. The image of $\mathcal{S}^{\mathrm{c}\prime}_r$ under $-\psi_0^2$ contains, near 0, all directions in $(-\frac{\pi}{6}+0.2+o(1),\frac{7\pi}{6}-0.2-o(1))\supset[0,\pi]$. Since $b\mapsto\log b$ is biholomorphic near η, (1) follows except for the identification of the real segments. For real $\varepsilon\in(0,1)$, $\lambda_1\in(0,1)$ and Lemma 2.1 applies. For real $b>\eta$ the upper fixed point has multiplier $L_+\log b=\theta\cot\theta+\mathrm{i}\theta$ with $\theta=\operatorname{Im}(L_+\log b)\in(0,\pi)$ (the computation in the proof of Proposition 10.11), and $\lambda_1=1-\varepsilon$ gives $\varepsilon(\theta)$, whose argument lies in $(-\pi/2,-\pi/2+\theta)$. The conjugate fixed point $L_-=\overline{L_+}$ is the other one near e, so $\lambda_2=\overline{\lambda_1}$.

(2) $k_0(u)=(\mathrm{e}^u-1-u)/u^2=\frac12+\sum_{n\ge1}u^n/(n+2)!$ differs from $\frac12$ by less than 0.1 on $D_{1/2}$. Division of $f_\varepsilon(u)-u$ by $(u-u_1)(u-u_2)$ (Cauchy's formula on $|u|=1$) gives joint holomorphy and continuity at $\varepsilon=0$. Differentiating the product at u_j gives the multipliers.

(3) $\operatorname{Log}(\mathrm{e}^{\mu(1+u)-1})=\mu(1+u)-1$ when $|\operatorname{Im}(\mu(1+u))|<\pi$. □

10.2. **The logarithmic coordinate.** Put $q=(u-u_1)/(u-u_2)$ and $\chi=\log q$, so that

$$u=u(\chi)=u_2+\frac{u_1-u_2}{1-\mathrm{e}^\chi}.$$

The points $\chi\in2\pi\mathrm{i}\mathbb{Z}$ (the *holes*) correspond to $u=\infty$. Let $\mathcal{G}^{\mathrm{c}}=\{\chi:|\chi-2\pi\mathrm{i}k|\ge\frac12\text{ for all }k\in\mathbb{Z}\}$, and put $A_1=1/\operatorname{Log}\lambda_1=-\varepsilon^{-1}(1+O(\varepsilon))$ and $Z=A_1\chi$.

Lemma 10.2 (The step in the logarithmic coordinate)**.** *For* $0<|\varepsilon|<r_0$ *the following hold (the proofs of (1)–(5) do not use the sector).*

(1) *On* $\mathcal{G}^{\mathrm{c}}$, $|1-\mathrm{e}^\chi|\ge\frac16\max(1,|\mathrm{e}^\chi|)$; *hence* $|u(\chi)|\le C|\varepsilon|$ *and* $|u-u_1||u-u_2|\le C|\varepsilon|^2\min(|\mathrm{e}^\chi|,|\mathrm{e}^{-\chi}|)$. *On* $\{|\chi|\le1\}$, $|1-\mathrm{e}^\chi|\ge0.63|\chi|$.

(2) *For* $u\in D_{\rho_0}\setminus\{u_1,u_2\}$,

$$\frac{q(f(u))}{q(u)}=1+\beta(u),\qquad\beta=\frac{(u_1-u_2)k(u)}{1+(u-u_1)k(u)},$$

and $\Delta(u):=\operatorname{Log}(1+\beta(u))=-\varepsilon(1+\vartheta(u))$ *with* $|\vartheta|\le C(|u|+|\varepsilon|)$ *and* $|\Delta'(u)|\le C|\varepsilon|$ *on* $D_{\rho_0/2}$. *Thus* $\chi(f(u))=\chi(u)+\Delta(u)$ *on the branch continued along the short segment from* $\chi(u)$, *and* $A_1\Delta=1+O(|u|+|\varepsilon|)$.

(3) *On* $\mathcal{G}^{\mathrm{c}}$, *regarded as functions of* χ, $|\vartheta|\le C|\varepsilon|$ *and* $|\partial_\chi\Delta|\le C|\varepsilon|^2$. *On* $\{|\chi|\le1\}\cap\{|u(\chi)|\le\rho_0/2\}$, $|\partial_\chi\Delta|\le C|\varepsilon|^2/|\chi|^2$.

(4) $\Delta(u_1)=\operatorname{Log}\lambda_1$ *and* $\Delta(u_2)=-\operatorname{Log}\lambda_2$. *For* $\chi\in\mathcal{G}^{\mathrm{c}}$, $|\Delta-\operatorname{Log}\lambda_1|+|\partial_\chi\Delta|\le C|\varepsilon|^2|\mathrm{e}^\chi|$ *if* $\operatorname{Re}\chi\le0$, *and* $|\Delta+\operatorname{Log}\lambda_2|+|\partial_\chi\Delta|\le C|\varepsilon|^2|\mathrm{e}^{-\chi}|$ *if* $\operatorname{Re}\chi\ge0$.

(5) $|\partial_\varepsilon\Delta|\le1+C|\varepsilon|$ *at fixed* $\chi\in\mathcal{G}^{\mathrm{c}}$.

All functions here are jointly holomorphic in ε *and* u *(or* χ*).*

Proof. (1) On the circle $|\chi| = \frac{1}{2}$ the minimum of $|e^{\chi} - 1|$ is $1 - e^{-1/2} > 0.39$, attained at $\chi = -\frac{1}{2}$. The function $1/(e^{\chi} - 1)$ is holomorphic on the part of the strip $|\operatorname{Im}\chi| \le \pi$ outside the disc, is bounded by 1 on its horizontal edges (where $|e^{\chi} - 1| = 1 + e^{\operatorname{Re}\chi}$), and tends to -1 and 0 at the two ends; the maximum principle and periodicity give $|e^{\chi} - 1| \ge 0.39$ on $\mathcal{G}^{c}$. For $\operatorname{Re}\chi \ge \log 2$, $|e^{\chi} - 1| \ge \frac{1}{2}|e^{\chi}|$, and for $0 \le \operatorname{Re}\chi \le \log 2$, $0.39 \ge \frac{1}{6}|e^{\chi}|$. The bounds on u follow from $u - u_2 = (u_1 - u_2)/(1 - e^{\chi})$ and $u - u_1 = e^{\chi}(u - u_2)$. The last bound is the minimum of $|(e^{\chi} - 1)/\chi|$ on the closed unit disc, $1 - e^{-1}$.

(2) The identity follows from $(f(u) - u_j)/(u - u_j) = 1 + (u - u_{3-j})k(u)$. By Lemma 10.1(2) and $u_1 - u_2 = -2\varepsilon(1 + O(\varepsilon))$, $\beta = -\varepsilon(1 + O(|u| + |\varepsilon|))$, and $\operatorname{Log}(1 + \beta) = \beta(1 + O(|\beta|))$. Cauchy's estimate on D_{ρ_0}, where $\Delta = O(|\varepsilon|)$, bounds Δ'.

(3) The first bound is (2) with (1). For the derivatives, $\partial_{\chi}\Delta = \Delta'(u)\, du/d\chi$ with $du/d\chi = (u_1 - u_2)e^{\chi}/(1 - e^{\chi})^2$, which is $O(|\varepsilon|)$ on $\mathcal{G}^{c}$ by (1) and $O(|\varepsilon|/|\chi|^2)$ on $|\chi| \le 1$.

(4) $1 + \beta(u_1) = 1 + (u_1 - u_2)k(u_1) = f'(u_1) = \lambda_1$, and $1 + \beta(u_2) = 1/(1 + (u_2 - u_1)k(u_2)) = 1/\lambda_2$; all logarithms are principal and close to 0. Since $\Delta'(u) = O(|\varepsilon|)$, $|\Delta(u) - \Delta(u_1)| \le C|\varepsilon||u - u_1| \le C|\varepsilon|^2|e^{\chi}|$ by (1) when $\operatorname{Re}\chi \le 0$, and $|\partial_{\chi}\Delta| = |\Delta'(u)|\,|u_1 - u_2|\,|e^{\chi}|/|1 - e^{\chi}|^2 \le C|\varepsilon|^2|e^{\chi}|$. The case of u_2 is symmetric, with $e^{-\chi}$ in place of e^{χ}.

(5) At fixed χ, $\Delta = -\varepsilon(1 + \vartheta)$ with ϑ holomorphic in ε and $O(|\varepsilon|)$; Cauchy's estimate on a disc of radius $|\varepsilon|/2$ gives $\varepsilon\,\partial_{\varepsilon}\vartheta = O(|\varepsilon|)$. □

10.3. **Tilted initial regions.** Write $\omega = e^{i\alpha}$. For $x_0 \in [-\frac{1}{2}, \frac{1}{2}]$ let

$$\ell = \ell_{\alpha, x_0} = \{i\pi + x_0 + t\omega : \ t \in \mathbb{R}\},$$

and use the coordinates $\chi = i\pi + x_0 + (t + i\varsigma)\omega$, $t, \varsigma \in \mathbb{R}$. We call ℓ *admissible for* ε if

$$\cos\alpha \ge \tfrac{3}{5}, \qquad -\operatorname{Im}(\varepsilon\bar{\omega}) \ge \tfrac{1}{2}|\varepsilon|. \tag{16}$$

For $\varepsilon \in \mathcal{S}_r^{c}$ put $\alpha(\varepsilon) = \frac{1}{2}(\arg\varepsilon + \frac{\pi}{2}) \in (-\frac{\pi}{24}, \frac{7\pi}{24})$. Then $\cos\alpha(\varepsilon) > \cos\frac{7\pi}{24} > 0.6$ and $-\operatorname{Im}(\varepsilon\bar{\omega})/|\varepsilon| = \sin(\frac{\pi}{4} - \frac{1}{2}\arg\varepsilon) > \sin\frac{5\pi}{24} > 0.6$. Hence, for $\varepsilon_0 \in \mathcal{S}_r^{c\prime}$, the line ℓ_{α, x_0} with $|\alpha - \alpha(\varepsilon_0)| \le \frac{1}{20}$ is admissible for every $\varepsilon \in \mathcal{S}_r^{c}$ with $|\arg\varepsilon - \arg\varepsilon_0| \le \frac{1}{20}$: then $\alpha \in (-\frac{\pi}{24}, \frac{7\pi}{24})$, so $\cos\alpha > \cos\frac{7\pi}{24} > \frac{3}{5}$, and $-\operatorname{Im}(\varepsilon\bar{\omega})/|\varepsilon| \ge \sin(\frac{5\pi}{24} - \frac{1}{10}) > \frac{1}{2}$. At $\varepsilon(\theta)$ (real $b > \eta$), $\alpha = 0$ is admissible; at real $\varepsilon > 0$, $\alpha(\varepsilon) = \frac{\pi}{4}$. For fixed ε the admissible pairs (α, x_0) form a rectangle. The distance from $2\pi i k$ to ℓ is $|(2k - 1)\pi\cos\alpha + x_0\sin\alpha| \ge \frac{3}{5}\pi - \frac{2}{5} > 1.48$, so the strip $\{|\varsigma| \le \frac{1}{4}\}$ lies in $\mathcal{G}^{c}$.

Lemma 10.3 (Tilted initial region)**.** *Let $\varepsilon \in \mathcal{S}_{r_0}^{c}$, let ℓ be admissible for ε, and let $f_{\chi}(\chi) = \chi + \Delta(\chi)$ be f_{ε} in the coordinate χ.*

1. *On $\mathcal{G}^{c}$ the normal component $\operatorname{Im}(\Delta\bar{\omega})$ lies in $[\frac{2}{5}|\varepsilon|, \frac{6}{5}|\varepsilon|]$, and f_{χ} is injective on every convex subset of $\mathcal{G}^{c}$.*
2. *$f_{\chi}(\ell)$ is the graph $\varsigma = g(t)$ of a real-analytic function with $\frac{2}{5}|\varepsilon| \le g \le \frac{6}{5}|\varepsilon|$ and $|g'| \le C|\varepsilon|^2$.*
3. *$M(t, \mathsf{r}) = i\pi + x_0 + t\omega + \mathsf{r}\Delta(i\pi + x_0 + t\omega)$ is an orientation-preserving real-analytic diffeomorphism of $\mathbb{R} \times [0, 1]$ onto the closed strip $\hat{H} = \{0 \le \varsigma \le g(t)\}$ between ℓ and $f_{\chi}(\ell)$.*
4. *$\chi \mapsto u(\chi)$ is injective on $\hat{H}$. Its image $H = H_{\varepsilon}^{\ell}$ together with u_1, u_2 is a closed Jordan region in $D_{C|\varepsilon|}$, bounded by the arcs $\gamma = u(\ell)$ and $f(\gamma) = u(f_{\chi}(\ell))$, which join u_2 ($t \to +\infty$) to u_1 ($t \to -\infty$) and meet only there.*
5. *f_{ε} maps a neighbourhood of γ biholomorphically onto a neighbourhood of $f(\gamma)$; in the coordinate χ it is f_{χ}, which is injective on the convex strip $\{|\varsigma| \le \frac{1}{4}\}$.*

Proof. (1) By Lemma 10.2(3), $\operatorname{Im}(\Delta\bar{\omega}) = -\operatorname{Im}(\varepsilon\bar{\omega}) - \operatorname{Im}(\varepsilon\vartheta\bar{\omega}) \in [\frac{1}{2}|\varepsilon| - C|\varepsilon|^2, |\varepsilon| + C|\varepsilon|^2]$. If χ, χ' lie in a convex subset of $\mathcal{G}^{c}$, then $|f_{\chi}(\chi) - f_{\chi}(\chi') - (\chi - \chi')| \le C|\varepsilon|^2|\chi - \chi'| < |\chi - \chi'|$.

(2) The t-coordinate of $f_\chi(\mathrm{i}\pi+x_0+t\omega)$ is $t+\operatorname{Re}(\Delta\bar\omega)$, whose derivative is $1+O(|\varepsilon|^2)>0$; it is therefore a real-analytic bijection of $\mathbb{R}$, and (1) and $|\partial_\chi\Delta|\le C|\varepsilon|^2$ bound g and g'.

(3) $\partial_t M=\omega(1+O(|\varepsilon|^2))$ and $\partial_s M=\Delta$, so the Jacobian is $\operatorname{Im}(\Delta\bar\omega)(1+O(|\varepsilon|^2))\ge\frac13|\varepsilon|>0$. M maps $\mathsf{r}=0$ onto ℓ and $\mathsf{r}=1$ onto $f_\chi(\ell)$ bijectively, each segment $t=\text{const}$ to a segment of length $\le 2|\varepsilon|$ joining them, and is proper. A proper local diffeomorphism that is a bijection on the boundary lines is a diffeomorphism onto the region between them.

(4) Points of the strip $\{|\varsigma|\le\frac14\}$ that differ by $2\pi\mathrm{i}k$, $k\ne0$, have ς-coordinates differing by $2\pi|k|\cos\alpha>1$; so exp is injective on $\hat H$, and the Möbius map $q\mapsto u$ is injective. Along ℓ, $\operatorname{Re}\chi=x_0+t\cos\alpha-\varsigma\sin\alpha$ tends to $\mp\infty$ as $t\to\mp\infty$, so $q\to0$ ($u\to u_1$) and $q\to\infty$ ($u\to u_2$). The two boundary curves of $\hat H$ are disjoint by (2), and their images are disjoint arcs with common endpoints u_1,u_2; the image of $\hat H$ is the closure of the complementary component they bound that contains $u(\operatorname{int}\hat H)$. The inclusion in $D_{C|\varepsilon|}$ is Lemma 10.2(1).

(5) By Lemma 10.2(2), f_χ represents f_ε in χ, and (1) applies to the convex strip. □

Let $X=X^\ell_\varepsilon$ be the Riemann surface obtained from H by identifying $u\in\gamma$ with $f_\varepsilon(u)\in f(\gamma)$, with charts near the seam given by Lemma 10.3(5). Since χ is injective on $\hat H$ we use it (or $Z=A_1\chi$) as the coordinate on H. We call the branch of χ obtained by applying $f_\chi^{\pm1}$ to points of $\hat H$ the *continued branch*.

Lemma 10.4 (Flat model)**.** *Let ℓ be admissible for $\varepsilon\in\mathcal{S}^{\mathrm{c}}_{r_0}$. In the coordinate Z let $\hat H_0$ be the straight strip between the line $A_1\ell$ and $A_1\ell+1$, and let $C_0=\hat H_0/(\zeta\sim\zeta+1)=\mathbb{C}/\mathbb{Z}$. The map*

$$\mathcal{H}\big(A_1(\mathrm{i}\pi+x_0+t\omega)+\mathsf{r}\big)=A_1M(t,\mathsf{r}),\qquad t\in\mathbb{R},\ 0\le\mathsf{r}\le1,$$

induces a quasiconformal homeomorphism $\mathcal{H}:C_0\to X$ with $|\mathcal{H}(\zeta)-\zeta|\le C|\varepsilon|$ on $\hat H_0$ and Beltrami coefficient $|\upsilon_{\mathcal{H}}|\le C|\varepsilon|$. Consequently:

1. *X is biholomorphic to $\mathbb{C}^*$, the ends at u_1 and u_2 corresponding to the punctures. Let $Z_X:X\to\mathbb{C}^*$ be a uniformisation with the u_1-end at 0, and $z=(2\pi\mathrm{i})^{-1}\log Z_X$. Crossing the seam from γ to $f(\gamma)$ increases z by 1. The inverse $z\mapsto u$, extended by $\tilde F(z+1)=f_\varepsilon(\tilde F(z))$ and by one application of f_χ^{-1} on the strip $\{|\varsigma|\le\frac14\}$, is a holomorphic solution of this equation on the region to the right of the lift of $f_\chi^{-1}(\ell)$.*
2. *If $\mathsf{p}\in X$ and the lift z is normalised at p, then for $|\operatorname{Im}(z-z(\mathsf{p}))|\le1$ and $|\operatorname{Re}(z-z(\mathsf{p}))|\le2$ the Z-coordinate $Z(z)$ of $\tilde F(z)$ on the continued branch satisfies $|Z(z)-Z(\mathsf{p})-(z-z(\mathsf{p}))|\le C|\varepsilon|$.*

Proof. $A_1\Delta=1+\vartheta_1$ with $|\vartheta_1|\le C|\varepsilon|$ on $\mathcal{G}^{\mathrm{c}}$ (Lemma 10.2(2),(3)). In the source coordinate $\zeta=A_1(\mathrm{i}\pi+x_0+t\omega)+\mathsf{r}$ one has $\mathcal{H}(\zeta)=\zeta+\mathsf{r}\vartheta_1$, so $|\mathcal{H}(\zeta)-\zeta|\le C|\varepsilon|$. The partial derivatives are $\partial_t\mathcal{H}=A_1\omega(1+O(|\varepsilon|^2))$ and $\partial_{\mathsf{r}}\mathcal{H}=1+\vartheta_1$, against $A_1\omega$ and 1 for the identity. The unit vectors $A_1\omega/|A_1\omega|$ and 1 make an angle whose sine is $|\operatorname{Im}(\omega\overline{\operatorname{Log}\lambda_1})|/|\operatorname{Log}\lambda_1|\ge\frac12-C|\varepsilon|$, by $\operatorname{Log}\lambda_1=-\varepsilon(1+O(\varepsilon))$ and (16). So, in this basis, $D\mathcal{H}=I+O(|\varepsilon|)$ and $|\upsilon_{\mathcal{H}}|\le C|\varepsilon|$. $\mathcal{H}$ is a homeomorphism of the strips by Lemma 10.3(3); it maps the edge $\mathsf{r}=0$ to $A_1\ell$ and $\mathsf{r}=1$ to $A_1f_\chi(\ell)$ with the same t, and hence respects the identifications $\zeta\sim\zeta+1$ and $u\sim f_\varepsilon(u)$. Across k seams, $\mathcal{H}(\zeta+k)$ is represented on the continued branch by $A_1f_\chi^k(\mathcal{H}(\zeta)/A_1)$, and $A_1f_\chi^k(\chi)-A_1\chi-k=O(|k||\varepsilon|)$ for $|k|\le3$ by Lemma 10.2(2).

(1) $C_0\cong\mathbb{C}^*$. Each end of X is the image of an end of C_0, a punctured disc; quasiconformal maps preserve the infinite modulus of an annular end [20], so each end of X is a punctured disc and $X\cong\mathbb{C}^*$. In C_0 the passage from the edge $\mathsf{r}=0$ to $\mathsf{r}=1$ adds 1. The u_1-end is $t\to-\infty$, where $\operatorname{Im}\zeta=t\operatorname{Im}(A_1\omega)+O(1)\to+\infty$ because $\operatorname{Im}(A_1\omega)=-\operatorname{Im}(\operatorname{Log}\lambda_1\bar\omega)/|\operatorname{Log}\lambda_1|^2<0$. Since $\mathcal{H}$ preserves orientation and ends, the uniformisation with the u_1-end at 0 has the same

properties. The last assertion is the usual lift of the uniformisation; the extra step f_χ^{-1} is allowed by Lemma 10.3(5).

(2) Put $\xi = \exp(2\pi\mathrm{i}(\zeta - \zeta_{\mathsf{p}}))$ on C_0, $\zeta_{\mathsf{p}} = \mathcal{H}^{-1}(\mathsf{p})$, and let υ be the Beltrami coefficient of $\mathcal{H}$ transported to $\hat{\mathbb{C}}$. For complex t with $|\mathsf{t}| < R := 1/(2\|\upsilon\|_\infty)$ let $\mathcal{W}_{\mathsf{t}}$ be the solution with coefficient $\mathsf{t}\upsilon$ fixing $0, 1, \infty$. By [2, Thm. 11], $\mathsf{t} \mapsto \mathcal{W}_{\mathsf{t}}(\xi)$ is holomorphic; $\mathcal{W}_0 = \mathrm{id}$; and the $\mathcal{W}_{\mathsf{t}}$ are 3-quasiconformal and normalised, hence uniformly bounded, by some M_0, on the annulus $A = \{\mathrm{e}^{-3\pi} \le |\xi| \le \mathrm{e}^{3\pi}\}$ [20, Ch. II]. Schwarz's lemma applied to $\mathsf{t} \mapsto \mathcal{W}_{\mathsf{t}}(\xi) - \xi$ on $|\mathsf{t}| < R$ gives $|\mathcal{W}_1(\xi) - \xi| \le 2(M_0 + \mathrm{e}^{3\pi})\|\upsilon\|_\infty \le C|\varepsilon|$ on A, since $R \ge 1$ for small $|\varepsilon|$. By a degree argument on the boundary circles of A, $\mathcal{W}_1^{-1}$ also moves points of $\{\mathrm{e}^{-2\pi} \le |\xi| \le \mathrm{e}^{2\pi}\}$ by $O(|\varepsilon|)$. $\mathcal{W}_1 \circ \exp(2\pi\mathrm{i}(\mathcal{H}^{-1} - \zeta_{\mathsf{p}}))$ is a uniformisation of X normalised at p, so it equals $\exp(2\pi\mathrm{i}(z - z(\mathsf{p})))$. Taking logarithms on the annulus, and combining with the bounds on $\mathcal{H}$ across at most three seams, gives (2). □

Lemma 10.5 (The base orbit and the marked point)**.** *There are $r_2 \le r_0$, a small $\rho_1 > 0$ and N_1 such that the following holds for $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$ and every line ℓ admissible for ε. Let $\chi^*_n = \chi(u^*_n)$, $n \ge N_1$, be the branch continued from the principal value at $n = N_1$ (so $\chi^*_{n+1} = f_\chi(\chi^*_n)$), and $Z^*_n = A_1\chi^*_n$.*

1. *For $N_1 \le n \le N_1 + 10/|\varepsilon|$, $Z^*_n = Z^*_{N_1} + (n - N_1) + \mathcal{E}_n$ with $|\mathcal{E}_n| \le C\log n + C|\varepsilon|n$, and $Z^*_{N_1}$ is within relative distance $\frac15$ of a real number $\ge 4/\rho_1$.*
2. *On that range the normal coordinate $\varsigma(\chi^*_n)$ is strictly increasing. Let n_0 be the first index with $\varsigma(\chi^*_{n_0}) \ge 0$. Then $n_0 \le N_1 + 8/|\varepsilon|$ and $\chi^*_{n_0} \in \hat{H} \setminus f_\chi(\ell)$.*
3. *For $N_1 \le n \le n_0 + 1$, the $3|\varepsilon|$-neighbourhood of χ^*_n (equivalently, approximately, the 3-neighbourhood of Z^*_n) lies in $\mathcal{G}^{\mathrm{c}} \cup \{|\chi| < 1\}$ and in $\{|u(\chi)| \le 2\rho_1\}$.*

The marked point *of X^ℓ_ε is the point represented by $\chi^*_{n_0} \in \hat{H}$.*

Proof. At $\varepsilon = 0$ the orbit $f_0^n(u^*)$ increases to 0 along the negative axis, with $f_0^n(u^*) = -2/n + O(\log n/n^2)$. Choose N_1 with $|f_0^{N_1}(u^*)| \le \rho_1/2$, and then r_2 so small that $|u^*_{N_1} - f_0^{N_1}(u^*)| \le \frac{1}{10}|f_0^{N_1}(u^*)|$ for $|\varepsilon| < r_2$. Since $q(u) = 1 + (u_2 - u_1)/(u - u_2)$, $Z^*_{N_1} = -2/u^*_{N_1}\,(1 + O(|\varepsilon|/\rho_1) + O(\varepsilon))$, which gives the statement about $Z^*_{N_1}$. By Lemma 10.2(2), $Z^*_{n+1} - Z^*_n = A_1\Delta(u^*_n) = 1 + e_n$ with $|e_n| \le C(|u^*_n| + |\varepsilon|)$ while $u^*_n \in D_{\rho_0}$. While $|\chi| \le 1$, Lemma 10.2(1) gives $|u| \le |u_2| + |u_1 - u_2|/(0.63|\chi|) \le 3.2/|Z| + C|\varepsilon|$; on $\mathcal{G}^{\mathrm{c}}$, $|u| \le C|\varepsilon|$. By induction $\operatorname{Re} Z^*_n$ increases by at least $\frac12$ per step, and $|Z| \ge |Z^*_{N_1}| - 3 \ge 3/\rho_1$ on the 3-neighbourhoods, so $|u| \le 1.1\rho_1 + C|\varepsilon|$ there; this gives (1) and the last part of (3). Multiplying by $1/A_1 = -\varepsilon(1 + O(\varepsilon))$, χ^*_n lies within $C|\varepsilon|\log(1/|\varepsilon|) + C|\varepsilon|/\rho_1$ of the ray $\{-\varepsilon r : r \ge 0\}$ in the range of (1).

The origin has $\varsigma(0) = -\pi\cos\alpha + x_0\sin\alpha \le -1.48$ and $|\varsigma(0)| \le \sqrt{\pi^2 + 1/4}$, and along the ray ς increases at rate $-\operatorname{Im}(\varepsilon\bar\omega) \ge |\varepsilon|/2$; so the ray meets ℓ at $r \le 6.4/|\varepsilon|$. Before that crossing the ray stays at distance $\ge 2\pi\sin\frac{5\pi}{12} > 6$ from $2\pi\mathrm{i}k$, $k \le -1$, and in $\{\varsigma < 0\}$, at distance ≥ 1.48 from $2\pi\mathrm{i}k$, $k \ge 1$. With the smallness of the deviation this gives (3). The normal component of each step is at least $|\varepsilon|(\frac12 - C\rho_1 - C|\varepsilon|) > 0$, so $\varsigma(\chi^*_n)$ is strictly increasing, and n_0 exists with the stated bound. Then $-\frac65|\varepsilon| \le \varsigma(\chi^*_{n_0-1}) < 0$, so $\chi^*_{n_0-1}$ lies in the convex strip $\{|\varsigma| \le \frac14\}$, on which each $\mathrm{id} + \mathsf{r}\Delta$, $0 \le \mathsf{r} \le 1$, is injective. This isotopy carries ℓ to $f_\chi(\ell)$ and the side $\{\varsigma < 0\}$ of ℓ into the corresponding side of $f_\chi(\ell)$; hence $\chi^*_{n_0} \in \hat{H} \setminus f_\chi(\ell)$. □

The physical point u^*_n may lie in H for several n, because H spirals around u_1 and u_2 when $\alpha \ne 0$. These other visits are points of X different from the marked point, and correspond to the translated roots c_j discussed after Definition 8.1. Only the first crossing on the continued branch is used.

Lemma 10.6 (Shadowing the base orbit)**.** *Let $W_0 = \{z : \operatorname{dist}(z, [0,1]) < \frac14\}$. Under the hypotheses of Lemma 10.5, let z be the lift of Lemma 10.4 normalised by $z = n_0$ at the marked*

point. Then for $0 \le k \le n_0 - N_1$ the branch g_ε^k is holomorphic on $\tilde F(n_0 + W_0)$, and the Z-coordinates of $g_\varepsilon^k(\tilde F(n_0+w))$ lie within $\frac{1}{10}$ of $Z^*_{n_0-k}+w$ ($w \in W_0$). For $|\varepsilon|$ small the N_1 further applications of the same branch g_ε (principal logarithm) are defined and jointly holomorphic on $g_\varepsilon^{n_0-N_1}(\tilde F(n_0+W_0))$ and its images, which stay in $\{\mathrm{Re}(1+v)>0\}$ near the real segment $[u^*,0]$.

Proof. By Lemma 10.4(2) with p the marked point, the Z-coordinates of $\tilde F(n_0+w)$ lie within $C|\varepsilon|$ of $Z^*_{n_0}+w$. In the coordinate Z, f_ε is $Z \mapsto Z + A_1\Delta$, whose derivative is $1+\partial_\chi\Delta$, and g_ε is its inverse g_Z, with $|g_Z' - 1| \le 2|\partial_\chi\Delta|$ evaluated at $g_Z(Z)$. By Lemma 10.2(3) and $|\chi| = |Z|/|A_1|$, $|g_Z'-1| \le \delta(Z) := C|\varepsilon|^2 + C/|Z|^2$ on the 3-neighbourhood of the orbit segment, where Lemma 10.5(3) applies and $|Z| \ge |Z^*_{N_1}| - 3$. Since $g_Z^k(Z^*_{n_0}) = Z^*_{n_0-k}$, the mean-value inequality gives, as long as the images stay in that neighbourhood,

$$|g_Z^k(Z) - Z^*_{n_0-k} - (Z - Z^*_{n_0})| \le \Big(\prod_{j<k}(1+2\delta(Z^*_{n_0-j})) - 1\Big)|Z - Z^*_{n_0}| \le (\mathrm{e}^{C|\varepsilon|+C\rho_1}-1)\cdot 2,$$

because on the 3-neighbourhood of $Z^*_{n_0-j}$ one has $\delta \le 2\delta(Z^*_{n_0-j})$, using $\sum_j |\varepsilon|^2 \le C|\varepsilon|$ over $n_0 \le N_1 + 8/|\varepsilon|$ steps and $\sum_j |Z^*_{n_0-j}|^{-2} \le C/|Z^*_{N_1}| \le C\rho_1$ (the real parts grow by at least $\frac12$ per step). For small ρ_1 and $|\varepsilon|$ the right side is below $\frac{1}{20}$, so by induction the images stay in the neighbourhood and the claim holds. These sets lie in $\{|u| \le 2\rho_1\} \subset D_{\rho_0}$, where $g_\varepsilon \circ f_\varepsilon = \mathrm{id}$ (Lemma 10.1(3)).

At $k = n_0 - N_1$ the set lies within $\frac{1}{10}$, in Z, of $Z^*_{N_1} + W_0$. As $\varepsilon \to 0$, $Z = A_1\chi$ converges on this region to $-2/u$ locally uniformly, so the set lies, for small $|\varepsilon|$, in a neighbourhood of the compact set $\overline{K_0}$, $K_0 = \{u : -2/u \in Z_0 + W_0^{1/5}\}$, where $Z_0 = -2/f_0^{N_1}(u^*)$ and $W_0^{1/5}$ is the $\frac15$-neighbourhood of W_0. At $\varepsilon = 0$ the attracting Fatou coordinate is $\Phi_{\mathrm{att}}(u) = -2/u + \frac13\log(-u) + O(u)$; on $\overline{K_0}$ the term $\frac13\log(-u)$ varies by $O(\rho_1)$, so $\Phi_{\mathrm{att}}(\overline{K_0}) \subset c_0 + N_1 + W_0^{1/5+C\rho_1}$ for a real constant c_0. The Fatou parametrisation $P_0 = \Phi_{\mathrm{att}}^{-1}$ extends to an entire function with $P_0(z+1) = f_0(P_0(z))$, and $g_0(P_0(z)) = P_0(z-1)$ near the real axis, where $1 + P_0(z-1)$ is close to $(0,1]$. Hence $g_0^{N_1}$ is holomorphic on a neighbourhood of $\overline{K_0}$, and all intermediate images lie in a compact subset of $\{\mathrm{Re}(1+v)>0\}$ near $[u^*,0]$. This is an open condition, and $g_\varepsilon \to g_0$ uniformly on compact subsets of $\{\mathrm{Re}(1+v)>0\}$; by compactness it persists for small $|\varepsilon|$, jointly holomorphically in ε. □

Definition 10.7. For $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$ and an admissible line ℓ, the *normalised tilted solution* is

$$F_\varepsilon^\ell(z) = g_\varepsilon^{n_0}(\tilde F(z+n_0)), \qquad z \in W_0,$$

with $\tilde F$ lifted so that the marked point has $z = n_0$ and g_ε the principal branch of Lemma 10.1(3), which is holomorphic along the way by Lemma 10.6. It is holomorphic on W_0, $F_\varepsilon^\ell(0) = u^*$, and $F_\varepsilon^\ell(z+1) = f_\varepsilon(F_\varepsilon^\ell(z))$ where both sides are defined; in $w = \mathrm{e}(1+u)$ it satisfies $F(z+1) = b^{F(z)}$ and $F(0) = 1$.

Lemma 10.8 (Change of line)**.** *Let ℓ, ℓ' be admissible for the same $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$. Then $F_\varepsilon^\ell = F_\varepsilon^{\ell'}$.*

Proof. Since the admissible pairs form a rectangle, it suffices to treat $|\alpha - \alpha'| \le \frac{1}{100}$ and $|x_0 - x_0'| \le \frac{1}{100}$. Let $\chi = \mathrm{i}\pi + x_0 + (t + \mathrm{i}\varsigma)\omega \in \hat H$ and let ς' be its normal coordinate for ℓ'. Then $|\varsigma'| \le \frac{1}{100}(1+|t|) + O(|\varepsilon|)$. Each application of $f_\chi^{\pm1}$ changes ς' by at least $\frac25|\varepsilon|$ in the same direction (Lemma 10.3(1)) and moves the point by at most $2|\varepsilon|$. So after at most $(1+|t|)/(40|\varepsilon|)$ steps, forward or backward, the orbit of χ on the continued branch reaches the strip $\hat H'$ of ℓ', within distance $(1+|t|)/20$ of χ. The distance from χ to the holes is at least $\max(1.48,\, 0.6|t| - 0.5)$, which exceeds $(1+|t|)/20 + \frac12$; so this part of the orbit stays in a disc about χ contained in $\mathcal{G}^{\mathrm{c}}$, on which f_χ is injective. As in the proof of Lemma 10.5(2), exactly one iterate $f_\chi^{n(\chi)}(\chi)$ lies

in $\hat{H}' \setminus f_\chi(\ell')$. The map $\chi \mapsto f_\chi^{n(\chi)}(\chi)$ is locally a fixed iterate of f_χ, it jumps by one iterate exactly across ℓ', where the two values are identified in $X^{\ell'}$, and it satisfies $n(f_\chi(\chi)) = n(\chi) - 1$ on ℓ. It therefore induces a holomorphic map $X^\ell \to X^{\ell'}$, and the same construction from ℓ' gives its inverse. It preserves the two ends. The continued base orbit χ_n^* is common to both lines, and both marked points are its first crossings; the orbit segment between them lies in the region just described, so the map sends the marked point $\chi_{n_0}^*$ of ℓ to $f_\chi^{n_0'-n_0}(\chi_{n_0}^*) = \chi_{n_0'}^*$, the marked point of ℓ'. In the lifts normalised at the marked points ($z = n_0$ and $z = n_0'$), and since f_χ corresponds to $z \mapsto z+1$, one gets $\tilde{F}'(z+n_0') = f_\varepsilon^{n_0'-n_0}(\tilde{F}(z+n_0))$ near $z = 0$. Applying the inverse branches of Definition 10.7 gives $F_\varepsilon^{\ell'} = F_\varepsilon^\ell$ on W_0. □

We write F_ε for the common normalised tilted solution.

10.4. **Holomorphic dependence.**

Proposition 10.9 (The tilted family is holomorphic)**.** *There is $r_3 \le \min(r_2, 1/240)$ such that $(\varepsilon, z) \mapsto F_\varepsilon(z)$ is jointly holomorphic on $\mathcal{S}_{r_3}^{c\prime} \times W_0$.*

Proof. Fix $\varepsilon_0 \in \mathcal{S}_{r_3}^{c\prime}$ and the disc $U_0 = \{|\varepsilon - \varepsilon_0| < |\varepsilon_0|/40\}$; it lies in $\mathcal{S}_{r_2}^{c}$ and $|\arg\varepsilon - \arg\varepsilon_0| < \frac{1}{20}$ on it. Choose $\alpha \neq 0$ with $|\alpha - \alpha(\varepsilon_0)| \le \frac{1}{20}$, and then $x_0 \in [-\frac{1}{2}, \frac{1}{2}]$, such that $\chi_{n_0}^*(\varepsilon_0)$ is not on ℓ or $f_\chi(\ell)$; for $\alpha \neq 0$ a change of x_0 moves the line in its normal direction by $x_0 \sin\alpha$, so all but finitely many small values of x_0 will do. By the admissibility remark above $\ell = \ell_{\alpha,x_0}$ is admissible for all $\varepsilon \in U_0$. Shrink U_0 so that n_0 is constant and $\chi_{n_0}^*(\varepsilon)$ stays in the interior of $\hat{H}_\varepsilon$. By Lemma 10.8 this choice does not affect F_ε.

The local-triviality argument of Proposition 3.5 is not available here, because at neutral parameters the ends of X_ε have no a priori holomorphic puncture coordinate; we use the Beltrami equation instead. Let M_ε be the diffeomorphism of Lemma 10.3(3) for ℓ. The map $\mathcal{J}_\varepsilon = M_\varepsilon \circ M_{\varepsilon_0}^{-1} : \hat{H}_{\varepsilon_0} \to \hat{H}_\varepsilon$ respects the seam identifications and induces a homeomorphism $X_{\varepsilon_0} \to X_\varepsilon$. At each point its derivative sends $v_1 = \partial_t M_{\varepsilon_0} = \omega(1 + O(|\varepsilon_0|^2))$ to $w_1 = \partial_t M_\varepsilon = \omega(1 + O(|\varepsilon_0|^2))$ and $v_2 = \Delta_{\varepsilon_0}$ to $w_2 = \Delta_\varepsilon$, with $|w_2 - v_2| \le 1.1|\varepsilon - \varepsilon_0| \le 0.03|\varepsilon_0|$ by Lemma 10.2(5). Writing it as $\zeta \mapsto \mathsf{a}\zeta + \mathsf{b}\bar{\zeta}$,

$$\mathsf{b} = \frac{(w_1 - v_1)v_2 - (w_2 - v_2)v_1}{2\mathrm{i}\,\mathrm{Im}(v_2\bar{v}_1)}, \qquad |\,\mathrm{Im}(v_2\bar{v}_1)| \ge \tfrac{2}{5}|\varepsilon_0|(1 - C|\varepsilon_0|),$$

so $|\mathsf{b}| \le 0.05$, $|\mathsf{a} - 1| \le 0.05$, and the Beltrami coefficient $\upsilon_\varepsilon = \mathsf{b}/\mathsf{a}$ of $\mathcal{J}_\varepsilon$ satisfies $|\upsilon_\varepsilon| \le \frac{1}{10}$. It is holomorphic in ε at each point, since a, b are affine in w_1, w_2; being uniformly bounded, it is holomorphic as an L^∞-valued function. Transport υ_ε to $\hat{\mathbb{C}}$ by a fixed uniformisation Z_{ε_0} of X_{ε_0}, and let $\mathcal{W}_\varepsilon$ be the normalised solution fixing $0, \infty$ (the ends) and 1. By [2, Thm. 11], $\mathcal{W}_\varepsilon(\xi)$ is holomorphic in ε for each ξ, and $Z_\varepsilon = \mathcal{W}_\varepsilon \circ Z_{\varepsilon_0} \circ \mathcal{J}_\varepsilon^{-1}$ is a uniformisation of X_ε with the ends at $0, \infty$.

In the interior of $\hat{H}_{\varepsilon_0}$, $\mathcal{J}_\varepsilon$ is a real-analytic diffeomorphism, holomorphic in ε at fixed source points, and υ_ε is real-analytic in the source point and holomorphic in ε. By the interior Schauder estimates for the Beltrami equation [1, Ch. V], applied to difference quotients in ε, $(\varepsilon, \xi) \mapsto \mathcal{W}_\varepsilon(\xi)$ is C^1 there. The maps $\mathcal{W}_\varepsilon$ and $\mathcal{J}_\varepsilon \circ Z_{\varepsilon_0}^{-1}$ have the same Beltrami coefficient υ (transported), i.e. Z_ε is conformal. Put $\xi_\varepsilon(\chi) = Z_{\varepsilon_0}(\mathcal{J}_\varepsilon^{-1}(\chi))$. Differentiating $\mathcal{J}_\varepsilon(Z_{\varepsilon_0}^{-1}(\xi_\varepsilon(\chi))) = \chi$ in $\bar{\varepsilon}$ at fixed χ, and using that $\mathcal{J}_\varepsilon$ is holomorphic in ε at fixed source points, gives $\partial_{\bar{\varepsilon}}\xi_\varepsilon + \upsilon\,\partial_{\bar{\varepsilon}}\overline{\xi_\varepsilon} = 0$. With $\partial_{\bar{\xi}}\mathcal{W}_\varepsilon = \upsilon\,\partial_\xi\mathcal{W}_\varepsilon$ and $\partial_{\bar{\varepsilon}}\mathcal{W}_\varepsilon = 0$ this gives $\partial_{\bar{\varepsilon}}[\mathcal{W}_\varepsilon(\xi_\varepsilon(\chi))] = \partial_\xi\mathcal{W}_\varepsilon\,(\partial_{\bar{\varepsilon}}\xi_\varepsilon + \upsilon\,\partial_{\bar{\varepsilon}}\overline{\xi_\varepsilon}) = 0$. So $(\varepsilon, \chi) \mapsto Z_\varepsilon(\chi)$ is jointly holomorphic in the interior of the strips.

With $\chi_{n_0}^*(\varepsilon)$ in the interior, $\zeta_0(\varepsilon) = (2\pi\mathrm{i})^{-1}\log Z_\varepsilon(\chi_{n_0}^*(\varepsilon))$ is holomorphic, and $\tilde{F}_\varepsilon(z + n_0) = u_\varepsilon\big(Z_\varepsilon^{-1}(\mathrm{e}^{2\pi\mathrm{i}(z+\zeta_0(\varepsilon))})\big)$ is jointly holomorphic wherever the point lies in the interior of $\hat{H}_\varepsilon$, by the

holomorphic inverse function theorem. Points of $n_0 + W_0$ whose images lie on the seam, or in the extra step of Lemma 10.4(1), are handled in the same way with finitely many further admissible angles α (by compactness of $\overline{W_0}$), for which they are interior; Lemma 10.8 shows that the two descriptions define the same function. Composing with the inverse branches of Lemma 10.6, which are jointly holomorphic, gives joint holomorphy on $U_0 \times W_0$. On overlapping discs the local families coincide by Lemma 10.8. □

10.5. **Abel coordinates on the tilted region.**

Lemma 10.10 (Koenigs times along orbits)**.** *Let $b < \eta$ be real, so $\varepsilon > 0$, and put $\Psi = A_1 \log(u - u_1) + A_2 \log(u - u_2)$, $A_2 = 1/\log \lambda_2$, and $F_b = \Psi \circ f_b - \Psi - 1$ as in Lemma 4.2. Let P be a path in the χ-plane such that for each $\chi \in P$ the forward orbit $f_\chi^k(\chi)$ stays in $\mathcal{G}^{\mathrm{c}}$ and $\operatorname{Re} f_\chi^k(\chi)$ decreases by at least $\varepsilon/2$ per step. Then, along P, the branch of R^{-1} continued along P satisfies*

$$R^{-1} = \Psi + \sum_{k \ge 0} F_b \circ f^k + c_+, \qquad \sum_{k \ge 0} |F_b(f^k(u))| \le C\varepsilon,$$

with a constant c_+. The same holds for S^{-1} with backward orbits (f_χ^{-k}, Re increasing by at least $\varepsilon/2$ per step) and $\Psi - \sum_{k \ge 1} F_b \circ f^{-k} + c_-$.

Proof. By Lemma 4.2 and Lemma 10.2(1), $|F_b(f^k(u))| \le MC\varepsilon^2 \min(|e^{\chi_k}|, |e^{-\chi_k}|) \le C\varepsilon^2 e^{-|\operatorname{Re}\chi_k|}$; since $\operatorname{Re}\chi_k$ changes by at least $\varepsilon/2$ per step, the sum is at most $C\varepsilon^2 \cdot \sum_k e^{-|\operatorname{Re}\chi_0 - k\varepsilon/2|} \le C\varepsilon$. The orbit converges to u_1 because $\operatorname{Re}\chi_k \to -\infty$. As in the proof of Theorem 4.4(1), $\lim_n [\Psi(f^n u) - n] = \Psi(u) + \sum_{k \ge 0} F_b(f^k u)$ exists and equals $A_1 \log \sigma_1(u)$ plus a constant, because $\sigma_1(f^n u) = \lambda_1^n \sigma_1(u)$ and $A_1 \log \lambda_1 = 1$; the constant depends only on the branches, which are continued along P. The backward case is identical with the reflected inverse family of Theorem 4.4(2). □

10.6. **Kneser's initial region.** The exterior end of the corridor uses the following parameter-dependent form of Kneser's construction.

Proposition 10.11 (Local base analyticity on the real exterior ray)**.** *For every real $b_0 > \eta$, Kneser's real-normalised solution admits a jointly holomorphic extension in (b, z) to $\Delta \times W$, where Δ is a complex disc about b_0 and W is a complex neighbourhood of $[0, 1]$. These local extensions agree on overlap, so the classical exterior branch has a jointly holomorphic germ along the entire real ray (η, ∞). In particular, every compact subinterval of that ray has a common complex base neighbourhood and a common complex height neighbourhood of $[0, 1]$ on which this family is jointly holomorphic.*

Proof. Write $\mathcal{L} = \log b$ and choose the principal fixed-point pair $L_+(b)$ and $L_-(b)$, conjugate when $b > \eta$ and characterised there by $0 < \mathcal{L} \operatorname{Im} L_+ < \pi$. They are simple fixed points and hence holomorphic in b near b_0. Set $u_b = (L_+ + L_-)/2$, $v_b = (L_+ - L_-)/(2\mathrm{i})$, and $\gamma_b(t) = u_b + \mathrm{i}v_b t$ for $-1 \le t \le 1$. At a real base, $u_b, v_b \in \mathbb{R}$, $v_b > 0$, and

$$E_b(\gamma_b(t)) = r_b \mathrm{e}^{\mathrm{i}\theta_b t}, \qquad r_b = \mathrm{e}^{\mathcal{L}u_b} = |L_+|, \quad \theta_b = \mathcal{L}v_b \in (0, \pi).$$

Since $v_b = r_b \sin\theta_b$, the multipliers satisfy $|E_b'(L_\pm)| = \mathcal{L}r_b = \theta_b / \sin\theta_b > 1$. The straight chord γ_b and this circular arc meet only at $L_\pm$ and bound the initial region $H_b = \{\operatorname{Re} w \ge u_b,\ |w| \le r_b\} \setminus \{L_+, L_-\}$. This is Kneser's initial region as described in [31, §4, Lem. 4]. The arcs depend holomorphically on b through their real-analytic parameter t. They remain disjoint away from their endpoints after shrinking a disc Δ about b_0; at the endpoints their tangents remain distinct because the fixed-point multipliers are nonreal. The bounded Jordan interiors vary continuously, so their union in $\Delta \times \mathbb{C}$ is open; the ambient coordinates (b, w) make its projection a holomorphic family. Identify thin analytic collars of the two arcs by $(b, w) \mapsto (b, E_b(w))$. The map is injective on the collar after shrinking Δ, since it is injective on the compact chord at b_0 and its derivative never vanishes. This gives a Hausdorff holomorphic family of cylinders with one end at each $L_\pm(b)$.

The ends can be filled holomorphically. Indeed, parameter-dependent Koenigs coordinates $\sigma_{\pm,b}$, normalised by $\sigma_{\pm,b}(L_\pm) = 0$ and $\sigma'_{\pm,b}(L_\pm) = 1$, linearise E_b near the two repelling fixed points. If $\ell_{\pm,b} = \log E_b'(L_\pm(b))$ is chosen continuously, then $\exp(2\pi\mathrm{i} \log \sigma_{\pm,b}/\ell_{\pm,b})$ (or its reciprocal) is a puncture coordinate on the corresponding glued end. Filling both punctures yields a proper holomorphic submersion of compact genus-zero surfaces over Δ, just as in the collar construction of Proposition 3.5.

We also have a holomorphic marked section represented by $w = 1$. For real $b > \eta$, $E_b(x) > x$ for every real x: the difference is positive at $x \to \pm\infty$ and has no real zero. The adjacent real intervals between the chord and its image are (u_b, r_b), with $r_b = E_b(u_b) > u_b$. The forward endpoints $E_b^n(u_b)$ increase to $+\infty$, since a finite limit would be a real fixed point. For any $x > 0$ below u_b, forward iteration eventually enters (u_b, r_b) or one of its boundary seams; for $x > r_b$, repeated real logarithms eventually do the same. Indeed, a monotone orbit staying on either side would have a finite real fixed-point limit. Thus these dynamical translates and their seam collars cover the positive real axis. In particular, 1 is in the interior of one such interval or on a seam. A finite chain of holomorphic dynamical charts represents it near b_0; at a seam the analytic collar gives the same section. The entire compact real segment from 1 to $b = E_b(1)$ is covered by finitely many of these charts.

The two filled ends and this marked point are three disjoint holomorphic sections. Fischer–Grauert local triviality [10] and the fibrewise three-point Möbius normalisation give a jointly holomorphic coordinate q_b on the compactified cylinder, taking the upper end to 0, the lower end to ∞, and $w = 1$ to 1. The upper puncture coordinate $\exp(2\pi\mathrm{i} \log \sigma_{+,b}/\ell_{+,b})$ shows that the logarithmic lift $z = (2\pi\mathrm{i})^{-1} \log q_b$, with $z(1) = 0$, gains $+1$ under the edge identification by E_b. On the real slice, reflection exchanges the two ends and fixes the marked point, so $q_b(\bar{w}) = 1/\overline{q_b(w)}$ in the real dynamical charts. Thus z is real there; its local injectivity and

$z(E_b(w)) = z(w) + 1$ make it strictly increasing along the positive axis. Inverting on the finite chart chain covering the real segment from 1 to b gives a jointly holomorphic superfunction on $\Delta \times W$, after shrinking both.

For real b, the lift restricted to H_b is injective and its integer translates cover the universal-cover plane. Choosing any $d \in H_b$ and subtracting its value gives the initial-region criterion of [31, Thm. 1]; the injectivity and tiling assertions for the Kneser generalisation are proved in [31, Thm. 5]. We normalise at d here, since 1 need not itself belong to H_b. Their Abel coordinates differ only by a constant on H_b, and continuation along the real dynamical charts to $w = 1$ makes that constant zero. Thus the inverse just constructed is the classical real-normalised Kneser branch. The marking fixes the local family uniquely, and the identity theorem patches the parameter discs on overlaps. A finite subcover of any compact real base interval, followed by intersection of its finitely many height neighbourhoods, gives the final uniformity assertion. □

10.7. **The two ends of the corridor.**

Proposition 10.12 (Exterior end)**.** *For $0 < \theta < \theta_1$ (real $b \in (\eta, \eta + r_1)$), $F_{\varepsilon(\theta)}$ is Kneser's real-normalised solution on W_0.*

Proof. The line $\ell = \mathrm{i}\pi + \mathbb{R}$ ($\alpha = 0$, $x_0 = 0$) is admissible for $\varepsilon(\theta)$, because $-\operatorname{Im}\varepsilon(\theta) = \theta$. Since $\chi \in \mathrm{i}\pi + \mathbb{R}$ means $q < 0$ and $u_2 = \overline{u_1}$, $\gamma = u(\ell)$ is the open segment between the conjugate fixed points, i.e. in $w = \mathrm{e}(1+u)$ the chord γ_b of Proposition 10.11, and $f(\gamma)$ is its image $E_b(\gamma_b)$. Hence H is the initial region H_b and X is the cylinder of that proposition, with the upper end at $L_1 = L_+$. There the uniformisation takes the upper end to 0 and is normalised so that $z(1) = 0$, with z real and increasing along the positive axis. For conjugate u_1, u_2 the real axis is $\{|q| = 1\}$, i.e. $\operatorname{Re}\chi = 0$, and the continued branch of the real base orbit moves up the imaginary χ-axis from 0; it crosses the chord between $u^*_{n_0-1}$ and $u^*_{n_0}$, and $u^*_{n_0} = E_b^{n_0}(1)$ has Kneser coordinate n_0. So our lift is Kneser's, and our inverse branches are the real logarithms along the orbit, as in Kneser's normalisation. By Lemma 10.8 the admissible line used in the family may be replaced by this one. □

Proposition 10.13 (Interior end)**.** *There is $r_4 > 0$ such that $F_\varepsilon = K^W_{b(\varepsilon)}$ on W_0 for real $\varepsilon \in (0, r_4)$. Here r_4 is chosen so that $b(\varepsilon) \in (b_1, \eta)$ with the closeness conditions of Theorem 6.3, and so that $0.39h + C_{\mathrm{im}} < d - D$, where C_{im} is the constant in (17) below and d, D are as in Lemma 6.2 and Theorem 6.3.*

Proof. Let $\varepsilon \in (0, r_4)$ be real, so $b < \eta$ is real, $\lambda_1 \in (0,1)$ and $u_1 < 0 < u_2$. Take $\alpha = \frac{\pi}{4}$, $x_0 = 0$, so that $\chi = \mathrm{i}\pi + (t + \mathrm{i}\varsigma)\mathrm{e}^{\mathrm{i}\pi/4}$ has $\operatorname{Re}\chi = (t-\varsigma)/\sqrt{2}$ and $\operatorname{Im}\chi = \pi + (t+\varsigma)/\sqrt{2}$. Put $h = 2\pi/|\log\lambda_1|$. The real line of the u-plane appears as follows: $u < u_1$ is $\{\operatorname{Im}\chi \in 2\pi\mathbb{Z},\ \operatorname{Re}\chi < 0\}$, the segment (u_1, u_2) is $\{\operatorname{Im}\chi \in \pi + 2\pi\mathbb{Z}\}$, and $u > u_2$ is $\{\operatorname{Im}\chi \in 2\pi\mathbb{Z},\ \operatorname{Re}\chi > 0\}$; the lower half-plane is $\{0 < \operatorname{Im}\chi < \pi\}$ modulo 2π, because $q \mapsto u$ has negative determinant $u_1 - u_2$. Since f_ε is real and Δ is real at u_1, u_2, Lemma 10.2 gives $\operatorname{Im}\Delta = O(\varepsilon^2)$ on $\mathcal{G}^{\mathrm{c}}$ and $\operatorname{Im}\Delta = O(\varepsilon^2|\mathrm{e}^{\mp\chi}|)$ near the two ends.

Marked point. The base orbit is real, $u^*_n < u_1$, so χ^*_n is real and moves to the left. On the real axis $\varsigma = -(\operatorname{Re}\chi + \pi)/\sqrt{2}$, so the marked point $p_c := \chi^*_{n_0}$ lies in $[-\pi - 2\varepsilon, -\pi]$, with $t \approx -\pi\sqrt{2}$.

The two charts. Let $U \subset X$ and $V \subset X$ be the images of $M(\{t < \pi\sqrt{2} - 1\})$ and $M(\{t > -\pi\sqrt{2} + 1\})$. They are open, and saturated for the seam identification, since $M(t,0) \sim M(t,1)$ preserves t. For $\chi \in M(\{t < \pi\sqrt{2} - 1\})$ the forward orbit $f^k_\chi(\chi)$ moves left by at least $\varepsilon/2$ per step, and its imaginary part drifts by $O(\varepsilon)$ in total (the sum of $O(\varepsilon^2)$ over $O(1/\varepsilon)$ steps in $|\operatorname{Re}\chi| \le 4$ plus the geometric tail at the end). Such points with $\operatorname{Re}\chi \ge -\frac{1}{2}$ have $\operatorname{Im}\chi \in (\pi - 1, 2\pi - 0.7)$, so these orbits stay in $\mathcal{G}^{\mathrm{c}}$. Symmetrically, backward orbits of points of $M(\{t > -\pi\sqrt{2} + 1\})$ move right, stay in $\mathcal{G}^{\mathrm{c}}$ and converge to u_2. So Lemma 10.10 applies on both sets, and the

attracting and repelling Koenigs coordinates σ_1 and σ_{rep} of Section 3 are defined and nonzero along these orbits. Let r_+ be the branch of $R^{-1} = \log(\sigma_1/\sigma_1(u^*))/\log\lambda_1$ on $M(\{t < \pi\sqrt{2}-1\})$ continued from p_c, where $r_+(p_c) = n_0$ is real ($\sigma_1(u^*_{n_0})/\sigma_1(u^*) = \lambda_1^{n_0} > 0$). By Lemma 10.10, $\partial_\chi r_+ = A_1 + O(1)$ on convex subsets at distance $\geq \frac{1}{2}+\frac{1}{10}$ from the holes. For $\chi \in \ell$, $\sigma_1 \circ f = \lambda_1\sigma_1$ gives $r_+(f_\chi(\chi)) - r_+(\chi) \in 1 + \mathrm{i}h\mathbb{Z}$ for the continuation along $\mathsf{r} \mapsto M(t,\mathsf{r})$, a segment of length $O(\varepsilon)$ on which r_+ changes by $O(1)$, far less than h. Hence $r_+ \circ f_\chi = r_+ + 1$ on ℓ, and r_+ descends to a holomorphic map $U \to \mathbb{C}/\mathbb{Z}$.

Let (t_c, ς_c) be the coordinates of p_c, so $0 \leq \varsigma_c < g(t_c)$, and follow the curve $\varsigma = \varsigma_c\, g(t)/g(t_c)$, which lies in $\hat{H}$, in the direction of increasing t. Since $|g'| \leq C\varepsilon^2$ and $g \geq \frac{2}{5}\varepsilon$, $\operatorname{Im}\chi = \pi + (t+\varsigma)/\sqrt{2}$ increases strictly along it, from 0 at p_c. Let $u_m \in (u_1, u_2)$ be the point where it equals π. Let r_- be the branch of S^{-1} on $M(\{t > -\pi\sqrt{2}+1\})$ that is real at u_m; it descends to V in the same way.

Imaginary parts. In Lemma 10.10, $\Psi = A_1\chi + (A_1+A_2)\log(u-u_2)$, with $A_1 = -h/2\pi$, $A_2 = \frac{h}{2\pi}(1+O(\varepsilon))$ and $A_1 + A_2 \to \frac{1}{3}$ (Lemma 4.1), and $\arg(u-u_2) = \arg(u_1-u_2) - \arg(1-\mathrm{e}^\chi)$. On U the continued $\arg(1-\mathrm{e}^\chi)$ is bounded, because $\operatorname{Im}\chi < 2\pi - 0.7$ wherever $\operatorname{Re}\chi \geq -\frac{1}{2}$ and $|\mathrm{e}^\chi| \leq \mathrm{e}^{-1/2}$ elsewhere; on $U \cap V$ the range of $\operatorname{Im}\chi$ is bounded. Hence, with the branches fixed above,

$$\left|\operatorname{Im} r_+ + \tfrac{h}{2\pi}\operatorname{Im}\chi\right| \leq C_{\mathrm{im}} \text{ on } U, \qquad \left|\operatorname{Im} r_- + \tfrac{h}{2\pi}(\operatorname{Im}\chi - \pi)\right| \leq C_{\mathrm{im}} \text{ on } U \cap V. \tag{17}$$

On $V \setminus U$, $\operatorname{Im}\chi > 2\pi - 1/\sqrt{2}$. There $\arg(1-\mathrm{e}^\chi) = \operatorname{Im}\chi - \pi + O(1)$ where $\operatorname{Re}\chi \geq 0$, and $A_1 - (A_1+A_2) = -A_2$, so $\operatorname{Im} r_- \leq -\min(|A_1|, A_2)(\operatorname{Im}\chi - \pi) + C$. Along the curve from p_c to u_m, $\operatorname{Im}\chi$ increases from 0 to π, so the path runs through the lower half-plane: $\arg(u-u_1)$ increases by π and $\arg(u-u_2)$ returns to its initial value. Hence $\operatorname{Im} r_+(u_m) = A_1\pi + O(\varepsilon) = -h/2 + O(\varepsilon)$. By Lemma 3.1, $\sigma_1(u_m)/\sigma_1(u^*) < 0$, so $\operatorname{Im} r_+(u_m) \in -h/2 + h\mathbb{Z}$; therefore $\operatorname{Im} r_+(u_m) = -h/2$ exactly. This is the seam branch of Proposition 3.4 and Definition 8.1: by item (c) in Section 4, the path from the base point to the segment passing above u_1 gives $+h/2$, and the one passing below gives $-h/2$. (The reflected branch $\overline{K_b^W(\bar{z})}$, denoted $K_b^{W,\#}$ in Corollary 10.51 below, would require the path above u_1, i.e. a line of negative slope at real ε, which is not admissible.)

Transition. On $U \cap V$, $R(r_+) = u = S(r_-)$, so $r_+ - T(r_-) \in \mathrm{i}h\mathbb{Z}$ is constant, with T the map of Section 3 (branch $\operatorname{Im} t_0 = h/2$). At u_m, r_- is real and $\operatorname{Im} T(r_-) = h/2$ (Lemma 3.1), while $\operatorname{Im} r_+ = -h/2$; hence $r_+ = T \circ r_- - \mathrm{i}h$ on $U \cap V$. By (17) and $|\operatorname{Im}\chi - \pi| \leq \pi - 1/\sqrt{2}$ on $U \cap V$, $r_-(U \cap V)$ lies in $\{|\operatorname{Im} w'| < 0.39h + C_{\mathrm{im}}\}$, which is inside the strip $\Sigma_q = \{|\operatorname{Im} w'| < d - D\}$ of the proof of Theorem 8.2 by the choice of r_4. There T is univalent, $T(\mathbb{R}) = \{\operatorname{Im} w = h/2\}$, and T maps $\{\operatorname{Im} w' \lesseqgtr 0\} \cap \Sigma_q$ below and above that line.

The map to the sewn surface. Let X^W be the surface of Proposition 3.4: $C_+ = \{\operatorname{Im} w \geq -h/2\}/\mathbb{Z}$ and $C_- = \{\operatorname{Im} w' \leq 0\}/\mathbb{Z}$ glued by $w = T(w') - \mathrm{i}h$. Define $\Theta : X \to X^W$ by $\Theta = r_+$ where $\operatorname{Im} r_+ \geq -h/2$ and $\Theta = r_-$ where $\operatorname{Im} r_- \leq 0$. By (17) and the estimate on $V \setminus U$, points of $U \setminus V$ have $\operatorname{Im}\chi < 1/\sqrt{2}$, hence $\operatorname{Im} r_+ > -0.12h - C > -h/2$, and points of $V \setminus U$ have $\operatorname{Im} r_- < -0.38h + C < 0$. On $U \cap V$, $\operatorname{Im} r_+ \geq -h/2$ if and only if $\operatorname{Im} r_- \geq 0$, by the previous paragraph. So every point of X is covered, and on the curve $\operatorname{Im} r_- = 0$ both definitions give the same point of X^W. Thus Θ is continuous, holomorphic off an analytic curve, hence holomorphic. As a point of X tends to the u_1-end, $t \to -\infty$ and $\operatorname{Im} r_+ \to +\infty$; as it tends to the u_2-end, $\operatorname{Im} r_- \to -\infty$. So Θ extends to a holomorphic map of the compactified spheres with $\Theta^{-1}(\mathrm{end}) = \mathrm{end}$.

It has local degree one at the u_1-end; in fact r_+ is injective on U. Let $x, x' \in U$ be represented by $\chi, \chi' \in \hat{H}$ with $r_+(\chi') - r_+(\chi) = m \in \mathbb{Z}$. By Lemma 10.10, $r_+ = A_1\chi + \phi_+(\chi)$ where ϕ_+ is bounded with bounded derivative on the $\frac{1}{10}$-neighbourhood of $\hat{H} \cap U$, which lies in $\mathcal{G}^{\mathrm{c}}$. Equal imaginary parts give $|\operatorname{Im}\chi - \operatorname{Im}\chi'| \leq C\varepsilon$; since $\hat{H}$ is a strip of normal width $\leq \frac{6}{5}\varepsilon$ along a

line of slope 1, this forces $|\chi - \chi'| \le C_1\varepsilon$, and then $|m| \le |A_1||\chi - \chi'|(1 + C\varepsilon) + C \le C_2$. On the convex disc N of radius $(C_1 + 3C_2)\varepsilon$ about χ, $\partial_\chi r_+ = A_1(1 + O(\varepsilon))$, so r_+ is injective on N; the points χ' and $f_\chi^m(\chi)$ lie in N, and $r_+ \circ f_\chi^m = r_+ + m$ gives $r_+(f_\chi^m(\chi)) = r_+(\chi')$, hence $\chi' = f_\chi^m(\chi)$. Since $\hat{H}$ is a fundamental strip for f_χ, $x = x'$ in X. Hence Θ has degree one and is biholomorphic.

Normalisation. Θ maps the marked point p_c to the point of C_+ with attracting time $r_+(p_c) = n_0$. In Proposition 3.4 the lift is fixed by $K^W(0) = R(0) = 1$ on the branch $w(0) = 0$, i.e. attracting time 0 at $z = 0$, hence time n_0 at $z = n_0$. So the lifts agree, and $\tilde{F}(z + n_0) = R(r_+) = K_b^W(z + n_0)$ near $z = 0$. Near $z = 0$ the points $K_b^W(z + k)$, $0 \le k \le n_0$, stay near u_k^*, where the inverse branches of Definition 10.7 invert f_ε (Lemma 10.1(3)); hence $F_\varepsilon = K_b^W$ near $z = 0$, and by the identity theorem on W_0. □

10.8. **The continuation theorem.**

Theorem 10.14 (Continuation around the cusp)**.** *There are $r_5 > 0$ and an open set $\Omega \supset D^+ \cup (\eta - r_5, \eta) \cup (\eta, \eta + r_5)$, where $D^+ = \{b : |b - \eta| < r_5,\ \operatorname{Im} b > 0\}$, with the following properties.*

1. *Kneser's real-normalised solution $\mathcal{K}_b$, $b \in (\eta, \eta + r_5)$, continues analytically in the base to a family on Ω that is jointly holomorphic on $\Omega \times W_0$, $W_0 = \{z : \operatorname{dist}(z, [0, 1]) < \frac{1}{4}\}$. In particular it continues across the arc of the Shell–Thron boundary contained in D^+, whatever the arithmetic of the neutral multiplier there.*
2. *On $(\eta - r_5, \eta)$ the continued family equals K_b^W on W_0.*
3. *Let N_W be the neighbourhood of $(1, \eta)$ of Proposition 3.5. Continuing the family from a neighbourhood of $(\eta - r_5, \eta)$ inside N_W, on a b-dependent neighbourhood of $z = 0$ contained in the regular domain of K_b^W, one obtains K^W. Hence for every $b \in (1, \eta)$, if $\mathcal{K}_{b+\mathrm{i}y}$ is continued first through D^+ and then inside $N_W \cap \{\operatorname{Im} b > 0\}$, the limit $\lim_{y \downarrow 0} \mathcal{K}_{b+\mathrm{i}y}$ exists near $z = 0$ and equals K_b^W; by the functional equation and continuation in z it equals K_b^W on its whole regular domain.*

Proof. (1) By Lemma 10.1, $\varepsilon \mapsto b(\varepsilon)$ maps $\mathcal{S}_{r_3}^{\mathrm{c}\prime}$ biholomorphically onto an open set Ω that contains D^+ and the two real segments, for r_5 small. By Proposition 10.9, $(b, z) \mapsto F_{\varepsilon(b)}(z)$ is jointly holomorphic on $\Omega \times W_0$. By Proposition 10.12 it equals $\mathcal{K}_b$ on $(\eta, \eta + r_5)$. Since Ω is connected, it is the analytic continuation of $\mathcal{K}$ along every path in Ω. The Shell–Thron boundary near η consists of $b = \exp(\lambda \mathrm{e}^{-\lambda})$ with $|\lambda| = 1$. Its arc in the upper half-plane is $\varepsilon = 1 - \mathrm{e}^{\mathrm{i}\phi} = -\mathrm{i}\phi + O(\phi^2)$ with small $\phi > 0$, since then $b - \eta$ is a positive multiple of $-\varepsilon^2 = \phi^2 + \mathrm{i}\phi^3 + O(\phi^4)$; it lies in $\mathcal{S}_{r_3}^{\mathrm{c}\prime}$.

(2) is Proposition 10.13.

(3) By Proposition 3.5, K^W is jointly holomorphic on the set of (b, z) with $b \in N_W$ and z in a neighbourhood of 0 that depends continuously on b; the upper chart contains $w = 0$, where R is regular. Shrinking the discs Δ_{b_0} of that proposition, we may assume $\Delta_{b_0} \subset \{|b - b_0| < (\eta - b_0)/2\}$; then $N_W \cap \{\operatorname{Im} b > 0\}$ is a union of upper half-discs centred on $(1, \eta)$, which is connected since consecutive ones overlap. Each $\Delta_{b_0} \cap D^+$ is convex, and if it is nonempty its closure meets $(\eta - r_5, \eta)$. On a neighbourhood of $(\eta - r_5, \eta)$ times a small disc about $z = 0$, the two families agree by (2) and the identity theorem; hence they agree on every $\Delta_{b_0} \cap D^+$, and the continuation from D^+ into $N_W \cap \{\operatorname{Im} b > 0\}$ is unambiguous and equals K^W. At every $b \in (1, \eta)$ the boundary value is therefore K_b^W near $z = 0$. Both sides satisfy $F(z + 1) = b^{F(z)}$, so they agree on the whole regular domain. □

Corollary 10.15 (Separation limits for the continued Kneser family)**.** *For the continuation of Kneser's solution in Theorem 10.14, $K_b = \lim_{y \downarrow 0} \mathcal{K}_{b+\mathrm{i}y}$ exists for $b \in (1, \eta)$ and equals K_b^W. Its separation coefficients satisfy $c_n/\Lambda^n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for every $n \ge 1$ as $b \to \eta^-$. In particular*

$|c_1|/\Lambda \to |B_1| = 0.0890584\ldots$, *with* $B_1 \neq 0$. *This confirms the limit suggested by the numerics of Section 11.1: the first separation mode of the continued Kneser family, scaled by* Λ*, tends to the first coefficient of the inverse horn map of* $\mathrm{e}^u - 1$.

Proof. Theorem 10.14 gives $K_b = K_b^W$; apply Corollary 1.1 and Proposition 2.4. □

Remark 10.16 (Lower half-plane, and what is not claimed)**.** (1) Complex conjugation of the whole construction uses the sector $\{\bar\varepsilon : \varepsilon \in \mathcal{S}_r^{\mathrm{c}}\}$, the lines $\overline{\ell_{\alpha,x_0}}$ through $-\mathrm{i}\pi \equiv \mathrm{i}\pi \pmod{2\pi\mathrm{i}}$ with slope $-\alpha$, and the conjugate admissibility condition $+\operatorname{Im}(\varepsilon\bar\omega) \geq \frac{1}{2}|\varepsilon|$. It shows that continuation of Kneser's solution through the lower half-disc yields the reflected branch $K_b^{W,\#}$ of Corollary 10.51 on $(\eta - r_5, \eta)$. By Corollary 10.51 below, the two continuations differ by at least $2c\Lambda$ at the height x_* of Corollary 10.49; this is the monodromy of Kneser's family around η.

(2) The proof does not use any arithmetic property of the multiplier on the Shell–Thron boundary. The ends of the tilted region are punctures for Siegel, Cremer and parabolic multipliers alike, because the only inputs are the step estimate $A_1\Delta = 1 + O(\varepsilon)$ and the transversality $\operatorname{Im}(\operatorname{Log}\lambda_1\bar\omega) > 0$. In particular no Brjuno condition, Siegel-disc estimate or boundary-value (Painlevé-type) argument is needed near η.

(3) The theorem asserts holomorphy in (b, z) for heights near $[0, 1]$, and by the functional equation on the regular domain reached from there. No claim is made about the behaviour of the continued solution as $\operatorname{Im} z \to \pm\infty$ at parameters with a neutral multiplier; the construction only uses that each end of the tilted region is a puncture of the quotient.

(4) The theorem concerns continuation along the stated paths. The width of N_W is not quantified; the restriction to paths near the real axis is removed in Theorem 10.22, and the relation to Paulsen's family is discussed in Remark 10.25.

10.9. **The whole upper Shell–Thron region.** Theorem 10.14 continues Kneser's solution into a corridor about the cusp. Here we show that the continued family extends, single-valued, to the whole upper half of the Shell–Thron region. The extension is a quasiconformal transport of one tilted initial region by a holomorphic motion that conjugates the dynamics; the new point is a comparison lemma identifying the transported quotient with the tilted one near the base parameter. In this subsection λ denotes the multiplier of the attracting fixed point, so $\lambda = \lambda_1 = 1 - \varepsilon$ in the notation above, and

$$\mathbb{H}_\lambda = \{\lambda : |\lambda| < 1,\ \operatorname{Im}\lambda > 0\}, \qquad b(\lambda) = \exp(\lambda \mathrm{e}^{-\lambda}), \qquad E_\lambda = E_{b(\lambda)}.$$

Lemma 10.17 (Global parameter)**.** *b is injective on the unit disc, E_λ has the fixed point $L_1(\lambda) = \mathrm{e}^\lambda$ with multiplier λ, and $b(\{|\lambda| < 1\})$ is the set of bases for which E_b has an attracting fixed point. The set $\mathcal{U}^+ = b(\mathbb{H}_\lambda)$ lies in $\{\operatorname{Im} b > 0\}$ and is simply connected.*

Proof. For $g(\lambda) = \lambda\mathrm{e}^{-\lambda}$ one has $\operatorname{Re}(\lambda g'/g) = \operatorname{Re}(1-\lambda) > 0$ on the unit disc, so g is starlike and injective. Since $|g| < \mathrm{e} < \pi$, equality $\mathrm{e}^{g(\lambda_1)} = \mathrm{e}^{g(\lambda_2)}$ forces $g(\lambda_1) = g(\lambda_2)$; so b is injective. If L is a fixed point of E_b with multiplier $\lambda = L\log b$ of modulus < 1, then $L = \mathrm{e}^\lambda$ and $\log b = g(\lambda)$ (Lemma 2.1). By symmetry and injectivity g maps the upper half-disc into $\{\operatorname{Im} > 0\}$ (check $\lambda = \mathrm{i}y$, $y > 0$ small); since $|\operatorname{Im} g| < \pi$, $\operatorname{Im} b > 0$ there. An injective image of a simply connected domain is simply connected. □

Fix a base parameter $\lambda_0 = 1 - \varepsilon_0$ with $\varepsilon_0 \in \mathcal{S}_{r_3}^{\mathrm{c}\prime}$, $\operatorname{Im}\varepsilon_0 < 0$ and $|\lambda_0| < 1$, such that $E_{\lambda_0}^n(1) \neq L_1(\lambda_0)$ for all n. Each of these equations is holomorphic in λ on the unit disc and not identically satisfied (for real $\lambda \in (0,1)$ the orbit of 1 increases strictly to L_1), so this excludes a countable set. Only $\sigma_0(p_0) \neq 0$ for the marked point p_0 below is used. We take $r_3 \leq 1/240$ in Proposition 10.9, which only shrinks the corridor. Write $E_0 = E_{\lambda_0}$, let B_0 be its attracting basin and σ_0 its Koenigs coordinate at $L_1(\lambda_0)$, normalised by $\sigma_0'(L_1) = 1$ and extended to B_0 by $\sigma_0 = \lambda_0^{-n}\sigma_0 \circ E_0^n$. Since E_0 has no critical points, $\sigma_0' \neq 0$ on B_0.

Proposition 10.18 (Transport of the dynamics)**.** *For $\lambda\in\mathbb{H}_\lambda$ put $\mathfrak{s}=\operatorname{Log}\lambda$, $\mathfrak{s}_0=\operatorname{Log}\lambda_0$ (so $\operatorname{Re}\mathfrak{s},\operatorname{Re}\mathfrak{s}_0<0$),*

$$\mathsf{b}(\lambda)=\frac{\mathfrak{s}-\mathfrak{s}_0}{2\operatorname{Re}\mathfrak{s}_0},\qquad \mathsf{c}(\lambda)=\frac{\mathsf{b}}{1+\mathsf{b}}=\frac{\mathfrak{s}-\mathfrak{s}_0}{\mathfrak{s}+\overline{\mathfrak{s}_0}},\qquad \phi_\lambda(\zeta)=\zeta|\zeta|^{2\mathsf{b}(\lambda)}.$$

Then $|\mathsf{c}(\lambda)|<1$ on $\mathbb{H}_\lambda$. Let $\nu_\lambda=\mathsf{c}(\lambda)\,\frac{\sigma_0}{\overline{\sigma_0}}\,\frac{\overline{\sigma_0'}}{\sigma_0'}$ on $B_0\setminus\sigma_0^{-1}(0)$ and $\nu_\lambda=0$ elsewhere, and let $\mathcal{Q}_\lambda$ be the quasiconformal homeomorphism of $\hat{\mathbb{C}}$ with Beltrami coefficient ν_λ fixing $0,1,\infty$. Then:

1. *$\lambda\mapsto\mathcal{Q}_\lambda(w)$ is holomorphic for each w, $\mathcal{Q}_{\lambda_0}=\mathrm{id}$, and $\mathcal{Q}_\lambda\to\mathrm{id}$ locally uniformly on $\mathbb{C}$ as $\lambda\to\lambda_0$;*
2. *$\mathcal{Q}_\lambda\circ E_0=E_\lambda\circ\mathcal{Q}_\lambda$, hence $\mathcal{Q}_\lambda(E_0^n(1))=E_\lambda^n(1)$ for all $n\ge0$;*
3. *$\mathcal{Q}_\lambda(B_0)$ is the basin of $L_1(\lambda)=\mathcal{Q}_\lambda(L_1(\lambda_0))$, and $\sigma_\lambda\circ\mathcal{Q}_\lambda=\mathsf{k}_\lambda\,\phi_\lambda\circ\sigma_0$ on B_0, with $\mathsf{k}_\lambda\neq0$ holomorphic in λ and $\mathsf{k}_{\lambda_0}=1$;*
4. *for λ near λ_0, $\mathcal{Q}_\lambda$ maps the repelling fixed point $L_2(\lambda_0)$ of Section 10 to $L_2(\lambda)$.*

Proof. $|\mathfrak{s}+\overline{\mathfrak{s}_0}|^2-|\mathfrak{s}-\mathfrak{s}_0|^2=4\operatorname{Re}\mathfrak{s}\operatorname{Re}\mathfrak{s}_0>0$, so $|\mathsf{c}|<1$, uniformly on compact sets. In $L=\log\zeta$, ϕ_λ is the real-linear map $L\mapsto L+2\mathsf{b}\operatorname{Re}L$, with determinant $\operatorname{Re}\mathfrak{s}/\operatorname{Re}\mathfrak{s}_0>0$; it fixes $2\pi\mathrm{i}$ and sends $\mathfrak{s}_0$ to $\mathfrak{s}$. So ϕ_λ is a quasiconformal homeomorphism with $\phi_\lambda(\lambda_0\zeta)=\lambda\phi_\lambda(\zeta)$ and Beltrami coefficient $\mathsf{c}(\lambda)\zeta/\bar\zeta$; ν_λ is its pull-back by σ_0, hence E_0-invariant (the basin is completely invariant and $\sigma_0\circ E_0=\lambda_0\sigma_0$). Item (1) is the Ahlfors–Bers theorem [2, Thm. 11], since ν_λ depends holomorphically on λ through the scalar $\mathsf{c}(\lambda)$, and continuity of normalised solutions in ν. The map $\mathcal{Q}_\lambda\circ E_0$ has Beltrami coefficient $E_0^*\nu_\lambda=\nu_\lambda$, so $G_\lambda=\mathcal{Q}_\lambda\circ E_0\circ\mathcal{Q}_\lambda^{-1}$ has vanishing dilatation and is holomorphic on $\mathbb{C}$ by Weyl's lemma; it is a covering of $\mathbb{C}^*$ (it omits $\mathcal{Q}_\lambda(0)=0$), so $G_\lambda(w)=\mathrm{e}^{Aw+D}$, and $G_\lambda(0)=\mathcal{Q}_\lambda(1)=1$ gives $D\in2\pi\mathrm{i}\mathbb{Z}$, which we drop. Near $L_1(\lambda_0)$, $\mathcal{Q}_\lambda\circ\sigma_0^{-1}\circ\phi_\lambda^{-1}$ is conformal, so G_λ is conformally conjugate to $\zeta\mapsto\lambda\zeta$ at $\mathcal{Q}_\lambda(L_1(\lambda_0))$: that point is a fixed point of e^{Aw} with multiplier λ, which forces it to be e^λ and $A=\lambda\mathrm{e}^{-\lambda}$. So $G_\lambda=E_\lambda$, proving (2). For (3), let $V=\sigma_0^{-1}(D_\varrho)$ be a small forward-invariant disc about $L_1(\lambda_0)$. The map $\psi=\sigma_\lambda\circ\mathcal{Q}_\lambda\circ\sigma_0^{-1}\circ\phi_\lambda^{-1}$ is defined on $\phi_\lambda(D_\varrho)=D_{\varrho^{\operatorname{Re}\mathfrak{s}/\operatorname{Re}\mathfrak{s}_0}}$, conformal off 0 (equal Beltrami coefficients) and continuous at 0, hence conformal; it fixes 0 and satisfies $\psi(\lambda\zeta)=\lambda\psi(\zeta)$. Comparing Taylor coefficients ($\lambda^{n-1}\neq1$ for $n\ge2$) gives $\psi(\zeta)=\mathsf{k}_\lambda\zeta$. The identity extends to B_0 by $\sigma_\lambda=\lambda^{-n}\sigma_\lambda\circ E_\lambda^n$, and $\mathsf{k}_\lambda=\sigma_\lambda(\mathcal{Q}_\lambda(w))/\phi_\lambda(\sigma_0(w))$ is holomorphic in λ, because $\phi_\lambda(\zeta)=\zeta\exp(\mathsf{b}(\lambda)\log|\zeta|^2)$ is. In particular, on B_0 near any point where σ_λ is locally injective, $\mathcal{Q}_\lambda=\sigma_\lambda^{-1}\circ(\mathsf{k}_\lambda\phi_\lambda\circ\sigma_0)$ is jointly real-analytic in the space variable, holomorphic in λ, with nonvanishing Jacobian. For (4), $\mathcal{Q}_\lambda(L_2(\lambda_0))$ is a fixed point of E_λ depending continuously on λ and equal to $L_2(\lambda_0)$ at λ_0; fixed points are isolated. □

Let $H_0\subset\mathbb{C}$ be the tilted region of Lemma 10.3 for E_0 and an admissible line ℓ, written in $w=\mathrm{e}(1+u)$, and $p_0=E_0^{n_0}(1)$ its marked point; choose ℓ with p_0 in the interior. For $\lambda\in\mathbb{H}_\lambda$ let X_λ^{tr} be $\mathcal{Q}_\lambda(H_0)$ with its edges $\mathcal{Q}_\lambda(\gamma)$ and $\mathcal{Q}_\lambda(E_0(\gamma))=E_\lambda(\mathcal{Q}_\lambda(\gamma))$ identified by E_λ, and $p_\lambda=\mathcal{Q}_\lambda(p_0)=E_\lambda^{n_0}(1)$.

Proposition 10.19 (The transported family)**.** *X_λ^{tr} is biholomorphic to $\mathbb{C}^*$, its ends at $L_{1,2}$. Let z be the lift of its uniformisation with the L_1-end at $+\mathrm{i}\infty$ and $z=n_0$ at p_λ, and let F_λ^{tr} be the germ at $z=0$ given by the local inverse of $E_\lambda^{n_0}$ at p_λ with value 1, applied to the inverse of z at $z+n_0$. Then $F_\lambda^{\mathrm{tr}}(0)=1$, the lifted inverse uniformisation $\tilde F$ satisfies $\tilde F(z+1)=E_\lambda(\tilde F(z))$, and $(\lambda,z)\mapsto F_\lambda^{\mathrm{tr}}(z)$ is jointly holomorphic near $\mathbb{H}_\lambda\times\{0\}$ and single-valued there.*

Proof. $\mathcal{Q}_\lambda$ induces a quasiconformal homeomorphism $J_\lambda:X_0\to X_\lambda^{\mathrm{tr}}$, because it conjugates the edge identifications; each end of X_λ^{tr} is the image of a punctured-disc end of $X_0\cong\mathbb{C}^*$ (Lemma 10.4), hence a punctured disc. Near the seam the complex structure is given by local inverses of E_λ, which is injective there because E_0 is. The Beltrami coefficient of J_λ is the descent

of ν_λ; transported to $\hat{\mathbb{C}}$ by a uniformisation Z_0 of X_0 normalised at the ends and p_0, it equals $\mathsf{c}(\lambda)\nu$ for a fixed measurable ν with $|\nu| \le 1$. Let $\mathcal{W}_\lambda$ be the normalised solution; by Ahlfors–Bers it is holomorphic in λ at each point, and $Z_\lambda = \mathcal{W}_\lambda \circ Z_0 \circ J_\lambda^{-1}$ is the uniformisation of X_λ^{tr} normalised at the ends and p_λ. Near p_0 the coefficient ν_λ is real-analytic in w, because $\sigma_0(p_0) \neq 0$. The $\partial_{\bar\lambda}$-cancellation of Proposition 10.9, with $\mathcal{Q}_\lambda$ in place of $\mathcal{J}_\varepsilon$, shows that $(\lambda, w) \mapsto Z_\lambda(w)$ is jointly holomorphic near (λ, p_λ). The inverse function theorem and the local inverse of $E_\lambda^{n_0}$, which is determined by the value 1 at p_λ and depends holomorphically on λ, give the germ. The C^1-dependence needed in the cancellation is explicit here: near p_0, $\mathcal{Q}_\lambda$ is given by the formula at the end of the proof of Proposition 10.18, and $\mathcal{W}_\lambda$ is C^1 in (λ, ξ) by interior Schauder estimates [1, Ch. V]. All objects are defined by a single formula for every $\lambda \in \mathbb{H}_\lambda$, so the family is single-valued. □

Lemma 10.20 (The lifted motion near the ends)**.** *For λ near λ_0 let $\chi_\lambda = \log \frac{u-u_1(\lambda)}{u-u_2(\lambda)}$ be the coordinate of Section 10 for E_λ, and let $\tilde{\mathcal{Q}}_\lambda$ be the lift of $\mathcal{Q}_\lambda$ to these coordinates (defined off the holes, commuting with $\chi \mapsto \chi + 2\pi\mathrm{i}$, continuous in λ and equal to the identity at λ_0). The lift satisfies $\tilde{\mathcal{Q}}_\lambda \circ f_{\chi,\lambda_0} = f_{\chi,\lambda} \circ \tilde{\mathcal{Q}}_\lambda$ exactly (no deck shift), and $\tilde{\mathcal{Q}}_\lambda \to \mathrm{id}$ uniformly on compact subsets of $\mathcal{G}^{\mathrm{c}}$. Moreover, for every $\delta > 0$ there are T and a neighbourhood U_δ of λ_0 such that for $\lambda \in U_\delta$ there are real-linear maps $A_\lambda^\pm$ and constants $\vartheta_\lambda^\pm$ with $\|A_\lambda^\pm - I\| + |\vartheta_\lambda^\pm| \le \delta$ and*

$$|\tilde{\mathcal{Q}}_\lambda(\chi) - A_\lambda^+\chi - \vartheta_\lambda^+| \le \delta \quad (\mathrm{Re}\,\chi \le -T), \qquad |\tilde{\mathcal{Q}}_\lambda(\chi) - A_\lambda^-\chi - \vartheta_\lambda^-| \le \delta \quad (\mathrm{Re}\,\chi \ge T).$$

The same holds for $\tilde{\mathcal{Q}}_\lambda^{-1}$.

Proof. This is a statement for λ near λ_0, where $u_{1,2}(\lambda)$ and χ_λ are defined. $\mathcal{Q}_\lambda$ maps $u_1(\lambda_0), u_2(\lambda_0), \infty$ to $u_1(\lambda), u_2(\lambda), \infty$ (Proposition 10.18(3),(4)), so in the cross-ratio $q = (u-u_1)/(u-u_2)$ it is an orientation-preserving homeomorphism of $\mathbb{C}^* \setminus \{1\}$ fixing $0, \infty$. Such a map acts trivially on the winding number about 0, so it lifts to $\mathbb{C} \setminus 2\pi\mathrm{i}\mathbb{Z}$ commuting with $2\pi\mathrm{i}$; joint continuity of $(\lambda, w) \mapsto \mathcal{Q}_\lambda(w)$ (the λ-lemma) lets us choose the lift continuously with $\tilde{\mathcal{Q}}_{\lambda_0} = \mathrm{id}$, and gives local uniform convergence. The lifted conjugacy $\tilde{\mathcal{Q}}_\lambda \circ f_{\chi,\lambda_0} = f_{\chi,\lambda} \circ \tilde{\mathcal{Q}}_\lambda + 2\pi\mathrm{i}k$ holds with an integer k that is continuous in (λ, χ) and 0 at λ_0, hence $k = 0$.

Attracting end. In $L = \log\sigma_0$ and $L' = \log\sigma_\lambda$, Proposition 10.18(3) says that $\mathcal{Q}_\lambda$ is the affine map $L' = L + 2\mathsf{b}\,\mathrm{Re}\,L + \log\mathsf{k}_\lambda$. Near u_1, σ_λ and q are both local coordinates vanishing at u_1, so $\chi_\lambda = L' + \psi_\lambda(\mathrm{e}^{L'})$ with ψ_λ holomorphic near 0 and continuous in λ; likewise for λ_0. Composing, with $A_\lambda^+ = I + 2\mathsf{b}\,\mathrm{Re}$ and $\vartheta_\lambda^+ = \log\mathsf{k}_\lambda + \psi_\lambda(0) - A_\lambda^+\psi_{\lambda_0}(0)$, the remainder $\tilde{\mathcal{Q}}_\lambda(\chi) - A_\lambda^+\chi - \vartheta_\lambda^+$ equals $[\psi_\lambda(\mathrm{e}^{L'}) - \psi_\lambda(0)] - A_\lambda^+[\psi_{\lambda_0}(\mathrm{e}^{L}) - \psi_{\lambda_0}(0)]$. Since $|\mathrm{e}^{L'}| \le C|\mathrm{e}^{L}|^{\mathrm{Re}\,\mathfrak{s}/\,\mathrm{Re}\,\mathfrak{s}_0}$ and $\mathrm{Re}\,\mathfrak{s}/\,\mathrm{Re}\,\mathfrak{s}_0 \ge \frac12$ near λ_0, it is at most $C\mathrm{e}^{-T/2}$ on $\mathrm{Re}\,\chi \le -T$, uniformly for λ near λ_0. Choose T with $C\mathrm{e}^{-T/2} \le \delta$, and then U_δ so that $\|A_\lambda^+ - I\| + |\vartheta_\lambda^+| \le \delta$.

Repelling end. Let $M = \log\sigma_{\mathrm{rep},0}$ and $M' = \log\sigma_{\mathrm{rep},\lambda}$ be logarithms of the repelling Koenigs coordinates at L_2, in which E_0, E_λ are the translations by $m_0 = \log\mu_0$, $m_\lambda = \log\mu_\lambda$, $\mathrm{Re}\,m > 0$. For λ near λ_0, $\mathcal{Q}_\lambda$ maps a linearising disc of E_0 into one of E_λ, and its lift G_λ to $\{\mathrm{Re}\,M < -R\}$ satisfies $G_\lambda(M - m_0) = G_\lambda(M) - m_\lambda$ and $G_\lambda(M + 2\pi\mathrm{i}) = G_\lambda(M) + 2\pi\mathrm{i}$ (the first because $\mathcal{Q}_\lambda$ conjugates the local inverse branches at L_2, with the integer fixed by continuity from λ_0; the second by degree). Let A'_λ be the real-linear map with $A'_\lambda m_0 = m_\lambda$, $A'_\lambda(2\pi\mathrm{i}) = 2\pi\mathrm{i}$. Then $D_\lambda = G_\lambda - A'_\lambda$ is invariant under $M \mapsto M - m_0$ and $M \mapsto M \pm 2\pi\mathrm{i}$, which preserve $\{\mathrm{Re}\,M < -R\}$. Every point of that half-plane is carried by these maps into the compact set $\{-R - \mathrm{Re}\,m_0 \le \mathrm{Re}\,M \le -R,\ 0 \le \mathrm{Im}\,M \le 2\pi\}$, so $\sup|D_\lambda|$ is its supremum there, which tends to 0 as $\lambda \to \lambda_0$. As at the other end, $\chi = -M + \mathrm{const} + \psi^-(\mathrm{e}^M)$ with ψ^- holomorphic and vanishing at 0, for both parameters, and the same two-step choice of T and U_δ gives the second estimate. The inverse is treated in the same way. □

Lemma 10.21 (Comparison)**.** *There is a neighbourhood U of λ_0 such that for $\lambda \in U$ the transported quotient X_λ^{tr} and the tilted quotient $X_{1-\lambda}^{\ell}$ of Section 10 (same admissible line ℓ) are biholomorphic by a map that preserves the two ends and sends p_λ to the marked point. Consequently $F_\lambda^{\mathrm{tr}} = F_{1-\lambda}$ near $z = 0$ for $\lambda \in U$.*

Proof. Let ℓ be the admissible line fixed in the proof of Proposition 10.9 for the disc about ε_0; it is the line used for H_0. Take U inside that disc, so that ℓ is admissible, n_0 is constant, and the marked point of $X_{1-\lambda}^{\ell}$ is interior. Take also $U \subset U_\delta$ of Lemma 10.20 with $\delta = 1/2000$, and $T' \geq (T+1)/0.6$, so that $|t| \geq T'$ implies $|\operatorname{Re}\chi| \geq T$ on the strips; for fixed λ_0 this is a genuine neighbourhood, although its size may be small compared with $\operatorname{Re}\varepsilon_0$. Work in the χ-plane of E_λ, where E_λ is $f_\chi = \mathrm{id} + \Delta$ and the tilted strip is $\hat{H}_\lambda$ between ℓ and $f_\chi(\ell)$, and put $\hat{T}_\lambda = \tilde{\mathcal{Q}}_\lambda(\hat{H}_0)$, the transported strip, bounded by $\Gamma = \tilde{\mathcal{Q}}_\lambda(\ell)$ and $f_\chi(\Gamma)$. Let ς, t be the coordinates of Section 10 relative to ℓ, and $\mathsf{W}_\rho = \{|\varsigma| \leq \rho(1+|t|)\}$.

Step 1: $\hat{T}_\lambda \subset \mathsf{W}_{1/100} \cap \mathcal{G}^{\mathrm{c}}$ for $\lambda \in U$, after shrinking U. $\hat{H}_0$ lies in $\{|\varsigma| \leq \frac{6}{5}|\varepsilon_0|\}$. On $\{|t| \leq T'\}$ this is compact and $\tilde{\mathcal{Q}}_\lambda \to \mathrm{id}$ uniformly. For $|t| \geq T'$ (so $|\operatorname{Re}\chi| \geq T$), Lemma 10.20 gives $|\tilde{\mathcal{Q}}_\lambda(\chi) - \chi| \leq \|A_\lambda^{\pm} - I\|\,|\chi| + |\vartheta_\lambda^{\pm}| + \delta$, and $|\chi| < |t| + 4 \leq 5(1+|t|)$ on $\hat{H}_0$; with $\delta = 1/2000$ the displacement is at most $6\delta(1+|t|) = 0.003(1+|t|)$ (also on $|t| \leq T'$, after shrinking U). The parameter t of the image point differs from that of the source by at most this amount. Since $|\varepsilon| \leq 1.025|\varepsilon_0| \leq 0.005$ on U (using $|\varepsilon_0| \leq 1/240$), every point of $\hat{T}_\lambda$ has $|\varsigma| \leq 0.005 + \frac{0.003}{0.997}(1+|t|) < 0.0082(1+|t|) \leq \frac{1}{100}(1+|t|)$, and its distance to the holes is at least $\max(1.48, 0.6|t| - 0.5) - 0.003(1+|t|) \geq \frac{1}{20}(1+|t|) + \frac{1}{2} + 0.2$, so that the orbit discs of Step 2, of radius $(1+|t|)/20$, stay in $\mathcal{G}^{\mathrm{c}}$. Symmetrically, $\tilde{\mathcal{Q}}_\lambda^{-1}(\hat{H}_\lambda) \subset \mathsf{W}_{1/100}$ in the χ-plane of E_0.

Step 2: orbit correspondence. Let $P \in \hat{T}_\lambda$ with parameter t. As in the proof of Lemma 10.8, each application of $f_\chi^{\pm 1}$ changes ς by at least $\frac{2}{5}|\varepsilon|$ in the same direction and moves the point by at most $2|\varepsilon|$, and $|\varsigma(P)| \leq \frac{1}{100}(1+|t|)$; so the orbit of P reaches $\hat{H}_\lambda$ after at most $(1+|t|)/(40|\varepsilon|)$ steps, forward or backward, inside a disc about P contained in $\mathcal{G}^{\mathrm{c}}$, on which f_χ is injective. Exactly one iterate $f_\chi^{n(P)}(P)$ lies in $\hat{H}_\lambda \setminus f_\chi(\ell)$, by the monotonicity of ς and the isotopy argument of Lemma 10.5(2). (The edges of $\hat{T}_\lambda$ are Γ and $f_\chi(\Gamma)$ by the exact lifted conjugacy of Lemma 10.20.) The map $P \mapsto f_\chi^{n(P)}(P)$ is locally a fixed iterate, jumps by one iterate exactly across ℓ, where the two values are identified, and satisfies $n(f_\chi(P)) = n(P) - 1$ for $P \in \Gamma$, matching the identification of the edges of $\hat{T}_\lambda$. It therefore induces a holomorphic map $\Theta : X_\lambda^{\mathrm{tr}} \to X_{1-\lambda}^{\ell}$. For the inverse, apply the same construction in the χ-plane of E_0 to the strips $\hat{H}_0$ and $\tilde{\mathcal{Q}}_\lambda^{-1}(\hat{H}_\lambda) \subset \mathsf{W}_{1/100} \cap \mathcal{G}^{\mathrm{c}}$ (Step 1), and conjugate back by $\tilde{\mathcal{Q}}_\lambda$: this gives a map Θ' from $X_{1-\lambda}^{\ell}$ to X_λ^{tr}, locally a fixed iterate of E_λ. Both $\Theta' \circ \Theta$ and $\Theta \circ \Theta'$ send each point to a point of its own orbit in the same half-open strip, so they are the identity. Since orbit segments have length at most $(1+|t|)/20$ in χ, Θ preserves the ends ($\operatorname{Re}\chi \to \mp\infty$).

Step 3: marked points. By Proposition 10.18(2) and continuity of the lift, $\tilde{\mathcal{Q}}_\lambda$ maps the continued base orbit $\chi_n^*(\lambda_0)$ of Lemma 10.5 to $\chi_n^*(\lambda)$. So p_λ is represented by $\chi_{n_0}^*(\lambda)$, which lies in $\hat{H}_\lambda \setminus f_\chi(\ell)$ because n_0 is the first crossing for all $\lambda \in U$. Hence $n(p_\lambda) = 0$ and $\Theta(p_\lambda)$ is the marked point.

A biholomorphism of $\mathbb{C}^*$ preserving both ends and a marked point is the identity in the normalised coordinates, so the two lifts z agree; both germs at $z = 0$ are obtained by the local inverse of $E_\lambda^{n_0}$ with value 1 at the marked point, so they agree. □

Theorem 10.22 (Continuation to the upper Shell–Thron region)**.** *Let $\Omega = b(\mathcal{S}_{r_3}^{\mathrm{c}\prime})$ be the corridor of Theorem 10.14. The continuation of Kneser's solution from $(\eta, \eta + r_5)$ into $\Omega \cap \mathcal{U}^+$ extends analytically along every path in $\mathcal{U}^+ = b(\mathbb{H}_\lambda)$, the upper half of the Shell–Thron region, to a*

single-valued family of germs at $z = 0$, jointly holomorphic in (b, z). For every $b_0 \in (1, \eta)$ there is a disc D about $z = 0$ such that along every path in $\mathcal{U}^+$ ending at b_0 these germs converge uniformly on D to $K^W_{b_0}$; hence the boundary value is $K^W_{b_0}$ on its regular domain, by the functional equation.

Proof. By Proposition 10.19, $\lambda \mapsto F^{\mathrm{tr}}_\lambda$ is a single-valued holomorphic family of germs on $\mathbb{H}_\lambda$. By Lemma 10.21 it coincides with the cusp family $F_{1-\lambda}$ near λ_0. The part of the parameter domain of Theorem 10.14 that lies in $\mathbb{H}_\lambda$, $\{1-\varepsilon : \varepsilon \in \mathcal{S}'_{r_3},\ \operatorname{Im}\varepsilon < 0,\ |1-\varepsilon| < 1\}$, is connected (it is the part of a small sector lying inside the disc $|\varepsilon - 1| < 1$ and below the real axis) and contains λ_0, so the two families agree on it by the identity theorem. Hence F^{tr}, transferred to b by Lemma 10.17, is the analytic continuation of Kneser's solution along every path in $\mathcal{U}^+$ starting in $\Omega \cap \mathcal{U}^+$, and it is single-valued there.

For the boundary values, shrink the discs Δ_{b_0} of Proposition 3.5 to radius at most $(\eta - b_0)/2$, as in the proof of Theorem 10.14(3). For b in such a disc, $|\log b| \le \log b_0 - \log(1 - x/2) < \log b_0 + \log(1 + x) = \mathrm{e}^{-1}$ with $x = (\eta - b_0)/b_0$, and $g(\{|\lambda| < 1\}) \supset \{|y| < \mathrm{e}^{-1}\}$ because $\min_{|\lambda|=1} |\lambda \mathrm{e}^{-\lambda}| = \mathrm{e}^{-1}$; so E_b has an attracting fixed point and $N_W \cap \{\operatorname{Im} b > 0\} \subset \mathcal{U}^+$ (Lemma 10.17). This set is connected and meets $\Omega \cap \mathcal{U}^+$ near η, where K^W agrees with the cusp family (Theorem 10.14(3)); by the identity theorem $K^W = F^{\mathrm{tr}}$ on all of it. The z-domain of K^W in Proposition 3.5 contains a disc D about 0 for all b in a neighbourhood of b_0. Every path in $\mathcal{U}^+$ ending at b_0 eventually lies in $\Delta_{b_0} \cap \{\operatorname{Im} b > 0\}$, where $F^{\mathrm{tr}} = K^W$ is jointly holomorphic on a neighbourhood of the closure of that half-disc times D; so the germs converge uniformly on D to $K^W_{b_0}$. □

Proposition 10.23 (Crossing the attracting real slice)**.** *The family of Theorem 10.22 extends jointly holomorphically in (b, z) across every $b \in (\mathrm{e}^{-\mathrm{e}}, 1)$. The local extensions agree on overlaps and hence give a single-valued family of germs on a neighbourhood of this interval.*

Proof. Fix $x \in (-1, 0)$ and choose a simply connected neighbourhood V of x in $\{0 < |\lambda| < 1\}$ which meets $\mathbb{H}_\lambda$. Continue the branch $\mathfrak{s} = \operatorname{Log}\lambda$ from $V \cap \mathbb{H}_\lambda$ to V. The proof of Proposition 10.18 uses only $\operatorname{Re}\mathfrak{s} = \log|\lambda| < 0$ to obtain $|\mathsf{c}(\lambda)| < 1$. Thus its Beltrami coefficient, normalised solution $\mathcal{Q}_\lambda$, and all four conclusions hold on V by exactly the same formulas. The marked quotient and its uniformisation in Proposition 10.19 consequently give a jointly holomorphic family of germs on $V \times \{z : |z| < \delta\}$ for some $\delta > 0$. On $V \cap \mathbb{H}_\lambda$ this is the original transported family, since both use the same normalised Beltrami solution and marked quotient. Any two such local extensions agree on an upper-half-plane overlap, hence agree wherever both are defined by the identity theorem. Finally $b'(x) = b(x)(1 - x)\mathrm{e}^{-x} \neq 0$, and b maps $(-1, 0)$ increasingly onto $(\mathrm{e}^{-\mathrm{e}}, 1)$; local inversion transfers these germs to the base coordinate. □

Remark 10.24 (The punctured interior)**.** The same argument works globally in $\mathfrak{s}$. Every object in Propositions 10.18 and 10.19 is given by one formula in $\mathfrak{s} = \log\lambda$, and only $\operatorname{Re}\mathfrak{s} < 0$ and $|\mathrm{e}^{\mathfrak{s}}| < 1$ are used. So $\mathfrak{s} \mapsto F^{\mathrm{tr}}_{\mathfrak{s}}$ is holomorphic and single-valued on the whole half-plane $\{\operatorname{Re}\mathfrak{s} < 0\}$. This half-plane is the universal cover of the punctured interior $\mathcal{ST}^\circ \setminus \{1\}$ via $\mathfrak{s} \mapsto \exp g(\mathrm{e}^{\mathfrak{s}})$. Hence the continuation of Kneser's family continues along every path in $\mathcal{ST}^\circ \setminus \{1\}$, and $b = 1$ is its only possible singularity in the interior. A loop about $b = 1$ acts by $\mathfrak{s} \mapsto \mathfrak{s} + 2\pi\mathrm{i}$. Whether this monodromy is trivial is open. Numerically, the normalised modes $|c_1|/\Lambda$ of the boundary values K^W_b stay near 0.08 as $b \downarrow 1$ (0.0816 at $b = 1.01$), while $\Lambda \to 1$. So near $b = 1$ the separation from regular iteration is of order one.

Remark 10.25 (Relation to Paulsen's family)**.** Theorem 10.22 removes the restriction to paths near the real axis in Theorem 10.14: starting from $\mathcal{K}_b$ on $(\eta, \eta + r_5)$ and entering $\mathcal{U}^+$ inside Ω, every further path in $\mathcal{U}^+$ gives K^W on all of $(1, \eta)$. The family of [25] may enter $\mathcal{U}^+$ through a boundary arc far from η. It coincides with ours if its germ at one base in $\mathcal{U}^+$ equals F^{tr} there.

Theorems 10.33 and 10.34 show that Kneser's family crosses the entire open upper boundary arc holomorphically and enters with F^{tr}, and Theorem 10.40 shows that the whole continuation over the upper half-plane is single-valued, so this is Paulsen's family (Corollary 10.41). The nine-digit agreement at $b = \sqrt{2}$ (Section 11.1) is consistent with this.

10.10. **Crossing the boundary away from the cusp.** Theorem 10.14 crosses the Shell–Thron boundary only near η. Here we cross it along arcs far from the cusp, up to the multiplier angle $\alpha_1 = 0.9353\ldots\pi$. The tool is Kneser's own initial region: the straight chord between the two fixed points and its image. We show that it stays a crescent, with the same marked point, on a collar that crosses the boundary arc. This needs no linearisation of the neutral fixed point, as in Section 10.

Notation. For complex θ with $0 < \operatorname{Re}\theta < \pi$ put

$$\mu(\theta) = \theta\cot\theta, \quad \lambda_\pm = \mu \pm \mathrm{i}\theta, \quad r = \mathrm{e}^{\mu}, \quad \mathcal{L}(\theta) = \frac{\theta \mathrm{e}^{-\mu}}{\sin\theta}, \quad b(\theta) = \mathrm{e}^{\mathcal{L}(\theta)}.$$

We write $E_\theta(w) = \mathrm{e}^{\mathcal{L}(\theta)w}$; this is $E_{b(\theta)}$ with the logarithm of the base continued along paths in θ, and it is the map that enters the analytic continuation of $F(z+1) = b^{F(z)}$ in the base. $L_\pm = r\mathrm{e}^{\pm\mathrm{i}\theta}$ are fixed points of E_θ with multipliers $\mathcal{L}\,L_\pm = \lambda_\pm$, and $\mathcal{L} = \lambda_+\mathrm{e}^{-\lambda_+}$. So if $\lambda_+ \in \mathbb{H}_\lambda$, then $\mathcal{L} = \operatorname{Log} b$ is principal ($|\operatorname{Im}\mathcal{L}| < \pi$) and $b(\theta) = b(\lambda_+(\theta)) \in \mathcal{U}^+$ in the notation of Lemma 10.17. For real $\theta \in (0,\pi)$, $b(\theta) > \eta$ is real, $L_\pm$ is the conjugate pair of Proposition 10.11 with $\theta_b = \theta$, and $|\lambda_+| = \theta/\sin\theta > 1$. The affine coordinate

$$\zeta(w) = \frac{w/r - \cos\theta}{\mathrm{i}\sin\theta}$$

sends $L_\pm$ to ±1 and the chord $\gamma_\theta = [L_-, L_+]$ to $[-1,1]$, and a direct computation ($\mathcal{L}w = \mu + \mathrm{i}\theta\zeta$) gives

$$\zeta(E_\theta(w)) = \mathsf{s}_\theta(\zeta(w)), \qquad \mathsf{s}_\theta(\zeta) = \frac{\mathrm{e}^{\mathrm{i}\theta\zeta} - \cos\theta}{\mathrm{i}\sin\theta}.$$

s_θ is entire, $\mathsf{s}_\theta(\pm1) = \pm1$ and $\mathsf{s}'_\theta(\pm1) = \lambda_\pm$, so $g_\theta(t) = (\mathsf{s}_\theta(t) - t)/(1-t^2)$ is entire in t. Since $\mathsf{s}_\theta(\zeta_1) = \mathsf{s}_\theta(\zeta_2)$ only if $\theta(\zeta_1 - \zeta_2) \in 2\pi\mathbb{Z}$, s_θ is injective on every set of diameter $< 2\pi/|\theta|$.

The parameter region. Let $P_0 = \frac{1}{10}$, $P_1 = \frac{11}{5}$, let $V(p) = \frac{1}{4}$ for $p \le \frac{8}{5}$ and $V(p) = \frac{1}{4} - \frac{7}{60}(p - \frac{8}{5})$ for $p \ge \frac{8}{5}$, and

$$\Theta_{\mathrm{C}} = \{p(1+\mathrm{i}c) : 0 < p \le P_0,\ 0 \le c \le \tfrac{1}{10}\}, \qquad \Theta_{\mathrm{K}} = \{p + \mathrm{i}vp^2 : P_0 \le p \le P_1,\ 0 \le v \le V(p)\},$$

$\Theta = \Theta_{\mathrm{C}} \cup \Theta_{\mathrm{K}}$. Its vertical fibre over p is a segment $\{p + \mathrm{i}q : 0 \le q \le Q(p)\}$.

Lemma 10.26 (Certified inequalities)**.** *For $\theta = p + \mathrm{i}q \in \Theta$ and $-1 \le t \le 1$:*

1. $\operatorname{Im} g_\theta(t) < 0$ *and* $\operatorname{Im}(\overline{\mathsf{s}'_\theta(t)}\, g_\theta(t)) < 0$*;*
2. $\partial_q|\lambda_+(p+\mathrm{i}q)|^2 < 0$*,* $|\lambda_+| < 1$ *on the top edge $q = Q(p)$, and $\mathcal{L}'(\theta) \ne 0$;*
3. *on Θ_{C},* $\operatorname{Re}\cot\theta > 0$*;*
4. *for* $\operatorname{Re}\theta \le P_2 = \frac{19}{10}$*,* $|\operatorname{Im}\mathcal{L}(\theta)| < \pi$*.*

Proof. See Appendix A.1. □

By (1) at $t = 1$, $\operatorname{Im}\lambda_+ = -2\operatorname{Im} g_\theta(1) > 0$. By (2), on each fibre of Θ the multiplier λ_+ of the continued fixed point L_+ crosses the unit circle exactly once, at $q = q_{\mathrm{c}}(p)$, and

$$\Theta^+ = \{\theta \in \Theta : |\lambda_+(\theta)| < 1\} = \{p + \mathrm{i}q \in \Theta : q > q_{\mathrm{c}}(p)\}$$

is connected, with $\mathcal{L}$ principal and $b(\Theta^+) \subset \mathcal{U}^+$. (Below q_{c} the logarithm $\mathcal{L}$ need not be principal: for p near P_1, $\operatorname{Im}\mathcal{L}$ exceeds π on the fibre. By (4) this does not happen for $p \le P_2$.) The boundary point $\beta(p) = b(p + \mathrm{i}q_{\mathrm{c}}(p))$ has $\lambda_+ = \mathrm{e}^{\mathrm{i}\alpha(p)}$ with α continuous, $\alpha(p) \to 0$ as

$p \to 0$ and $\alpha(P_1) = \alpha_1 = 0.9353\ldots\pi$; so β runs through the whole upper boundary arc $\{\exp(e^{i\alpha}e^{-e^{i\alpha}}) : 0 < \alpha \le \alpha_1\}$.

Lemma 10.27 (The crescent)**.** *Let θ lie in a sufficiently small open neighbourhood Θ' of $\Theta\setminus\{0\}$.*

1. *$A_\theta = \mathsf{s}_\theta([-1,1])$ is a simple arc from -1 to 1, in the open lower half-plane except for its ends. Let H_θ be the closed Jordan region bounded by $[-1,1]\cup A_\theta$.*
2. *$N_\theta(t,\sigma) = t + \sigma(\mathsf{s}_\theta(t) - t)$ is a real-analytic diffeomorphism of $(-1,1)\times[0,1]$ onto $H_\theta\setminus\{\pm1\}$, with $N(t,0) = t$ and $N(t,1) = \mathsf{s}_\theta(t)$.*
3. *s_θ maps a neighbourhood of $[-1,1]$ biholomorphically onto a neighbourhood of A_θ, and the upper side of $[-1,1]$ to the side of A_θ facing H_θ.*
4. *Let X_θ be $H_\theta\setminus\{\pm1\}$ with $t\in(-1,1)$ glued to $\mathsf{s}_\theta(t)$. In the coordinates $x = \log\frac{1-t}{1+t}$ and $\chi = \log\frac{\zeta-1}{\zeta+1}$ the map $\Phi_\theta(x+\mathrm{i}\sigma) = \chi(N_\theta(t,\sigma))$ induces a quasiconformal homeomorphism from $C_0 = \mathbb{C}/\mathrm{i}\mathbb{Z}$ onto X_θ. Its Beltrami coefficient is holomorphic in θ at each fixed $x+\mathrm{i}\sigma$ and bounded by some $k<1$ locally uniformly in θ. Hence $X_\theta\cong\mathbb{C}^*$, with the ends at ±1 as punctures.*

Proof. All conditions below are strict inequalities that hold on Θ by Lemma 10.26, uniformly on compact parts, so they persist on a neighbourhood. $|\theta| < 2.45$ there (also on the rectangle used in Theorem 10.34), so s_θ is injective on every set of diameter $< 2\pi/2.45$, in particular on the disc of radius 1.28.

(1) $\operatorname{Im}\mathsf{s}_\theta(t) = (1-t^2)\operatorname{Im} g_\theta(t) < 0$ for $|t|<1$, and s_θ is injective on $[-1,1]$.

(2) $\partial_t N = (1-\sigma) + \sigma\mathsf{s}'$ and $\partial_\sigma N = (1-t^2)g$, so the Jacobian is

$$\operatorname{Im}(\overline{\partial_t N}\,\partial_\sigma N) = (1-t^2)\big[(1-\sigma)\operatorname{Im} g + \sigma\operatorname{Im}(\bar{\mathsf{s}}' g)\big] < 0$$

for $|t|<1$. N is continuous on the square $[-1,1]\times[0,1]$ and maps its boundary once around the Jordan curve ∂H_θ, the vertical sides to the points ±1. For $\zeta_0\notin\partial H_\theta$ all preimages are interior and count with the same sign, so their number is the winding number of ∂H_θ about ζ_0, namely 1 in H_θ and 0 outside. An interior point cannot map to ∂H_θ, since its image would have nearby points outside H_θ with preimages.

(3) $\mathsf{s}'_\theta \neq 0$, and s_θ is injective near $[-1,1]$. At $t\in(-1,1)$ the image of the upward direction is $\mathrm{i}\mathsf{s}'(t)$, and the direction from A_θ into H_θ is $-\partial_\sigma N(t,1) = -(1-t^2)g(t)$. Both lie on the same side of the tangent $\mathsf{s}'(t)$, because $\operatorname{Im}(\overline{\mathsf{s}'}\,\mathrm{i}\mathsf{s}') = |\mathsf{s}'|^2 > 0$ and $\operatorname{Im}(\overline{\mathsf{s}'}(-g)) > 0$.

(4) Near $t=1$ write $N-1 = (t-1)m$ with $m = 1-\sigma(1+t)g(t)$. For $t<1$, $m\neq0$ by (2); at $t=1$, $m = 1+\sigma(\lambda_+-1)$, whose imaginary part is $\sigma\operatorname{Im}\lambda_+ > 0$ for $\sigma>0$. Hence

$$\Phi_\theta = x + \mathrm{i}\pi + R_\theta(t,\sigma), \qquad R_\theta = \log m + \log(1+t) - \log(N+1),$$

with R real-analytic in (t,σ,θ) up to $t=1$; similarly at $t=-1$. Since $dt/dx = -(1-t^2)/2$, one has $\partial_x\Phi = 1 - \frac12(1-t^2)\partial_t R \to 1$ and $\partial_\sigma\Phi \to (\lambda_+-1)/(1+\sigma(\lambda_+-1))$ as $t\to1$, whose imaginary part is positive. So $D\Phi_\theta$ is continuous and nonsingular on $[-\infty,\infty]\times[0,1]$, with constant orientation. Its Beltrami coefficient $(\partial_x\Phi + \mathrm{i}\partial_\sigma\Phi)/(\partial_x\Phi - \mathrm{i}\partial_\sigma\Phi)$ is continuous there and in θ, of modulus <1, and holomorphic in θ because $\Phi_\theta(x+\mathrm{i}\sigma)$ is. $\Phi(x) = \chi(t)$ and $\Phi(x+\mathrm{i}) = \chi(\mathsf{s}_\theta(t))$, so Φ respects the two identifications. The rest is as in Lemma 10.4(1). □

The marked point. Let $w_{-1} = 0$ and $w_n = E_\theta^n(1)$, $n\ge0$, and write $\zeta_n = \zeta(w_n)$. We say w lies *above the chord* if $\operatorname{Im}\zeta(w) > 0$; such points are not in H_θ. If $w_{-1},\ldots,w_{n-1}$ lie above the chord and $\zeta_n\in H_\theta\setminus A_\theta$, we call $n = n_\theta$ the *entry index* and the class $\mathsf{m}_\theta\in X_\theta$ of ζ_n the *marked point*. It is defined intrinsically.

Lemma 10.28 (First entry on the cone)**.** *For $\theta\in\Theta_{\mathrm{C}}$ the entry index exists.*

Proof. An analytic estimate of the orbit in the coordinate $\log\frac{\zeta-1}{\zeta+1}$; see Appendix A.1. □

Lemma 10.29 (The marked point on Θ)**.** *The entry index exists for every $\theta \in \Theta$, and $\theta \mapsto \mathsf{m}_\theta$ is a holomorphic section of the family X_θ near Θ: near each θ_0 it is represented by a fixed iterate $\zeta_n(\theta)$, either in the interior of H_θ or in the seam chart given by s_θ.*

Proof. On Θ_{C} this is Lemma 10.28. On Θ_{K} the orbit is enclosed on parameter boxes and the entry is certified box by box (Appendix A.1).

Near θ_0: if $\zeta_n(\theta_0)$ is in the interior of H, $n_\theta = n$ nearby. If $\zeta_n(\theta_0) \in (-1, 1)$, then $n_\theta \in \{n, n+1\}$ nearby, according to the side of the chord, and ζ_n and $\zeta_{n+1} = \mathsf{s}_\theta(\zeta_n)$ are the same point of X_θ in the seam chart. The conditions $\operatorname{Im}\zeta_m > 0$, $m < n$, are open. So m_θ is locally the class of the single holomorphic function $\zeta_n(\theta)$. □

Definition 10.30. For θ near Θ let $Z_\theta : X_\theta \to \mathbb{C}^*$ be the uniformisation taking the L_+-end to 0, the L_--end to ∞ and m_θ to 1, and let $z = (2\pi\mathrm{i})^{-1}\log Z_\theta$ be lifted so that m_θ has $z = n_\theta$. With $\tilde{F}_\theta$ the inverse, extended by $\tilde{F}(z+1) = E_\theta(\tilde{F}(z))$ as in Lemma 10.4(1), put

$$F_\theta^{\mathrm{far}}(z) = G_\theta(\tilde{F}_\theta(z + n_\theta)),$$

where G_θ is the branch of $E_\theta^{-n_\theta}$ along the orbit $w_{n_\theta} \mapsto \cdots \mapsto w_0 = 1$ (for $n_\theta = -1$, $G_\theta = E_\theta$). Then $F_\theta^{\mathrm{far}}(0) = 1$, and $F(z+1) = \mathrm{e}^{\mathcal{L}(\theta)F(z)}$ holds as an identity of germs, with the corresponding branches along $w_{n_\theta+1} \mapsto \cdots \mapsto w_1$.

The sign of the lift, i.e. that crossing the seam from the chord to A_θ adds $+1$, is locally constant in θ. At real θ it is Kneser's, by Proposition 10.32 below.

Proposition 10.31 (Holomorphic dependence)**.** *$(\theta, z) \mapsto F_\theta^{\mathrm{far}}(z)$ is jointly holomorphic on a neighbourhood of $(\Theta \setminus \{0\}) \times \{0\}$.*

Proof. This is the proof of Proposition 10.9, with M_ε replaced by N_θ in the coordinates of Lemma 10.27(4). Fix θ_0. Then $\mathcal{J}_\theta = \Phi_\theta \circ \Phi_{\theta_0}^{-1}$ is quasiconformal, respects the seams, and is holomorphic in θ at fixed source points. Its Beltrami coefficient υ_θ is holomorphic in θ and, by the uniform continuity in Lemma 10.27(4), is bounded by $\frac{1}{10}$ for θ near θ_0. Ahlfors–Bers and the $\partial_{\bar\theta}$-cancellation give a uniformisation Z_θ that is jointly holomorphic in (θ, ζ) in the interior of H_θ. Near a point ζ_0 of the chord, which does not move with θ in these coordinates, Z_θ is given by Z_θ below the chord and by $Z_\theta \circ \mathsf{s}_\theta$ above it. Both are continuous up to the chord, and they agree there. So $\theta \mapsto Z_\theta(\zeta)$ is a locally uniform limit of holomorphic functions for each ζ, and $Z_\theta(\zeta)$ is holomorphic in ζ by Morera's theorem. Hartogs' theorem then gives joint holomorphy near the chord. The marked point is holomorphic by Lemma 10.29. The inverse branches G_θ are finitely many logarithms along an orbit that avoids 0, holomorphic in θ. □

Proposition 10.32 (The real slice)**.** *For real $\theta \in (0, P_1]$, F_θ^{far} is Kneser's real-normalised solution $\mathcal{K}_{b(\theta)}$ near $z = 0$.*

Proof. For real θ, $\zeta^{-1}([-1, 1])$ is the chord γ_b and $\zeta^{-1}(A_\theta) = E_b(\gamma_b)$, so H_θ is Kneser's initial region H_b and X_θ is the cylinder of Proposition 10.11, with the upper end at L_+ sent to 0 in both. For real w, $\operatorname{Im}\zeta(w) = (\cos\theta - w/r)/\sin\theta$, which is positive exactly when $w < u_b = r\cos\theta$, and $H_b \cap \mathbb{R} = [u_b, r_b)$ with $r_b = E_b(u_b)$. The real orbit $0 = w_{-1} < 1 = w_0 < w_1 < \dots$ increases to $+\infty$. Let n be the first index with $w_n \geq u_b$. If $n \geq 0$ then $w_{n-1} < u_b$, so $w_n < E_b(u_b) = r_b$. If $n = -1$ then $u_b \leq 0 < r_b$. Either way $w_n \in [u_b, r_b)$, so $n = n_\theta$, and m_θ is the point that Proposition 10.11 represents by $w = 1$ through the real dynamical charts. Its Kneser coordinate is n_θ. The normalisations of the lifts agree, and the branches G_θ are the real logarithms. □

Theorem 10.33 (Crossing the upper boundary away from the cusp)**.** *Let $p \in (0, P_1]$, and continue Kneser's real-normalised solution $\mathcal{K}_b$ at $b = b(p) > \eta$ analytically in the base along $q \mapsto b(p + \mathrm{i}q)$, $0 \leq q \leq Q(p)$, with $\log b = \mathcal{L}(p + \mathrm{i}q)$ continued along the path.*

(1) *The continuation exists along the whole path and is holomorphic in a neighbourhood of it. The multiplier λ_+ of the continued fixed point crosses the unit circle once, at $q = q_c(p)$, where $b = \beta(p)$ lies on the upper Shell–Thron boundary arc; near $\beta(p)$ the continuation is a holomorphic family on a full neighbourhood of $\beta(p)$.*
(2) *For $q > q_c(p)$ the base lies in $\mathcal{U}^+$, $\mathcal{L} = \operatorname{Log} b$, and the continuation equals $F^{\mathrm{tr}}_{b(p+\mathrm{i}q)}$, the family of Theorem 10.22.*

In particular, along these paths Kneser's family crosses every point of the upper boundary arc $\{\exp(\mathrm{e}^{\mathrm{i}\alpha}\mathrm{e}^{-\mathrm{e}^{\mathrm{i}\alpha}}) : 0 < \alpha \le \alpha_1\}$, $\alpha_1 = 0.9353\ldots\pi$, holomorphically, and it enters the upper Shell–Thron region with the germ F^{tr}. Its boundary values on $(1,\eta)$ are therefore K_b^W, by Theorem 10.22. For $p \le P_2$ ($\alpha \le 0.7531\ldots\pi$) the whole path lies on the principal sheet, $|\operatorname{Im}\log b| < \pi$.

Proof. By Proposition 10.31 and Lemma 10.26(2), b is locally biholomorphic on a neighbourhood Θ' of $\Theta \setminus \{0\}$ (as is $\mathcal{L}$), and F^{far} is a holomorphic family of germs there. By Proposition 10.32 it equals $\mathcal{K}$ on the real segment. So along every path in Θ' starting on that segment it is the analytic continuation of $\mathcal{K}$ in the base, with the logarithm $\mathcal{L}$ continued along the path.

Next, $\mathcal{L}(\theta) = \mathrm{e}^{-1}(1+\theta^2/2+O(\theta^4))$, so $b'(\theta) = b\,\mathcal{L}'(\theta)$ has positive real part for small p on Θ_{C}, and there $\mathcal{L}$ is principal. Hence $\operatorname{Im} b(p+\mathrm{i}q) = \int_0^q \operatorname{Re} b'(p+\mathrm{i}q')\,dq' > 0$ for $q > 0$, and for p small the vertical fibre is carried into $D^+ \cup (\eta, \eta + r_5) \subset \Omega$. By Theorem 10.14(1) the continuation of $\mathcal{K}$ along it agrees with the family on Ω. By Theorem 10.22 that family equals F^{tr} on $\Omega \cap \mathcal{U}^+$. So $F^{\mathrm{far}} = F^{\mathrm{tr}} \circ b$ on a nonempty open subset of the connected set Θ^+ (interior), and hence on all of it, by the identity theorem. The remaining assertions are the description of Θ^+ and β after Lemma 10.26. □

Theorem 10.34 (Completion of the open upper boundary arc)**.** *For every $0.935\pi \le \alpha < \pi$, put*

$$\beta(\alpha) = \exp(\mathrm{e}^{\mathrm{i}\alpha}\mathrm{e}^{-\mathrm{e}^{\mathrm{i}\alpha}}).$$

Kneser's real-normalised family admits analytic continuation, on the principal logarithm sheet, from a real base $b > \eta$ to a full neighbourhood of $\beta(\alpha)$. This continuation crosses the upper Shell–Thron boundary holomorphically and agrees with F^{tr} on its inner side. Consequently the same assertion holds at every point of the open upper boundary arc $0 < \alpha < \pi$. The endpoint $\beta(\pi) = \mathrm{e}^{-\mathrm{e}}$ is excluded.

Corollary 10.35 (Entering from outside)**.** *There is an open neighbourhood N of the open upper boundary arc $\{\beta(\alpha) : 0 < \alpha < \pi\}$ and a single-valued holomorphic family of germs on $N \cup \mathcal{U}^+$ that equals F^{tr} on $\mathcal{U}^+$. On $N \setminus \overline{\mathcal{U}^+}$ it is the continuation of Kneser's solution along paths that stay outside the closed Shell–Thron region on the principal sheet. Thus for every $0 < \alpha < \pi$ Kneser's family, continued from (η, ∞) through the exterior, reaches $\beta(\alpha)$ and enters $\mathcal{U}^+$ with the germ F^{tr}.*

Proof. Theorems 10.33 and 10.34 give, about each point of the open arc, a disc on which the continuation is holomorphic and equals F^{tr} on the inner side. Two such discs meet in a connected set containing inner points, so the local families agree on overlaps, and F^{tr} is single-valued on $\mathcal{U}^+$. Take N to be the union of the discs, shrunk so that $N \setminus \overline{\mathcal{U}^+}$ is a connected band along the arc. For p small the vertical path of Theorem 10.33 runs from the real exterior ray to this band outside $\overline{\mathcal{U}^+}$, on the principal sheet. Moving along the band then reaches the outer side of every $\beta(\alpha)$, and $|\operatorname{Im}\mathcal{L}| < \pi$ near the arc because $|\mathcal{L}| \le \mathrm{e}$ there. □

Proof. By Appendix A.2, λ_+ is injective on $R = [2.19, 2.31] + \mathrm{i}[0.71, 0.78]$, the branch $\theta(\alpha) \in R$ of $\lambda_+(\theta) = \mathrm{e}^{\mathrm{i}\alpha}$, $0.935\pi \le \alpha < \pi$, satisfies the inequalities of Lemma 10.26(1) on $(-1, 1)$ on a complex neighbourhood, and $w_{-1} = 0$ lies in the crescent, so the entry index is -1. Lemma 10.27

and Proposition 10.31 apply on a complex parameter neighbourhood of every $\alpha < \pi$. Also $\lambda'_+ \neq 0$ and $\mathcal{L}' = e^{-\lambda_+}(1-\lambda_+)\lambda'_+ \neq 0$, so $b = e^{\mathcal{L}}$ is a valid local parameter. These neighbourhoods carry one holomorphic family of marked crescent quotients: the inverse branch is unique on R, and their constructions agree on overlaps.

At $\alpha = 0.935\pi < \alpha_1$ this is exactly the crescent and the first mark of Theorem 10.33. The families therefore agree in an open parameter overlap. Analytic continuation along the connected arc, followed by the identity theorem on the inner overlaps, identifies the new family with F^{tr} wherever it meets $\mathcal{U}^+$.

Finally choose $p \le P_2$ in Theorem 10.33, follow its vertical path from the real exterior ray into $\mathcal{U}^+$, then follow a path in $\mathcal{U}^+$ to the inner side of $\beta(\alpha)$ and use the local crescent family to cross the boundary. The first segment is on the principal sheet by Lemma 10.26(4). On $\mathcal{U}^+$ the logarithm is $\mathcal{L} = \lambda e^{-\lambda} = \operatorname{Log} b$; at the target boundary $|\mathcal{L}| \le e < \pi$, so the last short segment is principal as well. This gives the asserted principal-sheet path and completes the proof. □

Remark 10.36. (1) The boundary point $b_c \approx 1.5217 + 0.0148\mathrm{i}$ ($\alpha = 2\pi(\sqrt{2}-1)/5$) tested numerically in Section 11.6 lies on the arc covered here. The smooth join observed there agrees with the holomorphic crossing proved here. (2) The endpoint $b = e^{-e}$ itself is treated in Section 10.12. There the crescent keeps its angle $\arg\lambda_+ \to \pi$ at L_+; only the interpolation used here degenerates. (3) The identification with the family of [25] is completed in Section 10.11 (Corollary 10.41).

10.11. **The whole upper half-plane.** Theorems 10.22, 10.33 and 10.34 describe the continuation inside the upper Shell–Thron region and across its boundary arc. Here we show that Kneser's straight chord also carries the family over the whole exterior part of the upper half-plane. So the continuation is a single holomorphic family on $\mathbb{H} = \{\operatorname{Im} b > 0\}$, and there is no path dependence.

Write $\overline{\mathcal{ST}}$ for the closed Shell–Thron region, $g(\lambda) = \lambda e^{-\lambda}$, and

$$\mathcal{E}^+ = \mathbb{H} \setminus \overline{\mathcal{ST}}, \qquad \mathbb{H} = \mathcal{E}^+ \cup \{\beta(\alpha) : 0 < \alpha < \pi\} \cup \mathcal{U}^+.$$

On $\mathbb{H}$ we use the principal logarithm $\mathcal{L} = \operatorname{Log} b$, $0 < \operatorname{Im}\mathcal{L} < \pi$.

Lemma 10.37 (The exterior parameter)**.** *$\mathcal{E}^+$ is simply connected. The fixed point L_+ of E_b continued from the real exterior ray (the one with $\operatorname{Im} L_+ > 0$ there) is single-valued and holomorphic on $\mathcal{E}^+$. Its multiplier $\lambda(b)$ maps $\mathcal{E}^+$ biholomorphically onto a domain Λ, with inverse $\lambda \mapsto \exp g(\lambda)$, and*

$$\Lambda \subset \{\lambda :\ |\lambda| > 1,\ 0 < \operatorname{Im}\lambda < \pi,\ 0 < \operatorname{Im} g(\lambda) < \pi,\ g(\lambda) \notin g(\overline{\mathbb{D}})\}.$$

Proof. $K = \overline{\mathcal{ST}} \cap \overline{\mathbb{H}}$ is a closed topological disc: it is bounded by the Jordan arc β and the segment $[e^{-e}, \eta]$. The complement of $\mathcal{E}^+$ in the Riemann sphere, $(\hat{\mathbb{C}} \setminus \mathbb{H}) \cup K$, is connected, so $\mathcal{E}^+$ is simply connected. A fixed point of E_b is multiple only if its multiplier is 1, i.e. $\mathcal{L} = 1/e$, which does not occur for $0 < \operatorname{Im}\mathcal{L} < \pi$. So L_+ continues without monodromy. $L_+ = e^{\lambda}$ and $\mathcal{L} = g(\lambda)$ (Lemma 2.1), so $b = \exp g(\lambda)$ and λ is injective. All fixed points of E_b, $b \in \mathcal{E}^+$, are repelling, so $|\lambda| > 1$ and $g(\lambda) = \mathcal{L} \notin g(\overline{\mathbb{D}})$. If $\operatorname{Im}\lambda = 0$ then $\mathcal{L}$ is real; if $\operatorname{Im}\lambda = \pi$ then $\operatorname{Im}\mathcal{L} = -\pi e^{-\operatorname{Re}\lambda} < 0$. Neither happens on $\mathcal{E}^+$, and near the real ray $\lambda = \theta\cot\theta + \mathrm{i}\theta$ with small $\theta > 0$; by connectedness $0 < \operatorname{Im}\lambda < \pi$. □

For $\lambda \in \Lambda$ let $\theta = \theta(\lambda) \neq 0$ be a root of $(\lambda - 2\mathrm{i}\theta)e^{2\mathrm{i}\theta} = \lambda$. (The root $\theta = 0$ is always present and is excluded.) For $\theta \neq 0$ the equation is equivalent to $\lambda_+(\theta) = \lambda$. Then $\lambda - 2\mathrm{i}\theta$ is the multiplier of a second fixed point $L_- = e^{\lambda - 2\mathrm{i}\theta}$, so the chord construction of Section 10.10 is defined with this θ. g has no critical point on Λ, so λ is a local holomorphic parameter.

Lemma 10.38 (Scope of Section 10.10)**.** *Let Θ'' be an open set of θ with $0 < |\theta| < \pi$ on which the inequalities of Lemma 10.26(1) hold for $-1 < t < 1$, $\operatorname{Im}\lambda_+ > 0 > \operatorname{Im}\lambda_-$, and the entry index of the marked point exists. Then Lemma 10.27, the local statement of Lemma 10.29, Definition 10.30 and Proposition 10.31 hold on Θ''.*

Proof. Those proofs use only these properties and the injectivity of s_θ on a neighbourhood of $[-1,1]$. The injectivity follows from $|\theta| < \pi$: $\mathsf{s}_\theta(\zeta_1) = \mathsf{s}_\theta(\zeta_2)$ forces $\theta(\zeta_1 - \zeta_2) \in 2\pi\mathbb{Z}$, and $|2\pi/\theta| > 2$. The bound $|\theta| < 2.45$ in Lemma 10.27 was used only for this. At the ends, $\operatorname{Im}\lambda_\pm$ play the roles of $\operatorname{Im} g(\pm 1)$ in part (4). □

Proposition 10.39 (Certified covering of Λ)**.** *There is a continuous choice of $\theta(\lambda)$ on Λ, real on the image of the real exterior ray, such that for every $\lambda \in \Lambda$ the inequalities of Lemma 10.26(1) hold on $(-1,1)$, $\operatorname{Im}(\lambda - 2\mathrm{i}\theta) < 0$, $|\theta| < \pi$, and the entry index of the marked point exists. The certificate* `certify_upper_halfplane.py` *has four parts (tails, core, cusp, interfaces), described in Appendix A.3.*

Proof. The listed checks are outward-rounded Arb computations. Every point of Λ lies in a certified box, because of the inclusion in Lemma 10.37. The roots are simple, since ∂_θ of the equation is $2\mathrm{i}\mathrm{e}^{2\mathrm{i}\theta}(\lambda - 2\mathrm{i}\theta - 1)$ and $\lambda - 2\mathrm{i}\theta \neq 1$ (L_- is repelling). So the certified local roots, which agree on all overlaps, define one continuous branch. □

Theorem 10.40 (Kneser's tetration on the upper half-plane)**.** *Kneser's real-normalised solution $\mathcal{K}_b$, $b > \eta$, continues analytically along every path in $\mathbb{H} = \{\operatorname{Im} b > 0\}$, with the principal logarithm. The result is a single-valued family of germs at $z = 0$, jointly holomorphic in (b, z). It is given on $\mathcal{E}^+$ by the straight-chord construction with the marked point of Section 10.10, on $\mathcal{U}^+$ by F^{tr}, and it is holomorphic across the open upper boundary arc. In particular, its boundary values on $(1, \eta)$ are K_b^W, whatever path in $\mathbb{H}$ is used. The continuation through the lower half-plane is the complex conjugate family.*

Proof. By Proposition 10.39 and Lemma 10.38 (with λ as parameter), the chord construction with $\theta(\lambda)$ and the first-entry mark gives a holomorphic family on Λ, hence on $\mathcal{E}^+$. Near the image of the real ray it is Kneser's solution, by Proposition 10.32, since there θ is real. Since $\mathcal{E}^+$ is connected, the family is the analytic continuation of $\mathcal{K}$ along every path in $\mathcal{E}^+$. Here the image Γ of the real ray lies in $\partial\Lambda$. The certified cells straddle it, $\theta(\lambda)$ extends holomorphically across Γ, it is real there by the anchor in (I), and Proposition 10.32 applies. Shrink the neighbourhood N of Corollary 10.35 into $\mathbb{H}$, so that $N \cap \mathcal{E}^+$ is the connected outer band. By that corollary, near the open arc this continuation extends holomorphically across the arc and equals F^{tr} on the inner side; F^{tr} is single-valued on $\mathcal{U}^+$ by Theorem 10.22. The three pieces agree on their overlaps, so they define one holomorphic family on $\mathbb{H}$. Since $\mathbb{H}$ is simply connected, analytic continuation of $\mathcal{K}$ along any path in $\mathbb{H}$ gives it. The boundary values on $(1, \eta)$ are those of Theorem 10.22. For the lower half-plane use $\overline{\mathcal{K}_{\bar b}(\bar z)} = \mathcal{K}_b(z)$. □

Corollary 10.41 (Paulsen's family)**.** *Suppose the family of [25] is understood, as in its definition, as the analytic continuation in b of Kneser's solution from (η, ∞), along paths in the upper half-plane and with the principal logarithm. Then it is the family of Theorem 10.40. In particular it enters the Shell–Thron region with the germ F^{tr} across every point of the open boundary arc. Its boundary values on $(1, \eta)$ are K_b^W, and the separation limits of Corollary 10.15 hold for it.*

10.12. **The endpoint $b = \mathrm{e}^{-\mathrm{e}}$ and the real segment below 1.** At $b = \mathrm{e}^{-\mathrm{e}}$ the fixed point L_+ has multiplier $\lambda_+ = -1$. In Theorem 10.34 the straight crescent seemed to degenerate there, because $f_1(1) = -\frac{1}{2}\operatorname{Im}\lambda_+ \to 0$. The crescent itself does not degenerate: its angle at L_+ is $\arg\lambda_+ \to \pi$. What degenerates is the interpolation $t + \sigma(\mathsf{s}_\theta(t) - t)$, whose factor $1 + \sigma(\lambda_+ - 1)$

vanishes at $\sigma=\frac12$. In the coordinate of the pair of fixed points the end is harmless, and the crescent can be continued through $\lambda_+=-1$ and across the real axis.

Put $\chi=\log\frac{\zeta-1}{\zeta+1}$, so that the chord is $\{\operatorname{Im}\chi=\pi\}$, with $x=\log\frac{1-t}{1+t}$. The image arc is $\Gamma(x)=x+\mathrm{i}\pi+\varphi(x)$, where, with the entire function $S(w)=\sinh w/w$,

$$\varphi(t)=\mathrm{i}\theta+\log S\big(\tfrac{\mathrm{i}\theta(t-1)}{2}\big)-\log S\big(\tfrac{\mathrm{i}\theta(t+1)}{2}\big),$$

with principal logarithms where $\operatorname{Re}S>0$. Then $\mathrm{e}^{\varphi(1)}=\lambda_+$ and $\mathrm{e}^{-\varphi(-1)}=\lambda_-$, and under (b) below $\varphi(\pm1)$ are the logarithms with imaginary part in $(0,2\pi)$. This follows from $\mathsf{s}_\theta(t)\mp1=\mathrm{e}^{\pm\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta(t\mp1)}-1)/(\mathrm{i}\sin\theta)$. We write $\varphi_x=d\varphi/dx=-\frac12(1-t^2)\,d\varphi/dt$, which is regular at $t=\pm1$. The mark has the closed form $\chi(\zeta(0))=\log\frac{\mathrm{i}\cot\theta-1}{\mathrm{i}\cot\theta+1}=2\mathrm{i}\theta\pmod{2\pi\mathrm{i}}$.

Lemma 10.42 (The crescent in the pair coordinate)**.** *Let Θ'' be an open set of θ on which, for $-1\le t\le1$:*

(f) *$|\theta|<\pi$, and $\operatorname{Re}S>0$ at both arguments above;*
(a) *$\operatorname{Re}(1+\varphi_x)>0$;*
(b) *$0<\operatorname{Im}\varphi<2\pi$;*
(c) *$\operatorname{Im}(\overline{(1+\varphi_x)}\,\varphi)>0$;*
(d) *$\operatorname{Im}\varphi<\pi$ wherever $\operatorname{Re}\Gamma=0$;*
(e) *the point $2\mathrm{i}\theta$ ($=\chi(\zeta(0))$, $\zeta(0)=\mathrm{i}\cot\theta$), which must have imaginary part in $(\pi,3\pi)$, lies in the strip between $\operatorname{Im}\chi=\pi$ and Γ.*

Then the region H_θ between the chord and $\mathsf{s}_\theta([-1,1])$ is a Jordan region, and s_θ maps the outer side of the chord to the inner side of the arc. $M(x,\sigma)=x+\mathrm{i}\pi+\sigma\varphi(x)$ is a flat model of the quotient with the properties of Lemma 10.27(4) (with M in place of N_θ), $w_{-1}=0$ lies in H_θ (entry index -1), and Definition 10.30 and Proposition 10.31 apply on Θ''. Parts (1) and (2) of Lemma 10.27 are not asserted. The arc need not lie in the lower half-plane, and it may leave $\zeta=1$ on the side $\zeta>1$.

Proof. By (a) Γ is a graph over $\operatorname{Re}\chi$. By (b) and (c) the map M is a local diffeomorphism. Its Jacobian is $\operatorname{Im}\varphi+\sigma\operatorname{Im}(\overline{\varphi_x}\varphi)$, linear in σ, and M is bijective on the two boundary lines. So, as in Lemma 10.3(3), it is a diffeomorphism onto the strip between $\operatorname{Im}\chi=\pi$ and Γ. By (b) the strip embeds in $\mathbb{C}/2\pi\mathrm{i}\mathbb{Z}$. By (d) it avoids $\chi\equiv0$, which corresponds to $\zeta=\infty$. So its image under $\zeta=(1+\mathrm{e}^\chi)/(1-\mathrm{e}^\chi)$ is a bounded Jordan region bounded by the chord and the arc.

As $x\to\mp\infty$, $\varphi\to\log\lambda_+$ and $\varphi\to-\log\lambda_-$ exponentially fast. Both limits have imaginary part in $(0,2\pi)$ by (b), so M is asymptotically affine and nondegenerate at both ends. This holds also for $\lambda_+=-1$, where $\operatorname{Im}\log\lambda_+=\pi$. The quotient of the strip is therefore quasiconformally $\mathbb{C}/\mathrm{i}\mathbb{Z}$, with punctures at both ends, and M is holomorphic in θ at fixed (x,σ).

The seam map s_θ is injective near $[-1,1]$ because $|\theta|<\pi$. In the coordinate χ the map s_θ sends the outward normal $-\mathrm{i}$ of the chord to $-\mathrm{i}(1+\varphi_x)$. The direction from Γ into the strip is $-\varphi=-\partial_\sigma M$. By (c) both have $\operatorname{Im}(\overline{(1+\varphi_x)}\,v)<0$, so they lie on the same side of Γ: the outer side of the chord goes to the inner side of the arc. By (e), $w_{-1}=0$ lies in the crescent. Since -1 is the smallest index, the entry index is -1, and no condition on earlier points is needed. The proofs of Lemma 10.27(4), Definition 10.30 and Proposition 10.31 use only these facts. The quotient does not depend on the interpolation, so on overlaps with Section 10.10 the two constructions give the same family. □

Theorem 10.43 (Holomorphy at $b=\mathrm{e}^{-\mathrm{e}}$)**.** *The family of Theorem 10.40 extends holomorphically to a neighbourhood of the real segment*

$$[\exp(-12\mathrm{e}^{12}),\ \exp(-\tfrac15\mathrm{e}^{1/5})]\approx[\mathrm{e}^{-1.95\cdot10^6},\ 0.7833],$$

which contains $b = \mathrm{e}^{-\mathrm{e}}$. At every base of this neighbourhood, including $b = \mathrm{e}^{-\mathrm{e}}$, it is given by the straight-chord construction with the mark $w_{-1} = 0$. In particular its boundary values on this segment are real-analytic.

Proof. By Appendix A.4, (a)–(f) of Lemma 10.42 hold for all λ in the square $[-1.1, -0.9] \times [-0.1, 0.1]$ and the strip $[-12, -\frac{1}{5}] \times [-10^{-3}, 10^{-3}]$, with one branch $\theta(\lambda)$, which lies in $R' = [2.20, 2.45] \times [0.62, 0.95]$ on the square. All checks are strict, so they persist on an open neighbourhood of the closed square and strip. For $\lambda = \mathrm{e}^{\mathrm{i}\alpha}$ near -1 the root in R' is the one used in Theorem 10.34. Indeed, that theorem's contraction disc of radius 0.00019 about θ_* lies in $R \cap R'$, and λ_+ is injective on R'. So the two roots agree near $\alpha = \pi$, and by continuity on the whole arc inside the square. So on the upper part of the square the new family is that of Theorem 10.34. It therefore equals F^{tr} inside and the family of Theorem 10.40 outside. The whole strip cannot be compared with Theorem 10.40 in the base variable, because $\operatorname{Im} g(x + \mathrm{i}y) \approx y(1 - x)\mathrm{e}^{-x}$ is large for $x \ll 0$. Let U be the tube of points of the strip near $[-12, -\frac{1}{5}]$ with $|\operatorname{Im} g| < \pi$; its width is about $\pi \mathrm{e}^x/(1 - x)$. On the real segment g is real, strictly increasing and has $g' = (1 - \lambda)\mathrm{e}^{-\lambda} > 0$. So $b = \exp g$ is injective on U, after shrinking it, and maps it onto a neighbourhood of the stated b-segment, with $b(-1) = \mathrm{e}^{-\mathrm{e}}$. $U \cap \{\operatorname{Im} \lambda > 0\}$ is connected, maps into $\mathbb{H}$ with the principal logarithm, and contains the upper part of the square near $\lambda = -1$. By the identity theorem the new family agrees there with the family of Theorem 10.40. Composed with the inverse of $\exp \circ g$ on U, it gives the holomorphic extension. □

Remark 10.44. (1) The extension across the segment is the continuation of the upper family. It need not agree with the continuation through the lower half-plane, which is the complex-conjugate family. At real b in the segment the partner fixed point L_- is not real, so the construction is not symmetric under conjugation. (2) For $b \in (1, \eta)$ the partner is real: θ is purely imaginary, the image arc falls onto the chord, and the straight crescent collapses. There the tilted regions of Section 10 are needed. Near $b = 1$ ($\lambda_+ \to 0^-$) the root θ goes to infinity. The interval towards $b = 0$ is covered instead by transport through the attracting two-cycle (Theorem 10.45).

10.13. **The real slice below the period-doubling point.** The interval below $\mathrm{e}^{-\mathrm{e}}$ admits a different transport coordinate: the multiplier of the attracting two-cycle. This avoids a uniform straight-crescent estimate as $b \downarrow 0$.

Theorem 10.45 (Continuation across the real slice below η)**.** *The family of Theorem 10.40 extends jointly holomorphically in (b, z) across every $b \in (0, 1)$. Together with Theorem 10.22, it extends across every point of $(0, \eta) \setminus \{1\}$. The extensions are locally single-valued and agree wherever their domains overlap in the upper half-plane.*

Proof. Proposition 10.23 covers $(\mathrm{e}^{-\mathrm{e}}, 1)$, Theorem 10.43 covers $\mathrm{e}^{-\mathrm{e}}$, and Proposition 3.5 covers $(1, \eta)$ by the identification in the proof of Theorem 10.22. We construct the extension on $(0, \mathrm{e}^{-\mathrm{e}})$.

Write $b = \mathrm{e}^{-\mathcal{M}}$, $\mathcal{M} > \mathrm{e}$, and let $p < q$ be the real two-cycle of $E_b(w) = \mathrm{e}^{-\mathcal{M}w}$. Its multiplier $m = \mathcal{M}^2 pq$ has the parametrisation

$$\mathcal{M}p = u(\coth u - 1), \qquad \mathcal{M}q = u(\coth u + 1), \qquad \mathcal{M} = \frac{u}{\sinh u}\mathrm{e}^{u \coth u}, \qquad m = \left(\frac{u}{\sinh u}\right)^2, \quad u > 0.$$

Indeed $\mathcal{M}q/\mathcal{M}p = \mathrm{e}^{2u}$ and $\mathcal{M}q - \mathcal{M}p = 2u$, which give $p = \mathrm{e}^{-\mathcal{M}q}$ and $q = \mathrm{e}^{-\mathcal{M}p}$. Moreover

$$\frac{d \log \mathcal{M}}{du} = \frac{1}{u} - \frac{u}{\sinh^2 u} > 0, \qquad \frac{d \log m}{du} = 2\left(\frac{1}{u} - \coth u\right) < 0.$$

Thus $\mathcal{M}$ increases from e to $+\infty$, while m decreases from 1 to 0. In particular m is a real-analytic coordinate on the whole interval $(0, \mathrm{e}^{-\mathrm{e}})$, with nonzero derivative.

Choose $b_* = \exp(-2\mathrm{e}^2)$, which is in the certified segment of Theorem 10.43, and let m_* be its two-cycle multiplier. Denote by B_p, B_q the two completely invariant basins for $E_{b_*}^2$, distinguished by convergence to p, q, and by σ_p, σ_q their Koenigs coordinates, normalised to have derivative 1 at the cycle. They satisfy $\sigma_j \circ E_{b_*}^2 = m_* \sigma_j$ and have nonzero derivative on their respective basins. On a simply connected domain containing $(0,1)$ and contained in $\{0 < |m| < 1\}$, take the logarithm real on $(0,1)$ and set

$$c(m) = \frac{\operatorname{Log} m - \log m_*}{\operatorname{Log} m + \log m_*}, \qquad \nu_m = c(m) \frac{\sigma_j}{\overline{\sigma_j}} \frac{\overline{\sigma_j'}}{\sigma_j'} \quad \text{on } B_j \setminus \sigma_j^{-1}(0),$$

and $\nu_m = 0$ elsewhere. Since $\operatorname{Re} \operatorname{Log} m < 0$, $|c(m)| < 1$. The relation $\sigma_q \circ E_{b_*} = E_{b_*}'(p)\sigma_p$ on B_p, and its counterpart on B_q, show that ν_m is E_{b_*}-invariant. The normalised solution $\mathcal{Q}_m$ fixing $0, 1, \infty$ therefore conjugates E_{b_*} to an entire universal cover of $\mathbb{C}^*$ taking 0 to 1, hence to $E_{b(m)}$ for a holomorphic $b(m) = \mathcal{Q}_m(b_*)$. The radial-stretch calculation of Proposition 10.18, applied to $E_{b_*}^2$, shows that the transported two-cycle has multiplier m. For real $m \in (0,1)$ the coefficient is conjugation-symmetric, so $\mathcal{Q}_m$ preserves the real axis in order. In particular $\mathcal{Q}_m(p), \mathcal{Q}_m(q) \in (0,1)$ are a distinct real two-cycle and $b(m) > 0$. The parametrisation above then gives $b(m) = \exp(-\mathcal{M}(u(m))) \in (0, \mathrm{e}^{-\mathrm{e}})$ and $b'(m) \neq 0$.

Let H_* be the marked straight crescent at b_* from Theorem 10.43; its mark is $w_{-1} = 0$. The quotients of $\mathcal{Q}_m(H_*)$, with the edges identified by $E_{b(m)}$, are quasiconformally equivalent to the twice-punctured quotient of H_*. They have the transported fixed-point ends and the same mark 0. The Beltrami coefficient descends to the quotient as $c(m)$ times a fixed coefficient. Ahlfors–Bers uniformisation and the $\partial_{\bar{m}}$-cancellation of Proposition 10.19 give a jointly holomorphic family of normalised germs, locally in (m, z): give the mark height -1 and compose the inverse uniformisation with $E_{b(m)}$, as in Definition 10.30, so that $F_m(0) = E_{b(m)}(0) = 1$. At the mark the coefficient is real-analytic in space because 0 belongs to a two-cycle basin and $\sigma_j(0) \neq 0$.

These germs agree with the straight-crescent family near b_*. Here is the local comparison. At $m = m_*$ the two marked crescents are identical. On a compact part away from the fixed-point ends, their edges are close and the orbit-selection argument of Lemma 10.8 gives a conformal map of quotients by locally fixed iterates of $E_{b(m)}$. At either end, use a repelling linearising coordinate and its logarithm. A lift of $\mathcal{Q}_m$ commutes with the deck translation $2\pi\mathrm{i}$ and conjugates translation by $\log \lambda_j(b_*)$ to translation by $\log \lambda_j(b(m))$. Consequently it is a real-linear map, tending to the identity as $m \to m_*$, plus a bounded doubly periodic remainder tending uniformly to zero on a fundamental parallelogram. In the logarithmic coordinate the transported and straight edges differ by at most $O(|m - m_*|(1 + |\operatorname{Re} t|))$, whereas a dynamical iterate changes their normal coordinate by a quantity bounded away from zero. At most $O(|m - m_*|(1 + |\operatorname{Re} t|))$ iterates select the unique point in the other half-open crescent; the total tangential displacement is of the same order and remains within the linearising collar after shrinking $|m - m_*|$. This is the end-collar argument of Lemma 10.21, now with repelling rather than neutral ends. These orbit maps agree on overlaps, preserve the ends and the mark, and identify the normalised inverse uniformisations. Thus the two families agree on an open neighbourhood of b_*.

Finally, $b'(m) > 0$ on the real interval. Take overlapping complex neighbourhoods of its points on which $b(m)$ is biholomorphic and whose upper halves map into the upper base half-plane. The transported family and that of Theorem 10.40 agree on the first such neighbourhood by the preceding comparison, hence on every successive overlap by the identity theorem. This gives the asserted extension along the whole interval. □

The proof of Theorem 10.22 gives more than boundary values on $(1, \eta)$: on $N_W \cap \{\operatorname{Im} b > 0\}$ it identifies the upper family with the jointly holomorphic sewn family of Proposition 3.5. Thus the upper continuation already extends holomorphically across every base in $(1, \eta)$.

Proposition 10.46 (Obstructions at $b = 1$ and $b = 0$)**.** *Let $W \subset \mathbb{C}$ be a connected height domain with $0 \in W$ and $W + 1 \subset W$. No normalised family $F(b, z)$ satisfying $F(b, 0) = 1$ and $F(b, z+1) = \exp((\operatorname{Log} b)F(b, z))$ can be jointly holomorphic on a product of W with a disc about $b = 1$. Moreover, for any such family on upper-half-plane bases approaching $b = 0$, the height value $F(b, 2)$ has no holomorphic extension through $b = 0$.*

Proof. Suppose first that the family extended jointly holomorphically across $b = 1$. At $b = 1$ the functional equation gives $F(1, z + 1) = 1$ for $z \in W$. Since $W + 1$ is a nonempty open subset of the connected domain W, the identity theorem yields $F(1, z) \equiv 1$ on W. Put $H(z) = \partial_b F(1, z)$. Differentiating the functional equation at $b = 1$ now gives $H(z + 1) = F(1, z) = 1$. The identity theorem again gives $H \equiv 1$ on W, whereas $F(b, 0) = 1$ gives $H(0) = 0$, a contradiction.

For the second assertion, the functional equation and normalisation give $F(b, 1) = b$ and $F(b, 2) = \exp(b \operatorname{Log} b)$. Along $b = \mathrm{i}t$, $t \downarrow 0$, this value tends to 1, but its derivative $\exp(b \operatorname{Log} b)(1+\operatorname{Log} b)$ is unbounded. Hence the upper-half-plane germ has no holomorphic extension through 0. □

10.14. **Consequences at the cusp.** The following statements describe the sewn family for real $b < \eta$, mostly close to η, and hence, by Theorem 10.14(3), the boundary values K_b of the continued Kneser family.

Corollary 10.47 (Non-reality at fractional real heights)**.** *For every real $b < \eta$ sufficiently close to η, and every $\delta > 0$ sufficiently small, some real x with $0 < |x| < \delta$ satisfies $\operatorname{Im} K_b^W(x) \neq 0$. In particular the sewn solution is not real-valued on any real interval about $z = 0$.*

Proof. Suppose that $K_b^W(x)$ were real for all real x in a neighbourhood of 0. Shrink that neighbourhood so its values lie in a real interval about 1 on which the attracting Koenigs coordinate σ_b is holomorphic and real, with $\sigma_b(K_b^W(x))/\sigma_b(1) > 0$. The upper representation of Theorem 6.3 gives

$$\frac{\sigma_b(K_b^W(x))}{\sigma_b(1)} = \lambda_b^{x+P_b(x)}, \qquad P_b(0) = 0.$$

Since $\log \lambda_b$ is real and nonzero, continuity at $x = 0$ makes $\operatorname{Im} P_b(x) = 0$ on this interval. The function P_b is holomorphic and periodic on a strip containing the whole real line, so reflection and the identity theorem make it real on that line. But its Fourier expansion has only non-negative modes. A real-valued periodic function has conjugate positive and negative coefficients; therefore all its nonconstant coefficients vanish. This contradicts $c_1(b) \neq 0$ for b sufficiently close to η, established in Corollary 6.4. □

By Theorem 10.14, this non-reality statement applies to the continued Kneser family $K_b = K_b^W$ near η. The tiny non-real part observed numerically by Paulsen at $b = \sqrt{2}$ [25, §6] is reproduced by K^W in all nine quoted digits (Section 11.1); this is numerical, not a proof that his continuation path gives K^W.

Corollary 10.48 (Complex cusp profile and its imaginary trace)**.** *Let $A = B_1 \mathrm{e}^{2\pi \mathrm{i} a} \neq 0$ and put $f = F_\eta^{\mathrm{att}}$. On some fixed complex disk about $z = 0$, as $b \to \eta^-$ through real bases,*

$$\frac{K_b^W(z) - R_b(z)}{\Lambda(b)} \longrightarrow f'(z) A(\mathrm{e}^{2\pi \mathrm{i} z} - 1)$$

locally uniformly, hence with all z-derivatives. In particular, on a fixed real interval about 0,

$$\frac{\operatorname{Im} K_b^W(x)}{\Lambda(b)} \longrightarrow f'(x) \operatorname{Im}\Big[A(\mathrm{e}^{2\pi \mathrm{i} x} - 1)\Big]$$

uniformly on compact subintervals. The right-hand side is not identically zero on any interval about 0.

Proof. By Theorem 6.3(4), with $B = e^{2\pi(Y_0+D+2)}$, there is a constant C independent of b and n such that $|c_n(b)| \le C(\Lambda B)^n$ for all $n \ge 1$ and b sufficiently close to η. Here we used $|\tau_n| \le M e^{2\pi n(Y_0+1)}$ and absorbed the factor Λ in the error bound into C. Since $P_b(0) = 0$, its constant mode is $-\sum_{n\ge1} c_n$. Choose a disk in the common upper representation supplied by Corollary 6.5. The Fourier tail is geometric on every smaller disk once b is close enough to η, and there

$$P_b(z) = c_1(b)(e^{2\pi i z} - 1) + O(\Lambda^2)$$

locally uniformly. Theorem 6.3(4) and Corollary 4.5 give $c_1/\Lambda \to A$. On a fixed disk about 0, $R_b \to F_\eta^{\mathrm{att}}$ with its derivatives by Corollary 6.5 and its proof. Taylor's formula gives $K_b^W(z) = R_b(z) + R_b'(z)P_b(z) + O(\Lambda^2)$ on a smaller disk. This proves the complex limit, with derivative convergence by Cauchy's formula. For real x, $R_b(x)$ and $f'(x)$ are real, giving the imaginary trace. If its last factor vanished on an interval, both the first and second derivatives at $x = 0$ would vanish, forcing $\operatorname{Re} A = \operatorname{Im} A = 0$, contrary to $A \ne 0$. □

Corollary 10.49 (Obstruction at the real cusp)**.** *For every sufficiently small complex neighbourhood V of 0, one can choose a real $x_* \in V \setminus \{0\}$ and constants $c, \delta > 0$ such that*

$$|\operatorname{Im} K_b^W(x_*)| \ge c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

There is therefore no jointly holomorphic family $F(b, z)$ on a product of a disc about $b = \eta$ and V which is real at real heights for real $b > \eta$ and agrees with K_b^W for all real $b < \eta$ sufficiently close to η. In particular, direct holomorphic continuation of a real-normalised Kneser family through the real cusp cannot identify it with the inner sewn branch.

Proof. The limiting profile in Corollary 10.48 is not identically zero on any real interval about 0. Choose x_* where it is nonzero. Since $\Lambda(b) > 0$ for real $b < \eta$, the uniform limit gives the stated lower bound after reducing δ. For fixed real x_*, Schwarz reflection in the parameter says that a holomorphic function $b \mapsto F(b, x_*)$ which is real on a real interval above η is also real on the real interval below η in the same disc. This contradicts the lower bound if $F = K^W$ there. □

Proposition 10.50 (Logarithmic monodromy at the first backward singularity)**.** *Let U be a simply connected parameter disc on which $\mathcal{L}(b) = \log b \ne 0$. Suppose $F_b(z)$ is jointly holomorphic near $U \times \{-1\}$, $F_b(-1) = 0$, and $F_b'(-1) \ne 0$. Locally in (b, t) write $F_b(-1 + t) = t\, u_b(t)$ with $u_b(0) \ne 0$. On any simply connected sector $0 < |t| < r$, all holomorphic backward branches satisfying $\exp(\mathcal{L}(b)G_b(-2 + t)) = F_b(-1 + t)$ have the form*

$$G_b(-2 + t) = \frac{\operatorname{Log} t + \log u_b(t) + 2\pi i k}{\mathcal{L}(b)}, \qquad k \in \mathbb{Z}.$$

Continuation once around $t = 0$ adds $2\pi i/\mathcal{L}(b)$. Hence no single-valued holomorphic or meromorphic backward branch satisfying the equation exists on a full punctured disc about $z = -2$. For every real $b \in (1, \eta)$, the sewn solution K_b^W has a simple zero at $z = -1$, so this description applies to it and to its local holomorphic parameter family.

Proof. The quotient $u_b(t)$ extends jointly holomorphically through $t = 0$. After shrinking U and r, it has a joint holomorphic logarithm. Two logarithms of $tu_b(t)$ on a connected sector differ by a constant multiple of $2\pi i$, giving the formula. A turn about $t = 0$ changes $\operatorname{Log} t$ by $2\pi i$. If a single-valued backward branch existed on the punctured disc, integrating the logarithmic derivative of $\exp(\mathcal{L}G) = tu$ around a small circle would give both 0 and $2\pi i$. A meromorphic branch has no pole away from the centre, since its exponential would have an essential singularity there; it is therefore subject to the same contradiction.

For the sewn family, the upper chart of Proposition 3.4 contains $w = -1$ and its inverse uniformising coordinate $w(z)$ is conformal, with $w(-1) = -1$. For real b, $R_b'(0) = \sigma(1) \log \lambda/\sigma'(1) \ne$

0, since $\sigma(1) \neq 0$ and $\sigma' > 0$ on $(-\infty, L)$ (forward iteration), and differentiating the functional equation at $z = -1$ gives $R_b'(-1) = R_b'(0)/\mathcal{L}(b) \neq 0$. Thus $(K_b^W)'(-1) = R_b'(-1)w'(-1) \neq 0$. Proposition 3.5 preserves this simple zero on a sufficiently small complex parameter disc about each real base. □

Corollary 10.51 (Conjugate inner branches and their jump)**.** *For real $b \in (1, \eta)$ put $K_b^{W,\#}(z) = \overline{K_b^W(\bar{z})}$ on the reflected regular domain. This reflected family is locally jointly holomorphic in (b, z) near every such real base. Both branches obey the tetration equation, are normalised at $z = 0$, agree at every nonnegative integer height where defined, and have a simple zero at $z = -1$ with the same logarithmic monodromy $2\pi\mathrm{i}/\log b$ at $z = -2$. For x_* as in Corollary 10.49,*

$$|K_b^W(x_*) - K_b^{W,\#}(x_*)| \geq 2c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

More precisely, on a fixed real interval about 0 the quotient of their difference by $\Lambda(b)$ tends locally uniformly to $2\mathrm{i}(F_\eta^{\mathrm{att}})'(x)\,\mathrm{Im}[A(\mathrm{e}^{2\pi\mathrm{i}x} - 1)]$.

Proof. For complex b near a real base, define the reflected family by $K_b^{W,\#}(z) = \overline{K_{\bar{b}}^W(\bar{z})}$. Proposition 3.5 makes it jointly holomorphic. Reflection preserves the equation for real b and therefore, by the identity theorem, on its local complex parameter disc. The integer values follow from the equation and normalisation. The zero at -1 and its simplicity follow from Proposition 10.50, as does the monodromy. On real heights the difference is exactly $2\mathrm{i}\,\mathrm{Im}\,K_b^W(x)$, so the lower bound and the profile follow from Corollaries 10.49 and 10.48. □

Proposition 10.52 (The cusp splitting is beyond all algebraic orders)**.** *On a sufficiently small real interval J about 0, put*

$$M(x) = (F_\eta^{\mathrm{att}})'(x)\,\mathrm{Im}[A(\mathrm{e}^{2\pi\mathrm{i}x} - 1)].$$

The interval J may be chosen so that $M(x) \neq 0$ for every $x \in J \setminus \{0\}$. For every such x and every integer $q \geq 1$, the germ along $t > 0$ of

$$t \longmapsto K_{\eta - t^q}^W(x)$$

has no meromorphic extension through $t = 0$. In particular, no finite ramification of the base parameter makes this value meromorphic at the real cusp; it has no convergent Puiseux–Laurent germ there. If it admits a single-valued holomorphic continuation on a full punctured t-disc, the singularity at $t = 0$ is essential.

Proof. The profile M is real analytic, vanishes at 0, and, by Corollary 10.48, is not identically zero on any interval about 0. Thus 0 is an isolated zero, and we shrink J accordingly. Suppose such a meromorphic germ $G(t)$ existed. Its reflection $G^{\#}(t) = \overline{G(\bar{t})}$ would also be meromorphic, so $D(t) = G(t) - G^{\#}(t)$ would be meromorphic near 0. For positive real t, the imaginary profile and the saddle-node scale of Λ give

$$D(t) = 2\mathrm{i}\,\mathrm{Im}\,K_{\eta - t^q}^W(x) = O\Big(\exp(-c_0 t^{-q/2})\Big) = o(t^N) \qquad \text{for every } N \geq 0$$

with some $c_0 > 0$. A meromorphic Laurent germ with this decay on the positive axis is identically zero. This contradicts the nonzero limit $D(t)/\Lambda(\eta - t^q) \to 2\mathrm{i}M(x)$. □

Corollary 10.53 (A two-jet witness at the normalisation height)**.** *Put $f = F_\eta^{\mathrm{att}}$ and $A = B_1\mathrm{e}^{2\pi\mathrm{i}a} \neq 0$. As $b \to \eta^-$ through real bases,*

$$\frac{1}{\Lambda(b)}\begin{pmatrix} \mathrm{Im}[(K_b^W)'(0)] \\ \mathrm{Im}[(K_b^W)''(0)] \end{pmatrix} \longrightarrow \begin{pmatrix} 2\pi f'(0)\,\mathrm{Re}\,A \\ 4\pi f''(0)\,\mathrm{Re}\,A - 4\pi^2 f'(0)\,\mathrm{Im}\,A \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Consequently at least one of the first two z-derivatives at 0, with the same choice of derivative for every $q \geq 1$, has no meromorphic base germ through $t = 0$ after $b = \eta - t^q$. The corresponding

derivative of the reflected inner branch differs from that of K_b^W by at least $c\Lambda(b)$ in modulus for some $c > 0$ and all sufficiently close real $b < \eta$.

Proof. The complex limit of Corollary 10.48 and Cauchy's formula give convergence of the first two derivatives. The derivatives of R_b at 0 are real for real b, giving the displayed vector. Since $f'(0) \neq 0$ by Corollary 6.5, vanishing of its first entry forces $\operatorname{Re} A = 0$, and then vanishing of its second forces $\operatorname{Im} A = 0$, a contradiction. For a nonzero entry, the reflection argument in the proof of Proposition 10.52 applies to that derivative. Reflection doubles its imaginary part, which also gives the lower bound. □

Remark 10.54 (Conjugate bypasses of the cusp)**.** Thus the monodromy of Remark 10.16(1) is quantified by Corollary 10.51.

11. Numerical results

11.1. **Ladder computations of a two-sided numerical solution.** In this numerical section, K_b denotes the extrapolated values from a pointwise two-fixed-point construction; no base continuation is assumed. The solution is computed at complex bases $b + \mathrm{i}y$ by a two-fixed-point θ-mapping construction. The program represents it as $R_{b+\mathrm{i}y} \circ (\mathrm{id} + \theta_{\mathrm{up}})$ in the upper half-plane and $S_{b+\mathrm{i}y} \circ (\mathrm{id} + \theta_{\mathrm{dn}})$ in the lower one. The computation runs along ladders $y \downarrow 0$ at working precisions up to 52 digits and is extrapolated to $y = 0$. Then c_1 is read off from $D = K - R$. First-mode dominance is confirmed by the constancy of $q(z) = D(z)/(R'(z)(e^{2\pi \mathrm{i}z} - 1))$ over four points, including $z = \frac{1}{2} - \frac{\mathrm{i}}{2}$, where the first mode is amplified by e^{π} (Table 2). Omitting the constant mode $c_0 = -\sum_{n \geq 1} c_n$ produces a spread of 1.917 instead.

Table 2. The ratio $|q|$ at four points. The last column is $\max|q|/\min|q|$.

base	$z = \frac{1}{2}$	$z = -\frac{1}{2}$	$z = \frac{1}{4}$	$z = \frac{1}{2} - \frac{\mathrm{i}}{2}$	spread
$1.1 + 0.0005\mathrm{i}$	$1.987\,636 \times 10^{-9}$	$1.987\,636 \times 10^{-9}$	$1.987\,636 \times 10^{-9}$	$1.987\,637 \times 10^{-9}$	1.0000
$1.05 + 0.003\mathrm{i}$	$1.422\,053 \times 10^{-7}$	$1.422\,053 \times 10^{-7}$	$1.422\,053 \times 10^{-7}$	$1.422\,061 \times 10^{-7}$	1.0000

Table 3 lists $|\hat{c}_1| = |c_1|/\Lambda$. Over the range shown, $|D|$ varies by forty-five orders of magnitude while $|\hat{c}_1|$ varies by less than 5% and increases towards $|B_1|$. The row $b = \sqrt{2}$ uses Paulsen's value $\operatorname{Im} K_{\sqrt{2}}(\frac{1}{2}) \approx -1.18899697 \cdot 10^{-48}$, from a 120-digit contour computation [25]. It shares neither code nor method with ours. Solving the sewing equation directly at $b = \sqrt{2}$ (80 digits, 32 Fourier modes, residual $< 10^{-75}$; `paulsen_sqrt2_check.py`) gives $c_1 = -2.0924123 \cdot 10^{-50} + 1.4780488648 \cdot 10^{-48}\,\mathrm{i}$, $R'_{\sqrt{2}}(\frac{1}{2}) = 0.4022184246$, and $\operatorname{Im} K^W_{\sqrt{2}}(\frac{1}{2}) = \operatorname{Im}[R'(\frac{1}{2})P(\frac{1}{2})] + O(\Lambda^2) = -1.18899697185 \cdot 10^{-48}$. This matches Paulsen's value to relative accuracy $1.6 \cdot 10^{-9}$, the rounding of his quoted digits.

The ladder values agree with the Kneser-free prediction τ_1 and t_2/t_1^2, moduli and arguments, to the precision of the extrapolation (Table 1). Solving the sewing equation of Theorem 6.3 numerically gives $\hat{c}_1 = \tau_1(1 + \vartheta)$ with $|\vartheta| = 0.38\Lambda$ at $b = 1.10$ and 0.40Λ at $b = 1.30$. This matches the first-order formula $\vartheta \approx \Lambda[-4\pi^2|\tau_1|^2 - 2\pi\mathrm{i}\tau_1 - 4\pi\mathrm{i}\tau_2\bar{\tau}_1/\tau_1]$.

11.2. **Approach to the parabolic limit.** The transition map needs no ladder. Its modes are measured on the line $\operatorname{Im} z = -h_2/2 + Y$ of repelling time, near the gate carrying the positive modes, so the working precision needed is independent of h. Results for $Y = 1, 1.5, 2, 2.5, 3$ agree to all printed digits. Table 4 follows the invariants to $\eta - b = 2.7 \cdot 10^{-5}$.

TABLE 3. Ladder values of $|\hat{c}_1|$ and their ratio to $|B_1|$.

b	λ	Λ	$\lvert\hat{c}_1\rvert$	$\lvert\hat{c}_1\rvert/\lvert B_1\rvert$	source
1.05	0.051362	1.68×10^{-6}	0.085017	0.95462	ladder, 20 digits
1.10	0.105964	2.30×10^{-8}	0.086427	0.97045	ladder, 20 digits
1.15	0.164802	3.10×10^{-10}	0.087205	0.97919	ladder, 20 digits
1.20	0.229312	2.28×10^{-12}	0.087732	0.98511	ladder, 20 digits
1.25	0.301735	4.91×10^{-15}	0.088125	0.98952	ladder, 24 digits
1.30	0.385935	9.82×10^{-19}	0.088434	0.99299	ladder, 32 digits
$\sqrt{2}$	0.693147	1.66×10^{-47}	0.088941	0.99868	[25]

TABLE 4. Kneser-free invariants near η. Here $\delta_1 = 1-|\tau_1|/|B_1|$ and $p = |\log\lambda|\log\lambda_2$.

λ	$\eta-b$	$\lvert\tau_1\rvert/\lvert B_1\rvert$	δ_1/p	$\arg\tau_1$	$\lvert t_2/t_1^2\rvert$	$\lvert t_3/t_1^3 - B_3/B_1^3\rvert$
0.5	9.04×10^{-2}	0.99601457	0.0102076	1.581415	1.555683	0.148
0.8	1.21×10^{-2}	0.99952725	0.0102002	1.585920	1.572187	0.0177
0.9	2.84×10^{-3}	0.99989062	0.0101993	1.586391	1.573905	0.00410
0.95	6.87×10^{-4}	0.99997362	0.0101991	1.586499	1.574298	0.000989
0.98	1.08×10^{-4}	0.99999587	0.0101990	1.586528	1.574403	0.000155
0.99	2.67×10^{-5}	0.99999897	0.0101990	1.586532	1.574418	0.0000385
limit	0	1	0.0101990063	1.5865331	1.5744227	0

11.3. **The deficit law.** The natural parameter is the product of the logarithmic multipliers, $p = |\log\lambda|\log\lambda_2 = \varepsilon^2(1-\varepsilon/3+O(\varepsilon^2))$. Up to an analytic factor, this is the canonical parameter of the Mardešić–Roussarie–Rousseau normal form [21]. Fits of degree 2, 3 and 4 in p over twelve bases agree to eleven digits:

$$\log\frac{\tau_n}{B_n e^{2\pi i n a}} = \kappa^{(n)}p + O(p^2), \tag{18}$$

$$\kappa^{(1)} = -0.0101990063459 - 0.0132509561825\,\mathrm{i},$$
$$\kappa^{(2)} = -0.0510565562 - 0.0623157805\,\mathrm{i},$$
$$\kappa^{(3)} = -0.1400443481 - 0.1692244639\,\mathrm{i},$$

with p^2-coefficients of order 10^{-4} and constant terms below 10^{-16}. Hence $\delta_1 = Cp + O(p^2)$ with $C = 0.0101990063$. Theorem 5.6 proves the first-order statement and computes $\kappa^{(n)}$ independently of any fit (Section 5). By Theorem 6.3, the Kneser-side $\hat{c}_n$ inherit (18) up to $O(\Lambda)$, which is beyond all orders in p. The $\kappa^{(n)}$ are not linear in n, so the first-order effect of the unfolding is not a translation of Fatou coordinates.

11.4. **Identification with an independent Kneser-type implementation.** The sewing solution can also be compared with a completely independent two-fixed-point computation at individual bases. Sheldon Levenstein's PARI/GP program `fatou.gp` builds the "merged" two-fixed-point solution for complex bases. At each base it constructs, independently, the solution with a θ-mapping at each of the two fixed points: the two regular representations of [25, Prop. 1 and Lemma 1]. Its author regards this as the continuation of Kneser's solution in the base. We use the unmodified program (sha256 `70559daf...`): published reference values with 100 verified digits at $b = 0.8+0.4\mathrm{i}$ and $b = 1+\mathrm{i}$, and our own runs at `\p 40` at $b = 1.2+0.1\mathrm{i}$ and $b = 1.3+0.2\mathrm{i}$, near the interval $(1,\eta)$. In those runs $\mathrm{sexp}(0) = 1$ to 10^{-53}, and the functional equation holds to 10^{-54}.

We solved the sewing equation of Theorem 6.3 directly at these complex bases (script `weld_complex.py`). R is the regular solution at the attracting fixed point and S the one at the repelling fixed point; the seam lies in the band of holomorphy of $T = R^{-1} \circ S$, and the root is selected as in Definition 8.1. Table 5 shows the result. At the three bases where the band is not too narrow, the sewing solution agrees with the independent value *to the full working precision of* 40 *digits*. That is 27 to 35 orders of magnitude below the size of the separation $|\mathcal{K} - R|$ itself. At $1 + \mathrm{i}$ the band is narrowest and the agreement ($2 \cdot 10^{-25}$) is limited by the number of samples. The neighbouring roots $c_{\pm 1}$ reproduce R instead, and miss by the full separation. The θ-mapping builder used for the ladders does not converge at $0.8 + 0.4\mathrm{i}$, whereas the sewing iteration converges in a few dozen steps.

TABLE 5. Sewing solution versus the independent `fatou.gp` values of $\mathcal{K}_b(\frac{1}{2})$ (working precision 40 digits).

b	$\lvert\Lambda\rvert$	$\lvert\mathcal{K}_b(\frac{1}{2}) - R_b(\frac{1}{2})\rvert$	$\lvert K_b^W(\frac{1}{2}) - \mathcal{K}_b(\frac{1}{2})\rvert$	$\lvert\hat{c}_1\rvert$
$1.2 + 0.1\mathrm{i}$	1.91×10^{-11}	6.91×10^{-13}	4.6×10^{-40}	0.0869956
$1.3 + 0.2\mathrm{i}$	2.92×10^{-11}	1.81×10^{-12}	8.3×10^{-41}	0.0876095
$0.8 + 0.4\mathrm{i}$	1.05×10^{-3}	9.86×10^{-5}	5.5×10^{-39}	0.0856072
$1 + \mathrm{i}$	2.13×10^{-2}	4.41×10^{-3}	2.0×10^{-25}	0.0873850

The agreement identifies the sewing solution with an independent implementation of the two-sided regular representation, at four bases, two of them near $(1, \eta)$. It does *not* test the analytic continuation across the Shell–Thron boundary: both programs work at one base at a time. The continuation itself is proved near the cusp in Section 10.

11.5. **The jump data of [26].** For $1 < b < \eta$ the base-continued values from above and below the real axis are complex conjugate. Paulsen [26, (29)] lists $G(b^+) - G(b^-) = 2\mathrm{i}\,\mathrm{Im}\,\mathcal{K}_{b^+}(\frac{1}{2})$ at seven bases in $[1.175, \sqrt{2}]$ and fits them by $K(\eta - b)^m \exp(C/(b - \eta))$, with $m \approx 20.7$. By Corollary 10.41 these are values of K_b^W, and by Corollary 1.1 their scale is $\Lambda = \exp(4\pi^2/\log\lambda)$, which behaves like $\exp(-C_\eta(\eta - b)^{-1/2})$, not like $\exp(C/(b - \eta))$. The quotient by Λ ($\lambda = -W_0(-\log b)$) is

b	1.175	1.2	1.25	1.3	1.35	1.4	$\sqrt{2}$
$\log\Lambda$	−24.24	−26.81	−32.95	−41.47	−55.30	−86.87	−107.71
$\lvert\mathrm{Im}\,\mathcal{K}_b(\frac{1}{2})\rvert/\Lambda$	0.0273	0.0318	0.0410	0.0503	0.0596	0.0689	0.0716

Over this range $|\mathrm{Im}\,\mathcal{K}_b(\frac{1}{2})|$ changes by 36 orders of magnitude and the quotient by a factor 2.6. The exponent m of the fit absorbs the difference between the two laws.

11.6. **A pointwise test near the Shell–Thron boundary.** Before Theorem 10.40 was available, we tested the pointwise output of the unmodified `fatou.gp` (`\p 24`, and 50 digits in `boundary_multiheight.py`) at six bases along the normal $\lambda = re^{\mathrm{i}\phi}$ through the boundary point $b_c = \exp(\lambda_c e^{-\lambda_c}) \approx 1.5217 + 0.0148\mathrm{i}$, $\lambda_c = e^{\mathrm{i}\phi}$, $\phi = 2\pi(\sqrt{2} - 1)/5$: three inside ($r = 0.98, 0.99, 0.995$) and three outside ($r = 1.005, 1.01, 1.02$). At heights $z = \frac{k}{6}$, $k = 1, \ldots, 5$, a degree-4 polynomial in b through five bases predicts the sixth to $\le 1.7 \cdot 10^{-12}$, and quadratics fitted on the two sides differ at b_c by $\le 7.1 \cdot 10^{-9}$, which is the size of the truncation term (raw values in `data/boundary-multiheight.json`). The two sides use structurally different representations, so this was an independent check of the smooth join proved in Theorem 10.34.

12. Open problems

The boundary arc up to $\mathrm{e}^{-\mathrm{e}}$ and the identification with Paulsen's base-continued family, under the analytic-continuation interpretation of Corollary 10.41, are settled by Theorems 10.34 and 10.40. The following questions remain.

(1) *Endpoint asymptotics and uniqueness.* Theorems 10.45 and 10.22 extend the upper family across every $b \in (0, \eta) \setminus \{1\}$. Proposition 10.46 shows that $b = 1$ and $b = 0$ cannot be removable singularities of a normalised jointly holomorphic family on a common forward-invariant height domain. Paulsen [26] studies selected noninteger heights as the base approaches 1. Determine uniform noninteger-height asymptotics for the present upper continuation near $b = 1$ and $b = 0$. A related question is a global uniqueness theorem for Kneser-type solutions inside the Shell–Thron region. The condition by limits at $\pm\mathrm{i}\infty$ alone admits solutions of the form $R \circ (z + 2\pi\mathrm{i}mz/\log\lambda + \cdots)$.

(2) *The first-order coefficients.* Theorem 5.6 expresses $\kappa^{(n)}$ as a convergent series over parabolic orbits. Is there a closed form, for instance in terms of the horn-map data B_n, a and the iterative residue? Determine the radius of convergence and the optimal Gevrey order of the expansion in p. The $\frac{1}{2}$-summability results of [6, 27] are in a different normalisation.

(3) *Remaining certification.* The coordinate a, the moduli of the first three horn coefficients, and the two horn ratios used in the numerical comparisons have interval enclosures above. Give interval enclosures of the ladder and extrapolated two-fixed-point values. This would turn the remaining numerical comparisons into computer-assisted ones.

Code and data availability

The exploratory computations use the Python library `kneser` (mpmath); the interval certificates use python-flint/Arb. The transition-map invariants are computed by `transition_map.py`, `eta_approach.py`, `eta_sweep.py`, `deficit_param.py` and `deficit_modes.py`. The parabolic data come from `parabolic_horn.py` and `parabolic_horn_inverse.py`. The first-order series and the second-order check are `mrr_derivative.py` and `mrr_second.py`. The boundary comparison is reproduced by `boundary_multiheight.py` from the unmodified `fatou.gp`. The confluence check is `confluence_check.py`, and the sewing solution is computed by `weld_solve.py`. The complex-base sewing solution and its comparison with `fatou.gp` are computed by `weld_complex.py`. The ladder computations of K_b use the complex-base builder of the library together with `c1_invariant.py` and `first_mode_check.py`. The signs and branches used in Section 10 (normal component of the step, the first crossing of the base orbit, and the seam height $-h/2$ through the lower half-plane) are checked numerically by `cusp_tilted_check.py`. The interval certificates are `certify_far_collar.py` (boundary collar), `certify_last_boundary_arc.py` (last boundary arc), `certify_upper_halfplane.py` (upper half-plane), `certify_endpoint.py` (endpoint segment), `certify_parabolic_a.py` (the coordinate a) and `certify_horn_coefficients.py` (horn coefficients).

Code and data are available from the author on request.

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The proofs in Sections 4–8 and 10 were checked by AI-assisted adversarial review, and the resulting corrections are incorporated. The author thanks the tetration forum for the `fatou.gp` reference values.

Appendix A. Interval certificates

All certificates are Python programs using python-flint/Arb ball arithmetic with outward rounding; every strict inequality is decided on enclosures, and floating point is used only to choose centres of Krawczyk boxes. Krawczyk's operator $K = m - Yf(m) + (1 - Yf'(B))(B - m)$,

$K \subset \operatorname{int} B$, proves existence and uniqueness of a zero in B for every parameter in the given balls. Running times are on a laptop.

A.1. **The boundary collar.** *Lemma 10.26.* Interval arithmetic with outward rounding (`certify_far_collar.p` python-flint/Arb), on boxes in (p, c) and (p, v) and subintervals of t. On the cone the quantities in (1) are divided by p, and $|\lambda_+|^2 - 1$ on its top edge is divided by p; this makes them analytic at $p = 0$. On Θ_{K} each strip of width $(P_1 - P_0)/440$ is certified up to the larger value of V at its ends. The top-edge test is made at the smaller value; this suffices because $|\lambda_+|$ decreases in q. They are evaluated through the entire series $g_\theta = -\mathrm{i}\theta\,(\theta/\sin\theta)\,\mathsf{h}$, $\mathsf{h} = -\sum_{n\ge 2} \mathrm{i}^n \theta^{n-2} P_n(t)/n!$ with $P_n(t) = (t^n - t^{n \bmod 2})/(t^2-1)$, $|P_n| \le n/2$ on $[-1,1]$, and the Bernoulli series of μ, each with an explicit tail bound. For $p \to 0$ the limits are $\operatorname{Im} g/p \to -\frac12$ and $\operatorname{Im}(\bar{\mathsf{s}}' g)/p \to -\frac12$.

Proof of Lemma 10.28. $w_{-1} = 0$ lies above the chord: $\zeta(0) = \mathrm{i}\cot\theta$, and Lemma 10.26(3) applies. Put $q_n = (\zeta_n - 1)/(\zeta_n + 1)$, so that $\zeta_n = (1+q_n)/(1-q_n)$ and $\operatorname{Im}\zeta_n = 2\operatorname{Im} q_n/|1-q_n|^2$. From s_θ,

$$q_{n+1} = \frac{\mathrm{e}^{\mathrm{i}\theta\zeta_n} - \mathrm{e}^{\mathrm{i}\theta}}{\mathrm{e}^{\mathrm{i}\theta\zeta_n} - \mathrm{e}^{-\mathrm{i}\theta}} = \mathrm{e}^{\mathrm{i}\theta}\frac{\sin(\theta(\zeta_n-1)/2)}{\sin(\theta(\zeta_n+1)/2)} = q_n \mathrm{e}^{\mathrm{i}\theta}\frac{T(v_n - \mathrm{i}\theta)}{T(v_n)}, \qquad v_n = \frac{\mathrm{i}\theta}{1-q_n},$$

with $T(w) = \sinh w/w$, since $\theta(\zeta_n+1)/2 = -\mathrm{i}v_n$ and $\theta(\zeta_n-1)/2 = -\mathrm{i}(v_n - \mathrm{i}\theta)$. From the Bernoulli expansion of $\coth w - 1/w$ and $2\zeta(2k) \le \pi^2/3$: on $|w| \le \frac{9}{20}$, $|(\log T)'(w)| \le K|w|$ and $|(\log T)''(w)| \le K_2$, with $K = 1/(3(1-x)) < 0.341$, $K_2 = (1+x)/(3(1-x)^2)$ and $x = (9/20)^2/\pi^2$.

A series enclosure gives $\chi_0 = \log q_0$ with $3p < y_0 = \operatorname{Im}\chi_0 < 3.3p$ on the whole cone. Put $\chi_{n+1} = \chi_n + \mathrm{i}\theta + \log T(v_n - \mathrm{i}\theta) - \log T(v_n)$ and $y_n = \operatorname{Im}\chi_n$. The segment $v_n - r\mathrm{i}\theta = \mathrm{i}\theta(1/(1-q_n) - r)$, $0 \le r \le 1$, has modulus at most $|\theta|(1 + 1/|1-q_n|)$. Now $|1-q| \ge \sin y$ if $0 < y \le \pi/2$ and $|1-q| \ge 1$ if $\pi/2 \le y \le 3\pi/2$, whatever $\operatorname{Re}\chi$ is. So while $3p \le y_n \le \pi + 1.2p$ the modulus is at most $1.005p(1 + 1/\sin 3p) < \frac{9}{20}$, and the correction term has modulus at most $K|\theta|^2(1 + 1/\sin 3p) < 0.16p$. Hence $0.84p < y_{n+1} - y_n < 1.16p$. Let n_* be the first index with $y_{n_*} \ge \pi$. Then $y_{n_*} < \pi + 1.16p$, and $0 < y_n < \pi$, i.e. $\operatorname{Im} q_n > 0$, for $n < n_*$. So $w_0, \dots, w_{n_*-1}$ lie above the chord.

If $y_{n_*} = \pi$, then $\zeta_{n_*} \in (-1, 1)$. Otherwise let $S = \{x_{n_*-1} + \mathrm{i}y : y_{n_*-1} \le y \le \pi\}$ and $f(\chi) = \chi + \mathrm{i}\theta + \log T(v - \mathrm{i}\theta) - \log T(v)$, the map E_θ in the coordinate χ. On S one has $|1-q| \ge 1$ and $|\zeta| \le 1$, and $f' - 1 = [(\log T)'(v - \mathrm{i}\theta) - (\log T)'(v)]\, vq/(1-q)$ has modulus at most $2K_2|\theta|^2 < \frac{1}{10}$. So $\operatorname{Im} f$ increases along S, from $y_{n_*} > \pi$ to less than $\pi + 1.16p$. Thus $f(S)$ is a curve in the open lower ζ-half-plane. It ends at the point of A_θ that is the image of $x_{n_*-1} + \mathrm{i}\pi$, and it meets A_θ nowhere else, since s_θ is injective on the unit disc. S approaches the chord from above, so by Lemma 10.27(3) $f(S)$ leaves A_θ into H_θ. Hence $f(S)$ minus its end lies in the interior of H_θ, and so does ζ_{n_*}. The constants are checked at $p = P_0$ in `certify_far_collar.py`. The bounds $|\theta|(1 + 1/\sin 3p)$ and $K|\theta|^2(1 + 1/\sin 3p)/p$ increase with p, because $p/\sin 3p$ does, so the checks at P_0 cover the whole cone.

Lemma 10.29 on Θ_{K}. On Θ_{C} this is Lemma 10.28. On Θ_{K} the script encloses the orbit on parameter boxes. It iterates the exact step of the previous proof in mean-value form about the midpoint parameter. For each box it certifies $\operatorname{Im}\zeta_m > 0$ for $-1 \le m < n$, and then one of the following. Either a Krawczyk test proves $\zeta_n = N_\theta(t, \sigma)$ with $|t| < 1$, $0 < \sigma < 1$; so ζ_n lies in the interior of H_θ, by Lemma 10.27(2). Or $\zeta_{n'}$ ($n' = n$ or $n-1$) lies in a rectangle $D = [\mathsf{a}, \mathsf{b}] \times [-\rho, \rho]$ with $-1 < \mathsf{a} < \mathsf{b} < 1$, $\rho \le \frac14$, for which Krawczyk tests prove the following: every point of $D \cap \{\operatorname{Im} \le 0\}$ is $N_\theta(t, \sigma)$ with $|t| < 1$, $\sigma < 1$; and every point of $\mathsf{s}_\theta(D \cap \{\operatorname{Im} \ge 0\})$ is $N_\theta(t, \sigma)$ with $|t| < 1$, $\sigma > 0$, the root being unique in a Krawczyk box that contains the known roots $(t, 1)$, $t \in [\mathsf{a}, \mathsf{b}]$, of the real edge. In the first case $\sigma \ge 0$ is automatic, since $\operatorname{Im} N(t, \sigma) = \sigma(1 - t^2) \operatorname{Im} g > 0$ for $\sigma < 0$. In the second case the root is $(t, 1)$ on the real edge, by uniqueness; off the real axis $\sigma \ne 1$, since $\sigma = 1$ would put a point of $\mathsf{s}_\theta(D)$ on A_θ and

contradict injectivity. Since $D \cap \{\mathrm{Im} > 0\}$ is connected and, by Lemma 10.27(3), $\sigma < 1$ just above the real axis, $\sigma < 1$ there. Hence $\zeta_{n'} \in H_\theta$ if it is not above the chord, and $\zeta_{n'+1} \in H_\theta \setminus A_\theta$ otherwise. All 33 027 boxes pass (about six minutes on a laptop).

A.2. **The last boundary arc.** *Theorem 10.34.* We use the chord and the coordinate s_θ of Section 10.10. The interval certificate `certify_last_boundary_arc.py` validates a simple root

$$\theta_* = 2.29857900665128663858\ldots + 0.76604606099318995273\ldots\,\mathrm{i}$$

of $\lambda_+(\theta) = -1$. On the rectangle $R = [2.19, 2.31] + \mathrm{i}[0.71, 0.78]$ it verifies $\mathrm{Re}(\lambda_+'(\theta)/\lambda_+'(\theta_*)) > 4/5$. The Noshiro–Warschawski criterion makes λ_+ injective on R. Validated contraction boxes give its unique inverse branch $\theta(\alpha) \in R$ for $0.935\pi \le \alpha \le 0.9999\pi$; a contraction disk of radius 0.00019 about θ_* covers the remaining interval $0.9999\pi \le \alpha \le \pi$.

For $-1 \le t \le 1$ set $f_1 = \mathrm{Im}\, g_\theta(t)$ and $f_2 = \mathrm{Im}(\overline{\mathsf{s}_\theta'(t)} g_\theta(t))$. The certificate proves $f_1, f_2 < 0$ on $[-1, 19/20]$ and $\partial_t f_1, \partial_t f_2 > 0$ on $[19/20, 1]$ along this branch. The first part uses 79 parameter boxes up to 0.9999π and a fixed 0.0002 square about θ_* for the rest. The evaluations use the entire series for g_θ and its t derivative, with explicit tails and outward-rounded Arb intervals. At the endpoint of the chord there is the exact identity

$$f_1(1) = f_2(1) = -\tfrac{1}{2}\,\mathrm{Im}\,\lambda_+ = -\tfrac{1}{2}\sin\alpha < 0 \qquad (\alpha < \pi).$$

The positive derivatives therefore close the inequalities all the way to $t = 1$, even as their endpoint margin tends to zero when $\alpha \to \pi$. The same certificate gives, by Krawczyk inclusion, $\zeta_\theta(0) = \mathrm{i}\cot\theta \in \mathrm{int}\, H_\theta$ throughout the arc. Thus the first-entry index is always -1; no orbit tracking is needed.

A.3. **The upper half-plane.** The certificate consists of four parts (`certify_upper_halfplane.py`):

(T) *Tails*, $\mathrm{Re}\,\lambda \le -12$. With $\mathsf{w} = 1/\lambda$ and $\theta = \pi - \delta$, $D = \delta/\mathsf{w}$ is the fixed point of the contraction $D = -(\pi - \mathsf{w}D)\,\ell(2\mathrm{i}\mathsf{w}(\pi - \mathsf{w}D))$, $\ell(z) = -\log(1-z)/z$. After multiplication by $1/|\lambda|$ all quantities are analytic at $\mathsf{w} = 0$. F is a contraction on the fixed convex set $|\mathrm{Re}\, D + \pi|, |\mathrm{Im}\, D| \le \frac{3}{2}$, so D is unique there. The quantities are checked on boxes in $(1/|\mathrm{Re}\,\lambda|, \mathrm{Im}\,\lambda)$ that include the limit. The mark is $w_{-1} = 0$: $\zeta(0) = \mathrm{i}\cot\theta = N_\theta(t, \sigma)$ with (t, σ) near $(0, \frac{1}{2})$, by a scaled Krawczyk test. There are 6230 boxes.

(K) *Core*, $-12 \le \mathrm{Re}\,\lambda \le 5$, $|\lambda - 1| \ge 0.08$. Boxes are discarded when they certainly violate $|\lambda| \ge 1$ or $0 \le \mathrm{Im}\, g \le \pi$, or satisfy $g(\lambda) \in g(\mathbb{D})$. The last test uses that $g(\mathbb{D})$ is starlike with boundary $\rho = \mathrm{e}^{-\cos\phi}$ at argument $\psi = \phi - \sin\phi$. For $\mathrm{Re}\,\lambda \ge 5$, $|g| \le (5 + \pi)\mathrm{e}^{-5} < \mathrm{e}^{-1}$, so $g(\lambda) \in g(\mathbb{D})$. On each remaining box a Krawczyk test gives the unique root θ in a small box. Adjacent boxes are checked to carry the same root, by uniqueness in the hull of their θ-boxes. The chain starts at a real θ. Then (1) of Lemma 10.26 is checked, together with the endpoint identities $f_1(1) = f_2(1) = -\frac{1}{2}\mathrm{Im}\,\lambda$ and $f_1(-1) = f_2(-1) = \frac{1}{2}\mathrm{Im}(\lambda - 2\mathrm{i}\theta)$ and a sign of $\partial_t f$ near $t = \pm 1$. This handles the degenerate edge $\mathrm{Im}\,\lambda \to 0$ ($b \to (0, \mathrm{e}^{-\mathrm{e}})$). The mark is checked as in Lemma 10.29. A box whose θ-box lies in Θ or in its complex conjugate needs no new check. Indeed $\mathsf{s}_{\bar\theta}(t) = -\overline{\mathsf{s}_\theta(-t)}$ and $\zeta_{\bar\theta}(\bar w) = -\overline{\zeta_\theta(w)}$, so the crescent, the inequalities (with $t \mapsto -t$) and the entry index of $\bar\theta$ are those of θ reflected in the imaginary axis.

(P) *Cusp*, $|\lambda - 1| < 0.08$. Write $\lambda_+(\theta) = 1 + \mathrm{i}\theta + \theta^2 D(\theta^2)$ with $|D| \le 0.34$ for $|\theta| \le 0.09$ (certified). By Rouché, $\lambda_+(\theta) = \lambda$ has exactly one root with $|\theta| < 0.09$. In polar form $\theta = \rho\mathrm{e}^{\mathrm{i}\phi}$, $\rho \le 0.09$, the conditions $|\lambda| \ge 1$, $\mathrm{Im}\,\lambda > 0$, $\mathrm{Im}\, g > 0$, scaled by ρ, ρ, ρ^2, force $0 \le \phi \le \arctan\frac{1}{10}$. The scaled $\mathrm{Im}\, g$ vanishes identically along the real directions $\phi = 0$ and $\phi = -\pi/2$, and so does $\mathrm{Im}\,\lambda$ along $\phi = -\pi/2$; there the sign of ∂_ϕ is used. So θ lies in the cone Θ_{C}, where Lemmas 10.26 and 10.28 apply.

(I) *Interfaces and anchor.* The core roots at $\operatorname{Re}\lambda = -12$ coincide with the tail roots. In the tails the root is the unique fixed point of the contraction on the fixed set $|\operatorname{Re} D + \pi|, |\operatorname{Im} D| \le \frac{3}{2}$, so it is one continuous branch there. The core θ-boxes on cells meeting $|\lambda - 1| < 0.08$ lie in $0 < |\theta| < 0.09$, so they carry the Rouché root. At $\lambda = \cot 1 + \mathrm{i}$ the certified box contains the real root $\theta = 1$ and not 0.

A.4. **The endpoint segment.** The certificate `certify_endpoint.py` (python-flint/Arb, under a minute) checks (a)–(f) of Lemma 10.42 for all λ in the square $[-1.1, -0.9] \times [-0.1, 0.1]$ and in the strip $[-12, -\frac{1}{5}] \times [-10^{-3}, 10^{-3}]$.

- *The square.* $\operatorname{Re}(\lambda_+'(\theta)/\lambda_+'(\theta_*)) > 0$ on the convex rectangle $R' = [2.20, 2.45] \times [0.62, 0.95]$ (θ_* is the root of $\lambda_+ = -1$), so λ_+ is injective there (Noshiro–Warschawski). Krawczyk boxes inside R' give the root $\theta(\lambda)$. Conditions (a)–(f) are checked on t-intervals, and (e) by comparing $\zeta(0)$ with the arc.
- *The strip.* The first strip box lies in the square, and its root enclosure lies in R'; this anchors the strip to the square's branch. Adjacent boxes carry the same root: a Krawczyk test on the hull of their θ-boxes, for λ on the common edge, proves that the root is unique there. On t-intervals touching ± 1, where $x = \mp\infty$, conditions (d) and (e) use the one-sided bounds $x \le x_{\max}$ and $x \ge x_{\min}$.

A.5. **The parabolic constants.** *Proposition 2.6.* The certificate `certify_horn_coefficients.py` uses the first 12 formal Abel coefficients as exact rationals. For their truncation α_{12}, the Abel defect vanishes to order 14 and has modulus < 4 on $|u| = \frac{1}{2}$. Once the forward and backward orbits reach $\operatorname{Re}(-2/u) > 80$ and $\operatorname{Re}(2/u) > 80$, respectively, the defect estimate and the increase of these real parts by at least $3/4$ per step bound each Fatou-coordinate tail after 100 iterates by $2.25 \cdot 10^{-17}$, and after 1200 iterates by $4.08 \cdot 10^{-31}$.

Outward-rounded Arb arithmetic covers $0 \le \operatorname{Re} z \le 1$, $1.9 \le \operatorname{Im} z \le 2.1$ with 256×40 boxes. On each box a Krawczyk test gives a unique local inverse of Φ_{rep}; adjacent roots agree on their common edges, and the roots at $\operatorname{Re} z = 0, 1$ are related by $f(u) = e^u - 1$. The same boxes give $|D(z)| < 1$ for $D = \Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1} - \mathrm{id}$. A chain of 400 further boxes joins this branch to $u = 11\mathrm{i}/20$ in the upper gate, fixing the horn-map normalisation.

At $z_j = j/128 + 2\mathrm{i}$ the inverse and $D(z_j)$ are enclosed using 1200 iterates. For $n = 1, 2, 3$, multiply the discrete Fourier average by $e^{4\pi n}$. The analytic periodic trapezoid bound on the strip of half-width $\delta = 1/10$ makes its alias error at most

$$2e^{2\pi n(2+\delta)} \frac{e^{-2\pi(128)\delta}}{1 - e^{-2\pi(128)\delta}}.$$

Adding this bound to the Arb sample enclosures gives the stated intervals.

The coordinate a. `certify_parabolic_a.py` computes the first ten formal Abel coefficients as exact rationals. For their truncation α_{10}, the residual $\alpha_{10}(e^u - 1) - \alpha_{10}(u) - 1$ vanishes to order 12; Arb bounds its modulus by 0.438 on $|u| = \frac{1}{2}$. Put $u_0 = 1/e - 1$, $u_{k+1} = e^{u_k} - 1$, and $x = -u_{1000} < 0.002$. Since $-u_{k+1} \le (-u_k) - (-u_k)^2/3$ for $-u_k \le 1$, Cauchy's estimate bounds the tail after $\alpha_{10}(u_{1000}) - 1000$ by

$$0.438\, 2^{12} \left(x^{12} + \frac{3x^{11}}{11} \right) < 10^{-27}.$$

The orbit and truncated value are enclosed by Arb; together they give the stated interval.

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